



# Driftwood

## Setup & Mod Guide

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Minecraft 1.21.1 – NeoForge

Version 4.0.1

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# Driftwood — Setup & Mod Guide

## Wave -1 — Prerequisites

### JDK 21

| Tool                                         | Role                                                  |
|----------------------------------------------|-------------------------------------------------------|
| <a href="#">Eclipse Temurin JDK 21 (LTS)</a> | Java runtime and development kit for Minecraft 1.21.1 |

Install the **JDK** (not just JRE) — the JDK includes [javac](#) and [jlink](#) needed by NeoForge. Minimum version 21.0.0, any LTS build. On Windows, use the MSI installer and ensure `JAVA_HOME` is set to the install path (e.g., `C:\Program Files\Eclipse Adoptium\jdk-21.0.6.13-hotspot`). On Linux/macOS, use your package manager (`apt install temurin-21-jdk`, `brew install temurin21`).

Verify installation:

```
java -version
# Expected: openjdk version "21.0.x" ...
```

### Performance Tuning

Recommended JVM arguments (for XMCL → instance settings → Java → JVM Arguments) — community consensus for Java 21 + NeoForge (AllTheMods guide, CleanroomMC, Distant Horizons' own FAQ):

```
-Xms12G -Xmx12G -XX:+UseZGC -XX:+ZGenerational -XX:+AlwaysPreTouch -XX:
+PerfDisableSharedMem -XX:+DisableExplicitGC -XX:
+UseDynamicNumberOfGCThreads
```

| System RAM | Recommended allocation       |
|------------|------------------------------|
| 16 GB      | <code>-Xms8G -Xmx8G</code>   |
| 32 GB      | <code>-Xms12G -Xmx12G</code> |
| 64 GB      | <code>-Xms16G -Xmx16G</code> |

**Why ZGC:** Minecraft's default collector, G1GC, stops the world while collecting — that's the classic modpack stutter. ZGC's **generational** mode (JEP 439) collects concurrently with near-zero pauses and scales to large heaps; Distant Horizons' own FAQ recommends ZGC for Java 21+ and names G1GC as a stutter source. On Java 21 the generational mode must

be enabled explicitly with `-XX:+ZGenerational` — newer JDKs (23+) have it on by default. Temurin JDK 21 includes it.

**-Xms must equal -Xmx:** an equal initial/max heap stops the JVM from resizing the heap mid-game, which causes micro-stutters. `-XX:+AlwaysPreTouch` commits all heap pages at startup (slightly slower launch, no page-fault hitches later); `-XX:+PerfDisableSharedMem` skips `hsperfdata` writes; `-XX:+DisableExplicitGC` ignores `System.gc()` calls some mods make (with ZGC those force an expensive full collection); `-XX:+UseDynamicNumberOfGCThreads` lets GC threads scale with your CPU.

**Memory sizing:** don't over-allocate — a bigger heap means longer GC cycles even with ZGC. For a 200+ mod pack with Distant Horizons, 12–16 GB is the sweet spot on a 10+ core CPU (Distant Horizons' guidance: 6–8 GB for 4–8 cores, 12–16 GB for 10+ cores). ZGC also needs a little extra RAM headroom because it doesn't use compressed object pointers. `-XX:SoftMaxHeapSize` is optional: it caps ZGC's *typical* usage while allowing bursts up to `-Xmx` (e.g. `-Xms12G -Xmx16G -XX:SoftMaxHeapSize=12G`) — most players won't need it.

**Distant Horizons is CPU-bound, not RAM-bound** — if you see stutter while LODs generate, lower DH's CPU Load preset (Video Settings → DH menu) before raising the heap.

**Fallback — tuned G1GC** (if ZGC misbehaves on your hardware, e.g. an older CPU): keep the same `-Xms/-Xmx` from the table and swap the collector flags. G1GC is the safe default for 4–8 GB heaps but shouldn't be pushed past ~12 GB:

```
-XX:+UnlockExperimentalVMOptions -XX:+UseG1GC -XX:G1NewSizePercent=20 -XX:G1ReservePercent=20 -XX:MaxGCPauseMillis=50 -XX:G1HeapRegionSize=32M
```

**Sources:** [AllTheMods](#) — Java & Arguments, [CleanroomMC](#) — JVM Arguments, [Distant Horizons FAQ](#) — ZGC & stuttering

## X Minecraft Launcher (XMCL)

| Tool                 | Role                                     |
|----------------------|------------------------------------------|
| <a href="#">XMCL</a> | Mod manager and launcher for the modpack |

**XMCL** manages mod and modpack installation, updates, and launching for this modpack. It supports both **CurseForge** and **Modrinth** as mod sources in one launcher, handles NeoForge installation automatically, and manages Java versions per instance. Open source, free, with no premium tier.

**Setup:**

1. Download and install **XMCL** from [xmcl.app](https://xmcl.app) for your OS.
  - Windows: App Installer, AppX, or Zip (x64)
  - macOS: DMG (Intel or Apple Silicon)
  - Linux: Deb, RPM, AppImage, Flathub
  - Or via winget: `winget install CI010.XMinecraftLauncher`
2. Open XMCL and follow the **NeoForge** instructions below to create your instance.

Once the instance is running, install mods through the **Mods** tab from either CurseForge or Modrinth, or by dropping `.jar` files into the instance's `mods/` folder. XMCL uses hard links — mods are stored once globally and linked per instance, avoiding duplicates. All mod links in this document link to their CurseForge pages — XMCL can install from CurseForge directly via the Mods tab.

For texture packs, drop `.zip` files into the instance's **Resource Packs** tab or folder. For shaderpacks, drop `.zip` files into the instance's **Shader Packs** tab or folder. Access the instance folder from the instance settings.

## NeoForge

| Tool                     | Role                            |
|--------------------------|---------------------------------|
| <a href="#">NeoForge</a> | Mod loader for Minecraft 1.21.1 |

NeoForge is the mod loader powering this modpack. **XMCL installs NeoForge and Minecraft 1.21.1 automatically** when you create a NeoForge instance (see [XMCL](#) above). Follow the XMCL steps — it handles the Minecraft version, mod loader, and Java runtime in one step.

If you prefer the official launcher:

1. Run **Minecraft 1.21.1** vanilla at least once from the official launcher to generate the game directory.
2. Download the **NeoForge installer** for 1.21.1 from [neoforged.net](https://neoforged.net).
3. Run it: `java -jar neoforge-1.21.1-<version>-installer.jar` → select “Install client”.
4. Launch the new **neoforge** profile once to generate the `mods/` directory.
5. Verify the main menu shows “NeoForge X.X.X” in the bottom-left corner.
  - Your Minecraft directory: Windows `%APPDATA%\minecraft`, Linux `~/.minecraft`, macOS `~/Library/Application Support/minecraft`

**Important:** Do not add any mod `.jar` files yet. Wave -1 is complete when the vanilla 1.21.1 client runs under NeoForge with no mods installed.

Skip this if you’ve played Minecraft before.

## Controls (Default)

| Action         | Key         |
|----------------|-------------|
| Move           | W A S D     |
| Jump           | Space       |
| Sneak          | Shift       |
| Sprint         | Ctrl        |
| Attack / Break | Left-click  |
| Use / Place    | Right-click |
| Inventory      | E           |
| Drop           | Q           |
| Hotbar slots   | 1–9         |

## Your First 10 Minutes

1. **Punch a tree** — Hold left-click on a tree trunk until it breaks. Get 8+ logs.
2. **Make a crafting table** — Open inventory (E), place 1 log → 4 planks. Fill all 4 crafting slots with planks → crafting table.
3. **Craft tools** at the crafting table (3×3 grid):

| Tool           | Pattern                                  |
|----------------|------------------------------------------|
| Wooden Pickaxe | 3 planks top row, 2 sticks middle column |
| Wooden Axe     | 3 planks top-left, 2 sticks down         |
| Wooden Sword   | 2 planks middle column, 1 stick below    |

1. **Gather stone** — Find grey stone, mine 8–12 blocks with your pickaxe.
  2. **Build a shelter** — Dirt/wood walls + door (6 planks, 2 columns of 3) + torches (stick + coal/charcoal).
  3. **Craft a furnace** — 8 cobblestone in a ring (center empty). Use it to smelt ore, cook meat, make charcoal.
  4. **Survive the night** — Stay inside. Dig, expand, or sleep if you have a bed (3 wool + 3 planks).
  5. **Day 2 priorities:** Find food, get iron (dig staircase-down until you see brown speckled blocks), find a village, make a bed.
-

# Wave 0 — Foundation

The essentials that make the game run well. Performance, rendering, shaders, chunk loading, and texture packs. Install everything in this wave before launching.

## Performance & Rendering

| Mod                                               | Role                                                                                                                                                                                                            |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Sodium</a> 0.8.12                     | Rendering engine — pin 0.8.12+ (current 1.21.1 line; incompatible with the removed Sodium Options API chain — see STATUS). Fall back to 0.6.x only if DH/Entity Culling misbehave                               |
| <a href="#">Iris</a>                              | Shader loader (v1.8.14, NeoForge native)                                                                                                                                                                        |
| <a href="#">Lithium</a>                           | Server-side game logic optimization                                                                                                                                                                             |
| <a href="#">FerriteCore</a>                       | Memory usage reduction                                                                                                                                                                                          |
| <a href="#">Distant Horizons</a>                  | LOD rendering                                                                                                                                                                                                   |
| <a href="#">ModernFix</a>                         | All-in-one perf + bug fixes                                                                                                                                                                                     |
| <a href="#">C2ME</a> 0.4.0-alpha.0.120+1.21.1     | Parallelizes chunk generation/loading/IO across CPU cores — the biggest worldgen speedup in the pack. Server-side; pairs with Fast Noise + Lithium. ⚠ alpha backport devbuild (flagged)                         |
| <a href="#">Fast Noise</a> 1.0.13+1.21.1+neoforge | Vanilla-parity worldgen optimization — faster chunk gen (10–18% overworld, much more in Nether/End). Replaces NoisiumForked (archived — its successor)                                                          |
| <a href="#">ImmediatelyFast</a>                   | Immediate-mode rendering                                                                                                                                                                                        |
| <a href="#">Entity Culling</a>                    | Hide off-screen entities                                                                                                                                                                                        |
| <a href="#">Dynamic FPS</a>                       | Background FPS reduction                                                                                                                                                                                        |
| <a href="#">FastWorkbench</a> 1.21.1-9.1.3        | Caches crafting-table recipes — eliminates late-game crafting lag spikes                                                                                                                                        |
| <a href="#">MoreCulling</a> v1.0.9                | Block-face culling — skips rendering hidden faces (complements Entity Culling). Leaf culling: keep at <b>default</b> — Cull Leaves (Wave 0.5) owns leaves; enabling MoreCulling's leaf modes would duplicate it |

**ModernFix** (configurable in-game via Mod List → ModernFix → Config, or `config/modernfix-mixins.properties`):

- `mixin.perf.dynamic_resources = true` — Load block/item models lazily instead of all at startup. **Biggest single launch-time speedup** in the pack. Reduces

memory usage too. Disable if you see missing textures or broken models. **Fast Noise** — replaces NoisiumForked (adopted Jul 2026; NoisiumForked archived May 2026, its own maintainer recommends Fast Noise). Vanilla-parity worldgen optimization: only the block/biome storage paths in [NoiseChunkGenerator](#) are swapped, so terrain output, datapacks, and modded worldgen (Terralith/Tectonic) are unchanged. Zero deps. Risky mod-compat-affecting optimizations disabled by default. [VERIFY] at launch: loads cleanly with Lithium (load-order fix in the mod’s changelog) and the Terralith/Tectonic stack generates correctly. **C2ME (NeoForge)** — parallelizes chunk generation, loading, and IO across CPU cores (the single biggest worldgen speedup available). Server-side — runs in the single-player integrated server. Zero deps; the standard pairing with Fast Noise (block-storage opt) + Lithium (game logic) — Fast Noise’s incompatible list excludes C2ME, and C2ME’s own benchmarks run zfastnoise + Terralith. ⚠️ **Alpha backport devbuild** for 1.21.1 ([0.4.0-alpha.0.120](#), auto-uploaded GitHub Actions; “alpha” is the project’s standing versioning convention — Fabric original: 35M DL). [VERIFY] at launch: clean load with Lithium/Fast Noise; Terralith/Tectonic/Incendium terrain correct; no new world needed (parity).

## Shaderpack

| Shaderpack                    | Description                                                                                                                                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Solas Shader</a>  | Main shader — modern fantasy look by Septonious; colored lighting, ray-marched volumetrics, generated PBR + Distant Horizons support, iris-native                                                                                           |
| <a href="#">Photon Shader</a> | Alt — crisp realistic look (sixthsurge); native DH LOD support with own DH settings screen; LabPBR-only (no generated PBR — flat micro-surface on map-less Faithful 32x); ColorWheel via patcher only ([UNVERIFIED] live). Use <b>v1.3b</b> |

**Solas Shader** is the pack’s main shader (Septonious) — a modern fantasy-stylized look: smooth, saturated lighting, 3D volumetric clouds, volumetric light, colored block lighting, aurora/Milky Way skies, and an integrated generated-PBR system. Runs on Iris (NeoForge) with zero extra mods. Use **V3.7** — Distant Horizons support is native (dedicated LOD programs, no shader-side setup; V3.7 fixed DH/Voxy translucency blending). Quality profiles run Low → Medium → **High (default)** → Ultra; LabPBR/SEUS resourcepack PBR is supported alongside the internally generated PBR.

**PBR note:** Faithful 32x (the base layer) ships no PBR maps, so Solas’s **generated PBR** (on by default at High) provides the depth on vanilla blocks — the flat 32x look remains the intended baseline. Pack-provided

maps (Fire Rekindled's Normal/Specular, RAY's 3D Rails' shader support) only apply when **PBR → Material → Advanced Materials** is enabled (Material Format: **LabPBR 1.3**, the default). If you enable it, also turn **Generated Specular** OFF — the shader's own warning says not to use it with an external PBR resourcepack (only a handful of blocks carry maps, so this is a small, optional win).

**Recommended baseline (NVIDIA RTX 4080 SUPER, 16GB VRAM, 4K — target stable 70-80 FPS):** start with the **High** profile — the shader's own default — and keep its settings. If FPS dips below 70: reduce **Colored Lighting (Voxel Volume Size)** 192 → 128, then **Volumetric Light Samples** 8 → 7, then **Shadow Resolution** 2048 → 1024, then **Shadow Distance** 192 → 128 (see the table below).

### Solas Shader Settings — Recommended

Start from the **High** profile (the shader's default), then apply the table below. All options are in the shader's own settings screen: Options → Video Settings → Shaderpacks → Solas Shader → **Shader Settings** (profile picker at the top; screens: Atmospherics, Water, PBR, Lighting, Colors, Cinematic Effects & Post Processing, Other). Values are tuned for the pack's target (RTX 4080 SUPER, 4K, 70–80 FPS, DH LODs at 128–256).

| Settings screen       | Option                               | Recommended                            | Why                                                                                                                                                                                          |
|-----------------------|--------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (shader settings top) | <b>Profile</b>                       | <b>High</b> (default)                  | Solas's shipped default — volumetric clouds, volumetric light, generated PBR, aurora. Ultra's extras (entity shadows, 4096 shadow maps) blow the 70–80 FPS 4K budget for little visible gain |
| Lighting              | Colored Lighting (Voxel Volume Size) | 192 (default) → <b>128</b> if FPS < 70 | voxelization distance for colored lighting — the shader's own warning: “higher values greatly decrease                                                                                       |

| Settings screen                               | Option                    | Recommended                                 | Why                                                                                                                                                                           |
|-----------------------------------------------|---------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               |                           |                                             | performance".<br>First lever                                                                                                                                                  |
| Lighting →<br>Shadows Config                  | Shadow<br>Resolution      | 2048 (default) →<br><b>1024</b> if FPS < 70 | the classic first<br>shadow lever<br>(1024 = Medium<br>profile's value)                                                                                                       |
| Lighting →<br>Shadows Config                  | Shadow Distance           | 192 (default) →<br><b>128</b> if needed     | DH shades the<br>distance anyway<br>(128 = Medium's<br>value)                                                                                                                 |
| Lighting →<br>Shadows Config                  | Entity Shadows            | OFF (default)                               | Ultra-only feature;<br>the player's own<br>shadow always<br>renders<br>regardless                                                                                             |
| Lighting                                      | Screen Space<br>Shadows   | <b>ON</b> (default)                         | ⚠ the DH-<br>relevant one —<br>shadows keep<br>rendering past the<br>realtime distance;<br>the shader's own<br>comment: "Very<br>useful with Voxy<br>and Distant<br>Horizons" |
| Lighting                                      | Screen Space AO<br>(SSAO) | ON (default)                                | baseline<br>occlusion; AO<br>Strength 1.0                                                                                                                                     |
| Atmospherics →<br>Volumetric Light<br>Config  | Volumetric Light          | ON (default)                                | sun rays — Solas<br>signature                                                                                                                                                 |
| Atmospherics →<br>Volumetric Light<br>Config  | Samples                   | 8 (default) → <b>7</b> if<br>FPS < 70       | ray samples per<br>frame (7 =<br>Medium's value)                                                                                                                              |
| Atmospherics →<br>Volumetric Clouds<br>Config | Volumetric Clouds         | ON (default)                                | 3D clouds — the<br>pack's sky. Last<br>resort if FPS < 70:<br>turn OFF, or drop<br>Render Distance<br>400 → 200                                                               |

| Settings screen                     | Option                                                                    | Recommended    | Why                                                             |
|-------------------------------------|---------------------------------------------------------------------------|----------------|-----------------------------------------------------------------|
| PBR → Integrated PBR                | Generated Normals · Generated Specular · Generated Emission               | ON (defaults)  | procedural depth on Faithful's flat 32x — the intended baseline |
| Water                               | Water Reflections · Normals (Perlin-Worley) · Caustics · Light Refraction | ON (defaults)  | Solas water look; all cheap at 4K                               |
| Cinematic Effects & Post Processing | TAA                                                                       | ON (default)   | temporal SSAO / screen-space-shadows cleaner                    |
| Cinematic Effects & Post Processing | FXAA                                                                      | OFF (default)  | TAA supersedes it                                               |
| Cinematic Effects & Post Processing | Bloom · Lens Flare                                                        | ON (defaults)  | cheap; part of the signature glow                               |
| Cinematic Effects & Post Processing | Depth of Field · Motion Blur                                              | OFF (defaults) | keep off — build & mine clarity, and both cost frames at 4K     |

### Photon Shader Settings — Recommended

Start from the **High** profile, then apply the table below. Same screens path as Solas: Options → Video Settings → Shaderpacks → Photon → **Shader Settings**. Values tuned for the pack's target (RTX 4080 SUPER, 4K, 70–80 FPS, DH LODs at 128–256). Photon has **no generated PBR** — on map-less Faithful 32x, blocks get correct lighting but flat micro-surface; that is the expected look difference vs Solas, not a misconfiguration.

| Settings screen       | Option         | Recommended | Why                                                                                                                      |
|-----------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------------------------|
| (shader settings top) | <b>Profile</b> | <b>High</b> | VL + GTAO + SSRT shadows without Ultra's voxel colored lights (Ultra-only, Iris-only). Ultra's extras blow the 4K budget |

| Settings screen         | Option                             | Recommended                                                       | Why                                                                                                  |
|-------------------------|------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Lighting → Shadows      | Shadow Resolution                  | 2048 (default) → <b>1024</b> if FPS < 70                          | classic first shadow lever                                                                           |
| Lighting → Shadows      | Shadow Distance                    | 192 (default) → <b>128</b> if needed                              | DH shades the distance anyway                                                                        |
| Lighting → Shadows      | Screen Space Shadows (SSRT)        | ON                                                                | shades LOD terrain past the realtime distance — the DH-relevant one                                  |
| Mods → Distant Horizons | DH Overdraw Distance / Fade Length | <b>16 / 16</b> (defaults)                                         | LOD edge blending — leave at defaults unless LOD seams appear                                        |
| Mods → Distant Horizons | Noise on DH Terrain                | <b>ON</b>                                                         | high-frequency detail noise keeps far LODs legible — this is why Photon LODs read farther than Solas |
| Lighting                | Colored Lights (voxel)             | OFF (default below Ultra)                                         | Ultra-only feature — do not force-enable on High                                                     |
| Post                    | TAAU (temporal upscaling)          | OFF (default) → <b>ON at 0.75</b> as emergency lever              | reserve FPS lever before touching LOD distance; off by default so enable only if needed              |
| Clouds                  | Cloud steps                        | defaults → <b>reduce primary/lighting steps first</b> if FPS < 70 | volumetrics are Photon's biggest GPU cost after voxel lighting                                       |

**Photon FPS order:** cloud steps → SSRT steps → shadow 2048→1024 → **TAAU ON** → LOD distance down. Keep overdraw 16/noise ON throughout.

**Distant Horizons per shader:** Solas needs no shader-side setup — DH support is native (dedicated [dh\\_terrain](#)/[dh\\_water](#) LOD programs; V3.7 fixed DH/Voxy translucency blending). Two Solas options interact with LODs: **Screen Space Shadows ON** (above — shades LOD terrain past the

realtime shadow distance) and **PBR** → **Material** → **Specular & Reflections** → **LOD Reflection: Fancy (default)** — “determines the quality of DH and Voxy chunks in reflections”. **Photon** has its own DH screen (`mods` → `distant_horizons`): keep `DH_OVERDRAW_DISTANCE/`  
`DH_OVERDRAW_FADE_LENGTH` at 16, `NOISE_ON_DH_TERRAIN` ON. Both: DH cloud rendering stays OFF (each shader draws its own clouds). Full research: [docs/shader-comparison.md](#).

**General MC settings:** Graphics **Fancy**, Render Distance **10 chunks**, Simulation Distance **8 chunks**, Biome Blend **2 blocks**, Mipmap Levels **4x**, Entity Distance **100%**, Fullscreen ON, VSync OFF, Max Framerate **120**. Always Defer Chunk Updates **Enabled**, Use No Error Context **Enabled**.

Distant Horizons handles far rendering — keep vanilla render distance low. DH LODs render through the shader pipeline; start LOD distance at **128** and push to **256** if the FPS budget allows. Monitor VRAM usage (shaders + DH + high-res texture packs can peak past 12GB at 4K). If you experience dips below 70 FPS, first reduce Shadow Resolution to **1024**, then lower Shadow Distance to **8 chunks**.

## Texture Packs

| Pack                             | Role                                                                                                             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------|
| Improved Create 32x              | 32x32 texture extension for Create — items, blocks, GUI; designed to pair with Faithful 32x                      |
| Improved Applied Energistics 2   | 32x32 texture extension for AE2 — items, blocks, GUI; designed to pair with Faithful 32x                         |
| Upscaled Mekanism 32x v1.5       | Faithful-style 32x textures for Mekanism — blocks, items, GUIs; pairs with Faithful 32x like Improved Create/AE2 |
| Mod Descriptions                 | Field Guide companion — lore descriptions for 200+ mods’ entries (lang-only, order irrelevant)                   |
| Fresh Animations                 | Living entity animation overhaul — mobs blink, look around, show emotion                                         |
| Fresh Animations: Objects v2.1.2 | Official FA extension — animates chests, boats, minecarts, shulker boxes                                         |

| Pack                              | Role                                                                                                                                                                                                                             |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | (wobble, slide, open/close). Load above Fresh Animations                                                                                                                                                                         |
| Fresh Animations: Emissive v1.6.0 | Official FA extension — emissive textures for Drowned/Stray/Bogged, Iron Golem, Guardians, firing Ghast, angry Bee/Wolf, Zombie Villager, Piglins, Hoglins, Bat. Pairs with ETF + LambDynamicLights. Load above Fresh Animations |
| Enhanced Boss Bars                | Redesigned boss bars with themed designs for vanilla and modded bosses                                                                                                                                                           |
| Visual Traveler's Titles          | Handmade dimension logos for Traveler's Titles entry cards — Overworld/Nether/End + Aether, Twilight Forest, Northstar planets. Uncovered dims fall back to text                                                                 |
| Dark Coffee GUI v2.2.1            | Warm dark coffee-themed 16x GUI retexture — all inventories, hotbar, hearts/armor/food icons, buttons, toasts. <b>Replaces Mindful Darkness' GUI darkening (full swap, Aug 2026)</b>                                             |
| Modded Coffee GUI v1.15.2         | Dark Coffee GUI's official mod-compatible addon (same author) — coffee-styled screens for EMI, Create, Farmer's Delight, Waystones, Relics, Curios, Supplementaries, etc.; load above the base pack                              |
| RAY's 3D Rails                    | 3D block & item textures for all rail types                                                                                                                                                                                      |
| RAY's 3D Ladders                  | 3D block & item textures for ladders — vanilla + Create; separate pack from 3D Rails (same author)                                                                                                                               |
| Better Lanterns                   | Enhanced 3D look for lanterns and chains                                                                                                                                                                                         |
| (Bee's) Fancy Crops               | Reworked crop models with texture variations — compatible with Farmer's Delight                                                                                                                                                  |
| Rekindled CTM                     | Connected textures for Fire/Soul Fire + moss carpet overhangs — official Fire Rekindled addon (needs Continuity)                                                                                                                 |

| Pack                                                             | Role                                                                                                       |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <a href="#">Fire Rekindled</a>                                   | Animated fire, lava, water, and portal textures — includes Create mod support                              |
| <a href="#">Motsch's Better Leaves</a>                           | Bushier, denser leaves with explicit mod support — pair with Cull Leaves for FPS                           |
| <a href="#">Bushy Pink Petals, Wildflowers &amp; Leaf Litter</a> | 3D models for pink petals, wildflowers, and leaf litter on the ground                                      |
| <a href="#">Modded Omelet — Modern Spawn Eggs</a>                | 1.21.5-style spawn eggs backported + restyled for 204 mods — matches pack mob lineup                       |
| <a href="#">Mace Fusion</a>                                      | Cleaner 3D model + 32x texture for the vanilla mace                                                        |
| <a href="#">Fresh Waystones Texture</a>                          | Improved textures for the Waystones mod block + item                                                       |
| <a href="#">Overlay's</a>                                        | Dynamic CTM overlays — dirt moisture, cracks, smooth block transitions (requires Continuity + Connector)   |
| <a href="#">MissingSoundsFix</a>                                 | Suppresses vanilla empty-sound warnings (salmon.ambient, cod.ambient, etc.) — fixes log spam from MC-97521 |
| <a href="#">Faithful 32x</a>                                     | Full 32x base pack — complete coverage, cohesive palette; load at bottom of pack order                     |
| <a href="#">Create Immersive Aircrafts 3.0</a>                   | Create-styled textures for Immersive Aircraft items & vehicles — pairs with its Recipes data pack          |

- **Improved Create 32x** — 32x32 textures for Create (items, blocks, GUI); partly based on Faithful 32x assets — designed to pair with it. Coverage still incomplete; author requires top placement.
- **Improved Applied Energistics 2** — 32x32 textures for AE2 (items, blocks, GUI); same author + Faithful pairing as Improved Create 32x. Coverage still incomplete; author requires top placement.
- **Upscaled Mekanism 32x** — Faithful-style 32x textures for Mekanism (blocks, items, GUIs, cables/pipes); covers Generators, Additions, Tools, and JEI Mekanism Multiblocks. v1.5 (Aug 13 2026) supports 1.21.1 (verified via Modrinth API). Zero deps. **Load in the 32x tier above the Faithful base.**

Mekanism screens stay bright — neither this pack nor the Coffee packs darken them (see Modded Coffee GUI).

- **Mod Descriptions** — Field Guide companion (resource pack, lang-only): adds lore descriptions for modded entries in the catalog — 200+ mods, incl. full coverage for Farmer's Delight, Brewin' And Chewin', Supplementaries, Spice of Life: Carrot Edition, Explorer's Compass, Waystones and partial for Cataclysm, Simply Swords, Upgrade Aquatic. Requires the Item Descriptions mod (already in pack as a required Field Guide companion). Load position irrelevant.
- **Fresh Animations** — idle animations, directional looking, sleep/blink, emotions for all vanilla mobs. Resource pack overlay (not a mod) — place high in pack order.
- **Fresh Animations: Objects** — official FA extension (FreshLX, v2.1.2 Jun 2026, 1.21.1 , [dependencies: \[\]](#)): animates non-mob entities base FA doesn't touch — chests (open/close anims, custom trapped-chest button), boats (wobble, slide on turns, land anims, idle oar positions), minecarts (wheels + track wobble), shulker boxes, ender chests. **Load directly above Fresh Animations.** 16x-vanilla-styled like the base — inherits the 16x-on-32x tradeoff already accepted for FA.
- **Fresh Animations: Emissive** — official FA extension (FreshLX, v1.6.0 Apr 2026, 1.21.1 , [dependencies: \[\]](#)): emissive textures for Drowned, Stray, Bogged, Iron Golem, Guardian & Elder Guardian, firing Ghast, angry Bee, angry Wolf, Zombie Villager, Piglins, Hoglins, Bat. Rendered via ETF (already in pack) and *lit* via LambDynamicLights — glowing mobs that cast real light (a firing Ghast lighting the cave is a genuine gameplay read). **Load directly above Fresh Animations.**
- **Enhanced Boss Bars** — themed boss bar designs for vanilla + Twilight Forest. For Cataclysm/Aether bars (custom rendering), also install the companion [Enhanced Boss Bars mod](#). **On install, set Jade's boss-bar options** (Main Menu → Mods → Jade → Config → Mod Settings):
  - **Display Bosses** → **OFF** (Jade's vanilla boss-bar overlay would fight the pack's themed bars)
  - **Boss Bars Overlap** → **Push Down Tooltip** (Jade tooltips push below the bar instead of covering it)
- **Visual Traveler's Titles** — handmade dimension logos for the Traveler's Titles mod's entry cards (un\_roman, 12.6M DL, co-authored with YUNGNIKYOUNG). Covers Overworld/Nether/End plus **Aether, Twilight Forest, and Northstar planets** (Mars/Mercury/Moon/Venus) — the pack's dimension lineup. Uncovered dims (Bumblezone, Deeper and Darker, Cataclysm arena) fall back to text titles, gracefully. No conflicts — title-card overlay touches nothing Dark Coffee/Faithful/CEM packs own. Standard Visual variant (Legacy's extra compat unneeded for our dims).

- **Dark Coffee GUI** — warm dark coffee-themed retexture of the vanilla GUI (16x): all inventories, a brand-new hotbar, remade hearts/armor/food icons, buttons, toasts. **Replaces Mindful Darkness (full swap, Aug 2026)** — the pack IS the dark UI now; the Mindful Darkness config override was deleted (running both would double-darken the already-dark textures). v2.2.1 supports 1.21.1 (verified via Modrinth API). Ships a **NOSHADERS variant** — use it if the pack’s shader-based text recolor looks off under Iris. **Modern UI compatible:** its documented conflicts are tooltip/text-engine mods, not GUI-texture packs (the instance already proves the surface with Faithful 32x); no shader-pipeline fight (unlike the old Dark Mode Everywhere watch). The pack’s text recolor follows Minecraft’s JSON font definitions, which Modern UI honors. Tooltips unaffected — Obscure Tooltips owns the surface (Modern UI tooltips off), and the pack’s custom tooltip frame is a 1.21.2+ feature (N/A on 1.21.1). **Load at the top** of pack order so its GUI wins over Faithful 32x’s 32x GUI and the 32x mod packs’ GUI portions (their item/block textures are untouched).
- **Modded Coffee GUI** — Dark Coffee GUI’s official mod-compat addon (same author): coffee-styled screens for **EMI, Create, Farmer’s Delight, Waystones, Relics, Curios, Supplementaries, AppleSkin, Overflowing Bars, Spell Engine, Immersive Aircraft, Immersive Overlays**, Sophisticated Storage/Backpacks (partial — scrollbar/search stay grey) and more. **Not covered — stay bright vanilla:** Mekanism GUIs (blocks/items now 32x via Upscaled Mekanism 32x), AE2, MineColonies, dimension mods (Mindful Darkness darkened every GUI uniformly; that coverage is the loss). **Load directly above Dark Coffee GUI.** Main test versions per author: 1.20.1 / 1.21.1.
- **RAY’s 3D Rails** — 3D rail models; shader support since v3.0. Compatible with Steam ‘n’ Rails and Create rails. Rails only — ladders are covered by the separate **RAY’s 3D Ladders** pack below.
- **RAY’s 3D Ladders** — 3D ladder models (vanilla + Create); shader support (LabPBR) since v2.0. Same author as RAY’s 3D Rails, but a separate pack — Rails does **not** cover ladders. Updated Aug 2026.
- **Better Lanterns** — 3D lanterns + connected chain models. Compatible with Arcane Lanterns and Thin Air.
- **(Bee’s) Fancy Crops** — vanilla crop rework with texture variations; built to work with Farmer’s Delight (Wave 4).
- **Rekindled CTM** — official Fire Rekindled addon: connected textures for fire and soul fire (plus moss-carpet overhangs). Requires Continuity — already in the pack (Wave 0.5, for Overlay’s), so zero new mods. **Load above Fire Rekindled.** Pairs well with Incendium’s Nether (connected soul fire across the transformed biomes). If fire textures look desynced from their neighbors, disable Sodium’s **“Animate Only Visible Textures”**. Slime/honey connected textures live in a separate pack (Viscous CTM) — not added.

- **Fire Rekindled** — animated fire, lava, water, portal textures with Normal/Specular maps for PBR shaders; supports Create (honey, steam particles). CTM addon: Rekindled CTM (previous bullet).
- **Motsch's Better Leaves** — extra layers on leaf blocks for denser, bushier foliage; explicit mod support (Aether, Deep Aether, Twilight Forest, Deeper and Darker, Upgrade Aquatic). Pair with [Cull Leaves](#) (Wave 0.5) for the FPS win. v9 ships static overlays — recompile the custom version if the 16x-on-32x leaf mismatch bothers you.
- **Bushy Pink Petals, Wildflowers & Leaf Litter** — 3D models for ground-cover flora; denser, more natural look.
- **Modded Omelet — Modern Spawn Eggs** — backports the 1.21.5-style spawn eggs to 1.21.1 and restyles modded eggs to match; 204 mods supported, incl. ~9 in this pack (Aether, Twilight Forest, Cataclysm, Hybrid Aquatic, Upgrade Aquatic, Darker Depths, MineColonies, Northstar Redux, Guard Villagers). Zero dependencies.
- **Mace Fusion** — refined 3D model + 32x texture for the vanilla mace; subtle, vanilla-like, no conflicts.
- **Fresh Waystones Texture** — improved textures for the Waystones block + item (Waystones is Wave 2); no conflicts, no dependencies.
- **Overlay's** — dynamic OptiFine-format CTM overlays: dirt moisture, cracks, smooth block transitions. Requires [Continuity](#) + [Sinytra Connector](#) (Forgified Fabric API already in pack). Fills the CTM gap Patrix couldn't deliver under Iris.
- **MissingSoundsFix** — silences vanilla empty-sound log spam ([salmon.ambient](#), etc. — MC-97521) with dummy sound mappings; no audio changed.
- **Faithful 32x** — full 32x base pack with complete coverage (blocks, items, mobs, GUI, fonts). Replaces Patrix 32x: no PBR normal/specular maps, but lighter, complete, and cohesive. Load at the **bottom** of pack order as the base layer. BetterGrassify samples the active pack at runtime — connected grass works at 32x automatically.
- **Create Immersive Aircrafts** — 16x retexture of Immersive Aircraft's crafting items, upgrade items, and vehicles in Create style (also renames some items in US English). Only touches IA assets — any position above the base pack works. Pairs with its Recipes data pack (Wave 2 Mobility / [datapacks/README.md](#)): textures and recipes are designed to match.

**Installation:** Install via XMCL — drop [.zip](#) files into the instance's Resource Packs tab, or copy them into the instance's [resourcepacks/](#) folder (access via instance settings).

## Resource Pack Load Order

Top = highest priority. Packs only win where they have content, so packs touching different assets (armor vs. blocks vs. items vs. language files) are order-

independent; the constraints that matter are called out per item. The 32x base pack sits at the **bottom** so every overlay above it wins.

1. Dark Coffee GUI — GUI theme (16x coffee); wins the GUI surface pack-wide (vanilla + covered-mod screens; see Texture Packs). Above Improved Create/AE2: those 32x packs only overlap the GUI portions — item/block textures unaffected; move below them if you prefer the 32x mod GUIs
2. Modded Coffee GUI — Dark Coffee GUI's mod-compatible addon; load directly above it
3. Improved Create 32x — author requires top placement; only overlaps the base pack
4. Improved Applied Energistics 2 — author requires top placement; only overlaps the base pack
5. Upscaled Mekanism 32x — Faithful-style Mekanism coverage; 32x tier, only overlaps the base pack
6. Mod Descriptions — Field Guide companion, lang-only; any position works
7. Fresh Animations
8. Fresh Animations: Objects (directly above FA — author requirement)
9. Fresh Animations: Emissive (directly above FA — author requirement)
10. Enhanced Boss Bars
11. Visual Traveler's Titles (title-card overlay — order-flexible, grouped with UI tier)
12. RAY's 3D Rails (3D rails)
13. RAY's 3D Ladders (3D ladders — pair with 3D Rails; disjoint assets)
14. Better Lanterns
15. (Bee's) Fancy Crops
16. Rekindled CTM (Fire Rekindled addon — load above it)
17. Fire Rekindled
18. Motsch's Better Leaves (pair with Cull Leaves mod)
19. Bushy Pink Petals, Wildflowers & Leaf Litter
20. Modded Omelet (spawn egg items)
21. Mace Fusion (mace item model)
22. Fresh Waystones Texture (Waystones block + item)
23. Overlay's — needs Continuity + Sinytra Connector mods
24. MissingSoundsFix — no visual impact, order irrelevant
25. Create Immersive Aircrafts (IA vehicles + items; above the base pack)
26. Faithful 32x — base layer (overlays above win)

## Infrastructure

| Mod           | Role                                                                     |
|---------------|--------------------------------------------------------------------------|
| Chunk Loaders | Keep chunks loaded across dimensions — essential for Phase 2+ automation |

| Mod                                          | Role                                      |
|----------------------------------------------|-------------------------------------------|
| <a href="#">SuperMartijn642's Config Lib</a> | Config library (Chunk Loaders dependency) |
| <a href="#">SuperMartijn642's Core Lib</a>   | Core library (Chunk Loaders dependency)   |

## Install

Use **XMCL** — create a 1.21.1 NeoForge instance, install all Wave 0 mods via the Mods tab. See [Wave 0](#) for the full list.

Wave 0.5 (Player Experience) can be installed at the same time — they're independent mods that don't conflict.

## First Launch Checklist

### 1. Video Settings

| Setting             | Recommended | Notes                                                                                        |
|---------------------|-------------|----------------------------------------------------------------------------------------------|
| Graphics            | Fancy       | Sodium makes this fast; Fancy (not Fabulous) avoids transparency conflicts with Iris shaders |
| Render Distance     | 10 chunks   | Distant Horizons handles the rest                                                            |
| Simulation Distance | 8 chunks    | Fewer distant mobs ticking = better FPS                                                      |
| VSync               | OFF         |                                                                                              |
| Max Framerate       | Uncapped    |                                                                                              |

### 2. Distant Horizons (LOD Mod)

Open **Video Settings** → **[colored tiles icon]** next to the FOV slider.

| Setting                    | Value      |
|----------------------------|------------|
| Enable Rendering           | ON         |
| Enable Distant Generation  | ON         |
| LOD Render Distance Radius | 128–256    |
| Quality Preset             | Medium     |
| CPU Load                   | Aggressive |
| Enable Cloud Rendering     | OFF        |
| Show LOD Gen Progress      | ON         |

**Tip:** DH generates LOD data as you explore, in new chunks only. First visit to an area has temporary pop-in. Terralith + Tectonic terrain takes longer than vanilla. Monitor VRAM — shaders + DH at 4K can push past 12GB.

### 3. Shader Setup

Drop your chosen shader [.zip](#) into [shaderpacks/](#). In-game: Options → Video Settings → Shaderpacks → select it. All three work with Iris on NeoForge: **Solas V3.7** (main — native DH support, ColorWheel fixes), **Photon v1.3b** (alt — own DH settings screen, ColorWheel via patcher).

No shader needs add-on mods — Solas fully replaces the previous Complementary Unbound + Euphoria Patches setup.

Recommended baseline (RTX 4080 SUPER, 16GB VRAM): High profile, tune from there — see the per-shader settings tables in [Shaderpack](#) above (incl. each shader's DH rules). General MC: Graphics Fancy, Render Distance 10, Simulation 8, VSync OFF. DH CPU Load: Aggressive on 8+ cores, Balanced below.

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## Infrastructure

### Chunk Loaders

Keeps chunks loaded when you're far away or in another dimension.

| Type                | Coverage            | Use                       |
|---------------------|---------------------|---------------------------|
| Single Chunk Loader | 1 chunk             | One machine or farm       |
| Spotloader          | Handheld, temporary | Explore nearby            |
| Multi Chunk Loader  | 3×3 chunks          | Factory complex or colony |

**How:** Craft (check EMI — iron + redstone + ender pearl), place, right-click to toggle. Colored border shows loaded area.

**Placement priority** (late game):

1. Main base (ME drive, Mekanism induction matrix, Create power core) — Multi Chunk
2. MineColonies colony — Single or Multi
3. Mining outposts (Digital Miner) — Single
4. Other dimensions — Single per outpost

No performance concern with a dozen loaders on a modern CPU.

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# Wave 0.5 — Player Experience

All the visual polish, UI improvements, inventory tools, storage, travel, trading, and quality-of-life features that make the game feel great.

## Visual & Client QoL

| Mod                                             | Role                                                                                                                                               |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">LambDynamicLights</a> 4.8.11+1.21.1 | Dynamic lighting — pin 4.8.11 (Aug 2026 EOL bugfix line; fixes duplicate-keybind <code>options.txt</code> corruption). Client-only, zero new deps  |
| <a href="#">BetterF3</a>                        | Customizable debug HUD                                                                                                                             |
| <a href="#">Sound Physics Remastered</a>        | Realistic sound                                                                                                                                    |
| <a href="#">AmbientSounds</a>                   | Nature ambiance (birds, wind, water)                                                                                                               |
| <a href="#">The Immersive Music Mod</a>         | 147 new biome/structure-based songs, vanilla silence gaps preserved                                                                                |
| <a href="#">Modern UI</a>                       | Full UI framework — modernized screens, own text engine, fancy tooltips, Gaussian blur                                                             |
| <a href="#">Immersive Hotbar</a>                | Animated hotbar slot transitions on pickup/use/swap, XP bar + level-up effects, tooltip animations                                                 |
| <a href="#">Stylish Effects</a>                 | Status-effect HUD overhaul — compact icons with remaining-time timer + amplifier, in the HUD, inventory, and every container; requires Puzzles Lib |
| <a href="#">Traveler's Titles</a>               | RPG-style titles entering biomes/dimensions                                                                                                        |
| <a href="#">Better Modlist</a>                  | Enhanced mod list screen with badges/categories                                                                                                    |
| <a href="#">Better Third Person</a>             | Independent 360° camera rotation in third-person view                                                                                              |
| <a href="#">3D Skin Layers</a>                  | Renders player skin layers (jackets, hats) with 3D depth                                                                                           |
| <a href="#">Presence Footsteps</a>              | Block-aware footstep sounds for every surface type                                                                                                 |
| <a href="#">Particular Reforged</a>             | Ambient particles: 3D water splashes, fireflies, falling leaves, cave dust, waterfall cascades                                                     |

| Mod                                     | Role                                                                                                                                           |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Particle Rain                           | Replaces default weather with particle effects: angled rain, desert sandstorms, fog, wind                                                      |
| Swinging Lanterns                       | Subtle swinging movement on lanterns and attached chains — makes builds feel alive                                                             |
| Wakes Reforged                          | Realistic water wakes and splashes when traveling on water                                                                                     |
| Sounds                                  | 170+ new sound effects for UIs, items, blocks, actions                                                                                         |
| Better Days                             | Customizable day/night cycle length, enhanced sleep                                                                                            |
| Better Safe Bed                         | Prevents mobs that can't reach you from blocking sleep — chill-rule essential                                                                  |
| Beautiful Enchanted Books [Mod Edition] | Unique textures per enchantment book type — standalone mod, no resource pack needed                                                            |
| Overflowing Bars                        | Expanded health, armor, and toughness bars beyond vanilla limits                                                                               |
| AttributeFix 21.1.3                     | Lifts vanilla attribute caps (armor 30+, health) so Overflowing Bars displays real values — requires Prickle + Bookshelf (both auto-installed) |
| Leave My Bars Alone                     | Keeps food/health/armor bars visible while riding mounts — requires Puzzles Lib                                                                |
| Cherished Worlds                        | Favorite/pin worlds to top of list, prevent accidental deletion                                                                                |
| Eating Animation                        | First-person eating/drinking animation — food moves to mouth and shrinks                                                                       |
| Camera Overhaul                         | Camera tilting, screen shakes, mouselook roll, idle sway — satisfying movement feedback                                                        |
| [DLS] Dark Loading Screen               | Replaces the default bright-red loading screen with a dark version — easier on the eyes                                                        |

| Mod                                                 | Role                                                                                                                     |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Cosy Critters &amp; Creepy Crawlies</a> | Ambient atmospheric animals — birds, bugs, and critters that bring the world to life                                     |
| <a href="#">Foxified Dense Flowers</a>              | Renders multiple flowers per block in flower fields for denser, more natural-looking flora                               |
| <a href="#">Effectual</a>                           | Atmospheric effects and decorative particles — steam breath, footprints, cave dust, sparks                               |
| <a href="#">BetterGrassify</a>                      | OptiFine Fancy + Fast better grass — connected grass/podzol/path/nylium sides, better snow. Requires Forged Fabric API   |
| <a href="#">Auto HUD 8.11+1.21.1-neoforge</a>       | Auto-hides idle HUD elements (full health bar, status effects) for a cleaner screen; toggle the whole HUD with a keybind |
| <a href="#">[EMF] Entity Model Features</a>         | OptiFine-format custom entity models — powers Fresh Animations; requires ETF                                             |
| <a href="#">[ETF] Entity Texture Features</a>       | Emissive/random/custom entity textures — required by EMF                                                                 |
| <a href="#">Continuity</a>                          | CTM connected textures (OptiFine format) — enables Overlay's; runs on NeoForge via Connector + Forged Fabric API         |
| <a href="#">Cull Leaves</a>                         | Culls leaf interior faces — big FPS win in forests; recommended companion for Motsch's Better Leaves                     |
| <a href="#">Grassier Grass 1.4.5</a>                | Animated 3D grass blades with wind shader — lush swaying fields; Iris/DH compatible                                      |

## Inventory & UI

| Mod                                     | Role                            |
|-----------------------------------------|---------------------------------|
| <a href="#">EMI</a>                     | Recipe viewer with recipe trees |
| <a href="#">Jade</a>                    | Block info HUD                  |
| <a href="#">Jade Addons (Neo/Forge)</a> | Extra Jade info panels          |

| Mod                                  | Role                                                                                                                                                                                |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JourneyMap                           | Real-time minimap + full-screen world map, waypoints, entity radar                                                                                                                  |
| JourneyMap Integration               | Shows Waystones as icons on JourneyMap                                                                                                                                              |
| AppleSkin                            | Food hunger/saturation overlay                                                                                                                                                      |
| Detail Armor Bar Reconstructed 5.0.2 | Armor detail on the bar — armor-slot icons + durability colors, protection-enchament tints, low-durability flash, netherite bar, mending/thorns, trims                              |
| Inventory Essentials                 | Drag-to-transfer, single-item moves, sorting                                                                                                                                        |
| ClientSort                           | Sort (4 modes), Fill Stacks, Transfer Matching, customizable button positions — requires Cloth Config API                                                                           |
| Fancy Toasts 1.5.0                   | Modern advancement-popup presentation — clean unlock toasts; pairs with Fancy Notify (advancement popups)                                                                           |
| Fancy Notify 1.0.1+                  | Custom notification popups — recipe/system/music/screenshot/biome toasts with per-type filters; owns toast silencing (replaces Toast Control)                                       |
| Shulker Box Tooltip                  | Preview shulker box contents on hover                                                                                                                                               |
| Item Borders                         | Colored borders on items based on rarity                                                                                                                                            |
| Obscure Tooltips 4.2.4               | Stylized tooltips — animated effects/particles, rarity emphasis, labels/shadows, scrolling, armor + tool previews. Owns the tooltip surface (Modern UI tooltips off, Aug 2026 swap) |
| Loot Journal                         | Animated item pickup notifications                                                                                                                                                  |
| Better Advancements                  | Overhauled advancements UI with editor/pan/zoom                                                                                                                                     |
| Polymorph                            | Choose crafting result when recipes conflict                                                                                                                                        |
| JEED                                 | Status effects in EMI browser — descriptions, sources, colors for all modded effects                                                                                                |

| Mod                                      | Role                                                                                                          |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <a href="#">MEED</a>                     | Curated descriptions + effect sources for 600+ effects across 100+ mods — JEED/EMIeffect addon                |
| <a href="#">JustEnoughBreeding</a> 3.2.1 | EMI breeding lookup — food, egg output, tamed-flag for vanilla + modded mobs; client-only, zero required deps |

## Storage & Travel

| Mod                                     | Role                                                           |
|-----------------------------------------|----------------------------------------------------------------|
| <a href="#">Sophisticated Storage</a>   | Upgradable barrels/chests                                      |
| <a href="#">Sophisticated Backpacks</a> | Portable inventory                                             |
| <a href="#">Waystones</a>               | Fast travel between waypoints                                  |
| <a href="#">GraveStone Mod</a>          | Keep inventory on death                                        |
| <a href="#">Simple Recall Potion</a>    | Drink to teleport back to your spawn point — simple, stackable |

## Village & Trading

| Mod                                                                          | Role                                                                                                                               |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Dynamic Villager Trades</a> 1.4.0 (Fabric via Sinytra Connector) | Dynamically changing villager trades — villagers specialize in what you buy from them, reducing the need for massive trading halls |
| <a href="#">Improved Village Placement</a>                                   | Forces villages to spawn on flat terrain — prevents cliff-side and underwater villages                                             |

## Configuration — Wave 0.5

**Dynamic Villager Trades** ([config/dynamicvillagertrades.properties](#) — Fabric build runs via Sinytra Connector, already in pack):

- `global_randomness = 0.7` (recommended, default 1.0) — Lower values make villagers settle on specific trades faster; higher values increase time to get desired trades. Multiplied by group randomness in the datapack.
- `refresh_delay = 1` (recommended, default 0) — Number of restocks between trade changes. 1 means trades change every other restock, giving you time to use existing trades before they rotate. Trades always change on level-up regardless.

- `no_book_duplicates = true` — Prevents librarians from selling the same enchant at different levels simultaneously.
- `enchant_repair_compat = false` — Adds Curse of Vanishing to every book sold by librarians (for Enchant & Repair compatibility). Keep disabled unless that mod is added.

**Waystones** (`config/waystones-common.toml`):

- `inventoryButton = "any"` — Opens waystone menu from inventory without a keybind.
- `enableCosts = false` — Free teleportation.

**Loot Journal** (`config/obscuria/loot_journal-client.toml`):

- `trackItemPickups = true` — Logs all item pickups for per-session statistics.
- `trackXpPickups = true` — Logs all XP pickups for per-session statistics.

**Inventory Essentials** (`config/inventoryessentials-common.toml` — tracked override):

- `inventorySorting = "CONSOLIDATE_ONLY"` — IE's middle-click sort only merges partial stacks, never reorders. **ClientSort owns all reordering** (this is the “configured lower” resolution of the sort-feature overlap between the two mods). All transfer features stay at defaults (single transfer, bulk transfer, shift-drag, tool/stack refill). Value verified (enum name `CONSOLIDATE_ONLY` — Balm writes enum names to TOML; `consolidate_only` also parses case-insensitively).

**Effectual** (`config/effectual.json` — tracked override; JSON, not TOML — Effectual uses a Gson config file):

- `"caveDust" = false` — Cave dust is Particular Reforged's effect; disabling it here removes the only duplicate particle. Everything else stays at defaults (steam breath, bubbles, sand/gravel placement sparks, footprints, etc.).

**Modern UI + ImmediatelyFast** (`config/immediatelyfast.json` — tracked override):

Modern UI requires three ImmediatelyFast mixins disabled on 1.21.1 (Modern UI compat notes — otherwise random game crashes). The tracked `config/immediatelyfast.json` ships with these flipped; ImmediatelyFast's other options stay at defaults:

| Setting                          | Value              | Why                                                                                 |
|----------------------------------|--------------------|-------------------------------------------------------------------------------------|
| <code>hud_batching</code>        | <code>false</code> | Modern UI requirement — HUD text batching conflicts with its tooltip/text rendering |
| <code>font_atlas_resizing</code> | <code>false</code> | Modern UI requirement — font atlas resize conflicts with its text engine            |

| Setting                       |  | Value              | Why                                                                      |
|-------------------------------|--|--------------------|--------------------------------------------------------------------------|
| <code>fast_text_lookup</code> |  | <code>false</code> | Modern UI requirement — fast glyph lookup conflicts with its text engine |

**Tooltip surface (Aug 2026 swap): Obscure Tooltips owns tooltip rendering** — Modern UI’s tooltip feature is **disabled via the Action Center** ([Ctrl+K](#)) so the two don’t fight (both replace the vanilla tooltip renderer; Modern UI’s docs acknowledge the conflict). Don’t re-enable Modern UI tooltips while Obscure Tooltips is installed. **Legendary Tooltips** (ver 1.16 ~ 1.21.4) would fight Obscure Tooltips the same way — if it’s ever added, configure it lower or remove it. **GUI dark mode moved to a resource pack** (Aug 2026 full swap): Mindful Darkness removed → **Dark Coffee GUI + Modded Coffee GUI** (see Texture Packs). Modern UI compatible: its documented conflicts are tooltip/text-engine mods, not GUI-texture resource packs (the instance already proves the surface with Faithful 32x); no shader-pipeline fight (unlike the old Dark Mode Everywhere watch). The pack’s text recolor follows JSON font definitions (Modern UI honors them); NOSHADERS variant is the fallback if text looks off under Iris.

Modern UI itself runs on defaults. Tune it in-game via **Action Center** ([Ctrl+K](#)): fonts (Preferences), disable its zoom (Sodium owns [C](#)), background blur/colors. Its configs write to [config/ModernUI/](#).

## Visual & Client QoL — What to Expect

These mods change how the game looks and feels. All are client-side and work immediately.

- **LambDynamicLights** — held light sources emit light around you; off-hand works. Pro tip: torch in off-hand while mining. Toggle [Ctrl+L](#).
- **BetterF3** — cleaner debug HUD; toggle FPS, coords, biome, light level. Customize in the Mods screen → BetterF3 → Config.
- **Sound Physics Remastered** — sound muffled by walls, echoes in caves, directional. Works automatically.
- **AmbientSounds** — biome-aware ambience (birds, wind, water). No config.
- **The Immersive Music Mod** — 147 biome/structure-aware songs with natural silence gaps.
- **Immersive Hotbar** — animated slot contents on pickup/use/swap, animated XP bar. Configurable via YACL.
- **Modern UI** — full UI framework: modernized screens, own text engine (fonts via Action Center [Ctrl+K](#) → Preferences), fancy rounded tooltips, Gaussian blur + fade-in backgrounds, borderless fullscreen, GUI-scale slider. Ships an OptiFine-style [C](#) zoom — disable it in the Action Center since Sodium owns [C](#). Requires the ImmediatelyFast config override below.

- **Stylish Effects** — status-effect display overhaul (Fuzss): the vanilla HUD effect icons become compact icons with a remaining-time timer + amplifier, and the same display follows you into the inventory and every container (tooltip shows the effect name). Replaces Status Effect Bars Reforged (Aug 2026 swap — SEBR's duration bars are natively covered). ⚠️ **Auto HUD watch:** both hook the effects HUD element — Auto HUD's idle-hide targets the vanilla display, so if it doesn't catch Stylish's compact icons, reposition/resize them via Stylish's config or accept the display staying. Requires Puzzles Lib.
- **Traveler's Titles** — RPG-style title cards entering biomes/dimensions. No config.
- **Better Modlist** — mod list with badges (library/core/addon), categories, search.
- **Better Third Person** — independent camera in third-person; bind a toggle key in Controls.
- **3D Skin Layers** — renders player skin layers (jackets, hats) with 3D depth. ⚠️ **EMF watch:** both touch player-model rendering; no official incompatibility, but if skin layers ever render broken/misaligned, 3D Skin Layers' config **Compatibility Mode** is the lever (the pack's CEM pack — Fresh Animations = mobs — doesn't define player models, so the risk is low).
- **Presence Footsteps** — block-aware footstep sounds; works with Sound Physics.
- **Particular Reforged** — ambient particles (splashes, fireflies, leaves, cave dust, cascades). Toggle per-effect in [config/particular.json](#).
- **Particle Rain** — particle-based weather (angled rain, sandstorms, fog). / [particlerain](#) or Mod List → Config.
- **Swinging Lanterns** — physics-based sway on lanterns and chains. No config.
- **Wakes Reforged** — water wakes and splashes; works with Small Ships and boats. ⚠️ **Particular Reforged watch:** both spawn water-splash effects (Wakes = travel wakes, Particular = 3D fall/entry splashes) — harmless layering, but if splashes look doubled near water, disable Particular's splashes in [config/particular.json](#).
- **Sounds** — 170+ UI/item/block sound effects. Customize in Mod List → Sounds or [config/sounds/](#).
- **Better Safe Bed** — unreachable mobs no longer block sleep. Chill-rule essential.
- **Overflowing Bars** — health/armor/toughness bars expand past vanilla limits — useful for high-HP builds from Relics bonuses and enchanted gear. Requires **AttributeFix** (next bullet) for values beyond vanilla caps — without it the bar renders numbers the game silently clamps.
- **AttributeFix** — lifts vanilla's arbitrary attribute caps (Darkhax, 21.1.3 NeoForge 1.21.1 ✅ via Modrinth API, [dependencies: \[Prickle, Bookshelf\]](#))

both auto-installed, LGPL). Changes no stats by itself — only raises the ceiling so Overflowing Bars + Detail Armor Bar display real Phase 3 values. Zero config.

- **Leave My Bars Alone** — keeps your bars visible while riding mounts. Requires Puzzles Lib (already in pack).
- **Cherished Worlds** — pin worlds to top, prevent accidental deletion. Right-click in world list.
- **Eating Animation** — first-person eating/drinking animation; works with Farmer's Delight foods.
- **Camera Overhaul** — camera tilt, screen shake, mouselook roll, idle sway. Config: `config/cameraoverhaul-client.toml`.
- **[DLS] Dark Loading Screen** — dark loading screen. No config.
- **Cosy Critters & Creepy Crawlies** — ambient birds/bugs/critters; client-side, toggleable via `/cosycritters`. Zero perf impact.
- **Foxified Dense Flowers** — denser flower fields; client-side.
- **Effectual** — atmospheric particles (steam breath, bubbles, footprints, sparks, sand/gravel placement). **Cave dust is disabled** in `config/effectual.json` (`"caveDust": false` — duplicated by Particular Reforged's cave dust; tracked override, Aug 2026 review). Requires Architecture API, Cloth Config, TLib (all in pack).
- **Dark Coffee GUI + Modded Coffee GUI** — dark coffee-themed GUI retexture (replaces Mindful Darkness, Aug 2026 full swap). Dark Coffee GUI restyles all vanilla inventories + HUD icons (hearts, armor, food, hotbar) in a warm coffee palette; Modded Coffee GUI (load directly above it) extends the style to EMI, Create, Farmer's Delight, Waystones, Relics, Curios, Supplementaries, AppleSkin, Overflowing Bars, Spell Engine, Immersive Aircraft, Immersive Overlays, Sophisticated Storage/Backpacks (partial — scrollbar/search stay grey). **Not covered — stay bright vanilla:** Mekanism, AE2, MineColonies, dimension mods (Mindful Darkness darkened every GUI; that coverage is the loss). Modern UI compatible — no shader-pipeline fight; tooltip surface still Obscure Tooltips' (Modern UI tooltips off). v2.2.1 / v1.15.2, 1.21.1 verified; NOSHADERS variant if the text recolor glitches under Iris. Load order + details in Texture Packs.  **Modded Coffee GUI watch:** GUI-texture coverage of Detail Armor Bar Reconstructed unverified (cosmetic only — bar renders fine, palette may differ until a compat texture exists).
- **Obscure Tooltips** — stylized tooltips: animated effects/particles, rarity-tinted flair, labels/shadows, auto-wrap, scrolling, armor + tool previews. **Owens the tooltip surface** — Modern UI's tooltip feature is off (Action Center `Ctrl+K`). Config: `config/obscuria/` (Fragmentum).
- **BetterGrassify** — OptiFine-style connected grass: grass blocks, snowy grass, podzol, mycelium, dirt paths, farmland, and both nylium blend their side textures into surrounding terrain (Fancy, default) or use the top texture (Fast). Adds Better Snow + Better Snowy Grass for snowy biomes. Ships no textures — samples your active texture pack's sprites at runtime, so it works at any

resolution; resource-pack compatibility mode is on by default. Set mode in config — YACL GUI (already in pack) or [config/bettergrass.json](#). Requires Forged Fabric API.

- **Grassier Grass** — animated 3D grass blades with a wind shader (64/block, 160 radius defaults) — grass leans and ripples in gusts, sheds blade particles in strong wind. Iris + Distant Horizons compatible (DH fix in 1.4.0). **Flower clumps OFF** — Foxified Dense Flowers owns flowers. Pure client-side, zero deps. If FPS dips under shaders + DH, lower blades-per-block/render radius first.

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## Inventory & UI — Your Daily Tools

- **EMI** — your primary learning tool: hover any item, [R](#) for recipe, [U](#) for uses. Recipe trees break complex crafts (Create sequenced assembly, Mekanism ore chains) into every step.
- **Jade + Jade Addons** — block name, mod source, machine status, crop growth, chest contents; Addons adds Create (SU/RPM), Mekanism (energy/gas), AE2 (channel) panels. Keep enabled. Conflict fix: Jade → Config → Mod Settings → **Display Bosses** → OFF, **Boss Bars Overlap** → Push Down Tooltip (avoids Enhanced Boss Bars conflict).
- **JourneyMap + Integration** — minimap with entity dots, fullscreen map ([J](#)), waypoints ([B](#) to set), entity radar. Waystones appear as icons via the Integration mod. [J](#) → right-click sets a temporary waypoint. Compass to Map auto-pins Explorer's Compass finds.
- **AppleSkin** — hover food for exact hunger/saturation; dashed outlines preview a meal. No config.
- **Detail Armor Bar Reconstructed** — armor-slot icons with durability colors on the bar itself; protection-enchant tints, low-durability flash, netherite bar, mending/thorns animations, trim display. **Hides with Auto HUD** (draws inside the vanilla armor layer, tied to the armor component — idle-hide and the [H](#) toggle both cover it) and renders cleanly past 20 armor points with Overflowing Bars (dedicated OB compat — replaces OB's armor-bar draw via [toggleCompatibleHeartMod](#)). Replaces Immersive Armor HUD (Aug 2026 swap — IA-HUD floated over the hidden HUD and glitched above 20 bars; open upstream bug). Config: [config/detailarmorbar.json](#) (tracked).
- **Inventory Essentials** — drag-to-transfer and single-item moves: [Ctrl+click](#) moves one item, [Shift+Ctrl+click](#) moves all of a type. ⚠ **Sorting overlap — configured lower:** IE ships its own middle-click sorting (modes: consolidate-only / retain-order / alphabetical / creative — default creative) which would duplicate ClientSort. The tracked override [config/inventoryessentials-common.toml](#) sets `inventorySorting = "CONSOLIDATE_ONLY"` — middle-click

merges partial stacks without reordering, and **ClientSort owns all reordering** (4 modes + button).

- **ClientSort** — sorting with 4 modes (creative order, quantity, alphabetical, item ID) plus Fill Stacks, Transfer Matching, and a Button Editor. The pack's sort tool — IE's middle-click sort is set to consolidate-only so ClientSort owns reordering.
- **Shulker Box Tooltip** — hover a shulker for a contents preview. No config.
- **Item Borders** — rarity-colored borders (gray → gold) to spot drops instantly.
- **Loot Journal** — animated pickup notifications; session summary on disconnect. Config: `trackItemPickups`, `trackXpPickups`, `showSummaryOnDisconnect`.
- **Better Advancements** — overhauled advancements UI, open with `L`. Pairs with Fancy Toasts for popup presentation.
- **Fancy Toasts** — modern advancement-popup presentation (clean unlock toasts). Replaces Advancement Plaques (Aug 2026). No config; zero deps.
- **Fancy Notify** — customizable notification popups: recipes, systems, music, screenshots, weather, biomes, advancements, with per-type filters (Mods screen → Fancy Notify → Filters). Recipe-toast silencing lives here (Filters → Recipes) — replaces Toast Control (Aug 2026). **Biomes filter OFF** (Traveler's Titles owns biome entries); music toasts complement The Immersive Music Mod. Re-adopted after the 1.0.0 startup crash (MusicManager NPE) was fixed in 1.0.1 — [Bivrik/FancyNotify#3](#) closed Aug 26 2026; ships explicit Fancy Toasts compatibility (FT owns advancement popups). CurseForge-only.
- **Polymorph** — when recipes conflict, click the button next to the crafting output to cycle. Persists per ingredient combination.
- **JEED** — browse any status effect in EMI (description, source mod, colors) — a built-in wiki for Relics and pack mod effects. Complements Stylish Effects (EMI lookup vs HUD/container display; JEED also adds description tooltips to Stylish Effects' effect widgets).
- **MEED** — curated descriptions + effect providers (which items/mobs/blocks/fluids apply each effect) for 600+ effects across 100+ mods. Addon for JEED/EMIeffect. Covers pack effect-heavy mods: Aether, Deep Aether, Cataclysm, Relics, Simply Swords, Too Many Bows, Deeper and Darker, Ranged Weapon API.
- **JustEnoughBreeding** — EMI breeding lookup (food, egg output, tamed-flag) for vanilla + modded mobs. Client-only 3.2.1, zero required deps. Chill-living/farming fit; Jade owns growth info (different surface, no overlap).

## Storage & Travel

### Sophisticated Storage + Backpacks

- **Barrel** — craft: 1 chest + 8 planks. Upgrade with iron/gold/diamond/netherite upgrades for more slots, pickup, magnet, filter, composting
- **Backpack** — equip in chest armor slot, press `B` to open. Upgradable like barrels

- **Create Integration** (auto-installed) — Sophisticated barrels and backpacks work fully on Create contraptions: upgrades, automation, inventory settings all function while moving

## Waystones

Craft a waystone → right-click to claim. Use a Warp Stone (ender pearl + amethyst) to teleport from anywhere. **No XP costs** (`enableCosts = false` in config). Set keybind to `Ctrl+Shift+W`.

## GraveStone Mod

Die → grave spawns with all your items. Right-click the grave to retrieve. No item loss, no despawning, 24-hour protection. ⚠ [VERIFY] at next launch (Aug 2026 review watch item): die while wearing a Sophisticated Backpacks backpack — the backpack item (with contents) must sit inside the grave and restore intact. Both mods touch death handling; no issue confirmed for this GraveStone Mod (community reports of empty backpacks involve other grave mods), but verify once.

## Simple Recall Potion

Drink to teleport straight back to your **spawn point** — the classic Terraria-style escape potion, no save points to manage. Works within your current dimension by default; enable the mod's interdimensional config option if you want it to work across the Aether, Twilight Forest, etc. Zero dependencies.

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# Village & Trading

## Dynamic Villager Trades

Villager trades change dynamically — villagers specialize in what you buy from them. No more massive trading halls. The actual mod by **orlounge** (3.3M DL); the 1.21.1 build is Fabric-only, so it runs via **Sinytra Connector** (already in pack).

**Config** (`config/dynamicvillagertrades.properties`):

- `global_randomness = 0.7` — faster specialization
- `refresh_delay = 1` — trades change every other restock
- `no_book_duplicates = true` — no duplicate enchantment books
- `enchant_repair_compat = false` — no Curse of Vanishing on librarian books (for Enchant & Repair compatibility; keep off unless that mod is added)

## Improved Village Placement

Forces villages to spawn on flat terrain. No more cliff-side or underwater villages. No config.

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## Session Plan — Wave 0.5

| Session                            | Goal                                                                                                                                                                                                                              |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Session 1 — Setup</b>           | Launch, set keybinds, configure Distant Horizons (LOD 128-256, Medium quality, CPU Aggressive on 8+ cores). Install shader (Solas main; Photon alt). Enable Fresh Animations texture pack. Walk around spawn to generate LOD data |
| <b>Session 2 — Survival basics</b> | Punch trees, build a starter shack, find food, get iron. Place your first Waystone. Craft a Sophisticated Storage barrel. Craft a backpack                                                                                        |
| <b>Session 3 — Comfort</b>         | Explore nearby terrain. Find a village. Set up a bed. Light up your base perimeter. Start noticing Traveler's Titles when entering new biomes                                                                                     |

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# Wave 1 — Tech (Create + Mekanism + AE2)

The classic tech trio. Create handles mechanical automation, Mekanism handles industrial processing, AE2 handles digital storage and autocrafting.

## Core Tech

| Mod                              | Role                                                                                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Create</b>                    | Mechanical automation — rotational power, water wheels, windmills, gear trains, conveyor belts, deployers, sequenced crafters, trains |
| <b>Mekanism</b>                  | Industrial processing — ore multiplication (2x→5x), digital miner, superdense energy storage, MekaSuit (flight, invulnerability)      |
| <b>Mekanism Generators</b>       | Power generation — heat, wind, solar, gas-burning, bio generators, fission reactor, fusion reactor, turbine                           |
| <b>Mekanism: Ponders</b> 1.0.3   | Ponder scenes for Mekanism machines and multiblocks (fission/fusion reactors, turbine). Requires Mekanism                             |
| <b>Mekanism Guide Book</b>       | Patchouli-based in-game guidebook for Mekanism. Requires Patchouli (already in pack)                                                  |
| <b>MekaJadeUpgrades</b> 1.3-21.1 | Jade tooltips for Mekanism — shows installed upgrades (speed/energy/etc.) on machines. Requires Mekanism + Jade                       |
| <b>AE2</b> 19.2.17               | Digital storage — ME drives, autocrafting, P2P tunnels, spatial storage. The storage backbone                                         |
| <b>Applied Mekanistics</b>       | AE2-Mekanism integration — autocrafting for chemical/gas/fission/fusion recipes, ME chemical storage, P2P chemical tunnels            |

**Note:** Mekanism v10 for 1.21.1 NeoForge is split into separate modules.

**Mekanism** (core) provides machines, processing, cables, and armor. **Mekanism Generators** provides all power generation. Both must be installed.

## Create Addons

| Mod                                        | Role                                                                      |
|--------------------------------------------|---------------------------------------------------------------------------|
| <b>Create Crafts &amp; Additions</b> 1.6.0 | Electric motors, alternators, rolling mill, wires/rods, silver/electrum — |

| Mod                                        | Role                                                                                                                                                      |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | bridges kinetic SU and Forge Energy (FE)                                                                                                                  |
| Create: Power Grid 0.5.5.1                 | Physics-based electricity simulation — Ohm's law, AC/DC, generators, motors, cables, FE conversion. Separates Create power from SU                        |
| Steam 'n' Rails NeoForge 0.2.1             | Expanded train system — new tracks (spruce, monorail), semaphores, conductor mob, coupling/decoupling blocks                                              |
| Create: The Factory Must Grow (TFMG)       | Heavy engineering & oil — crude oil drilling, distillation, diesel/gasoline/LPG engines, steel, aluminum, electricity with voltage                        |
| Mekanism TFMG Compatibility                | 85+ recipe bridges — deduplicates lead/steel/sulfur, integrates aluminum/plastic into Mekanism chains                                                     |
| TFMG Energy Converter 26.03                | Bidirectional TFMG electricity ↔ FE/AE2 bridge — lets TFMG diesel engines/generators power non-TFMG machines (Mekanism, AE2). Requires Kotlin for Forge   |
| Create: Gunsmithing                        | Steampunk firearms integrated with Create — craftable using Create machines. Animated guns, attachments, ammo types. Optional Better Combat compatibility |
| Create Ornithopter Glider                  | Mechanical flapping glider with boost-assisted flight — integrates with Create rotational power                                                           |
| Sophisticated Storage Create Integration   | Full-featured Sophisticated Storage on Create contraptions — upgrades, automation, inventory settings work while moving                                   |
| Sophisticated Backpacks Create Integration | Full-featured Sophisticated Backpacks on Create contraptions — upgrades, automation, inventory settings work while moving                                 |

| Mod                           | Role                                                                                                                                                                                                   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create: Train Utilities 3.0.3 | Sliding doors with realistic sounds, platform blocks, station building blocks                                                                                                                          |
| Create Railways Navigator     | Display boards, route navigation, new schedule entries — search routes and manage train networks                                                                                                       |
| Create: Interiors 0.6.1       | Colored chairs with armrest toggles, tray tables, kelp seats — functional furniture for train cars                                                                                                     |
| Create Train Parts 0.4.1      | Crossing gates, sliding windows, train steps, slides — decorative and functional blocks for trains and stations                                                                                        |
| Create Better FPS             | Create rendering optimization — up to 50% FPS gain with shaders. Handles addon models, speed controllers, factory gauges, and schematic previews                                                       |
| ColorWheel                    | Flywheel + Iris compatibility — renders Create contraptions under shaders. Required for Create to work with Iris (shader-agnostic)                                                                     |
| Create Slice & Dice           | Create + Farmer's Delight automation — Slicer (auto Cutting Board), automatic FD Cooking Pot via heated mixing, Sprinkler for fluid-based farming                                                      |
| Create: Dragons Plus 1.11.2b  | Fan processing expansion — Bulk Coloring, Bulk Freezing, Bulk Ending, Bulk Sanding. Fluid Hatch for tanks. Adds Aether Bulk Enchanting. Library mod for Create addon devs                              |
| Create: Protection Pixel      | Create-themed power armor — brass/alloy tiers, per-piece active abilities (AoE, debuff clear, speed→damage), combustion reactor (blaze rods + water). Bridges gear between Mekanism Tools and MekaSuit |
| Create: Central Kitchen 2.6.0 | Mechanical Arm integration for Farmer's Delight blocks — automate                                                                                                                                      |

| Mod | Role                                                                                                           |
|-----|----------------------------------------------------------------------------------------------------------------|
|     | Cooking Pot, Cutting Board, Stove via Arms; extends to Brewin' And Chewin' kegs. Requires Create: Dragons Plus |

## Create Addon Details

- **Create Better FPS** — Create rendering optimization (addon models, speed controllers, gauges, schematic previews). Benchmarked 7→35 FPS with 512 waterwheels under shaders. Pairs with ColorWheel, which handles Flywheel-Iris compatibility.
- **ColorWheel** — required for Create contraptions to render under Iris shaders; install it before Create Better FPS.
- **Create Slice & Dice** — automates Farmer's Delight: Slicer (auto Cutting Board), FD Cooking Pot recipes as Create heated mixing, Sprinkler for fluid farming (water, lava, potions, Liquid Fertilizer). Meal production becomes fully pipeable in Phase 2.
- **Create: Dragons Plus** — fan-processing expansion (Bulk Coloring, Freezing, Ending, Sanding), Fluid Hatch for tank I/O, Aether Bulk Enchanting. Features toggleable in config; required by Create: Shimmer.
- **Create: Central Kitchen** — Mechanical Arms drive existing Farmer's Delight blocks directly (Cooking Pot, Cutting Board, Stove), plus Brewin' And Chewin' kegs and other Delight-ecosystem stations. Complements (not replaces) Create Slice & Dice: Slice & Dice converts recipes into new Create machines, Central Kitchen automates the FD blocks themselves. Requires Create: Dragons Plus (in pack).

## Create: Protection Pixel

Steampunk power armor built with Create materials. Each armor piece has a unique active ability powered by a waist-slot Combustion Reactor (blaze rods + water as fuel).

**Armor pieces** (brass → alloy tier progression):

| Piece                | Ability                                                          | Best for                       |
|----------------------|------------------------------------------------------------------|--------------------------------|
| <b>Plague Helmet</b> | Release steam to dispel blindness, darkness, weakness, slowness  | Exploration, debuff-heavy mobs |
| <b>Lancer Helmet</b> | Speed → damage scaling — the faster you move, the harder you hit | Mobility builds                |

| Piece                     | Ability                                                                           | Best for         |
|---------------------------|-----------------------------------------------------------------------------------|------------------|
| <b>Hunter Helmet</b>      | Sensors highlight nearby creatures while sneaking; +luck for finding treasure     | Dungeon crawling |
| <b>Breaker Chestplate</b> | +damage +attack speed while maintaining protection                                | DPS builds       |
| <b>Magnetic Storm</b>     | AoE magnetic burst when taking heavy damage or surrounded; pulls ranged attackers | Crowd control    |
| <b>Float Shield</b>       | Energy shell around user when charged — absorbs incoming damage                   | Tank builds      |

### How it works:

1. Craft armor pieces at Create machines (check EMI — requires brass-tier Create infrastructure)
2. Craft a **Combustion Reactor** (waist slot, check EMI) — this powers all equipped armor
3. Feed the reactor **blaze rods + water** (displayed in minutes in the GUI). More blaze rods = longer runtime
4. If reactor runs out: slowness + mining fatigue + weakness. Keep it fueled!

**Armor plates:** Stand on an **Armor Load Platform** (requires lava piped in — 100mB per assembly/disassembly). Right-click while sneaking to install/remove armor plates with different stat values. Plates also provide a repair method — no anvil needed.

**External devices:** Additional Create-themed accessories (back-mounted) that provide utility without consuming armor slots.

**Compatibility:** Works with Better Combat, Epic Fight, and backpacks (visual space reserved for backpack rendering).

**Dependencies:** [Create](#) (already in pack) + [Kleiders Custom Renderer API](#). For versions below 1.9.0, requires [Curios API](#) instead (already in pack). Currently on v2.1.9+ using Kleiders.

**Progression placement:** Brass-tier armor is early Phase 2 (requires Create brass). Alloy-tier is mid-Phase 2 (requires Create alloy infrastructure). MekaSuit remains the Phase 3 endgame upgrade. This fills the Phase 2 power-armor gap between basic Mekanism Tools and the MekaSuit.

## Mekanism Addons

| Mod               | Role                                                                                                                                                    |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mekanism: Weapons | Phase 3 energy weapons — Meka-Tana (katana + teleport), Meka-Bow (energy arrows), Meka-Gun (laser). Polonium/atomic-alloy gated. Better Combat built-in |
| Advanced Finders  | Ore detection devices — directional indicators for nearby ore veins, synchronized multiplayer support                                                   |
| Advanced Chimneys | Smoke ventilation systems — redirect factory emissions outdoors, customizable appearance and behavior                                                   |

**Dependencies:** Both Advanced Finders and Advanced Chimneys require [ForgeEndertech](#) library.

## Ores

| Mod                 | Role                                 |
|---------------------|--------------------------------------|
| Create (built-in)   | Zinc, brass, copper                  |
| Mekanism (built-in) | Osmium, tin, lead, uranium, fluorite |

## Configuration — Wave 1

**Mekanism** ([config/Mekanism/general.toml](#)):

- `allowChunkloading = false` — Disable Mekanism's built-in chunk loading. Prevents conflict with the dedicated Chunk Loaders mod and keeps chunk management unified.

**Ore generation:** All default rates. Osmium at iron-like frequency, tin/lead moderate, uranium/fluorite rarer. Default distribution aligns with the pack's progression curve.

### Interdependency notes for Wave 1:

- Mekanism's universal cables can power AE2 controllers via the Energy Acceptor. No adapter mod needed.
- **Create Crafts & Additions** electric motors and alternators create a direct kinetic↔FE bridge, replacing the Create Alternator workaround. Silver and electrum add new material tiers that integrate with Mekanism's ore processing line.
- **Steam 'n' Rails** expands Create's train system with new tracks, semaphores, conductor mobs, and coupling blocks. Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.
- Mekanism's 5x ore processing (Chemical Dissolution Chamber → Washer → Crystallizer) outclasses Create's Crushing Wheels (1.5x) and Millstone (1x) for ore multiplication. This is intentional — Create handles mechanical automation

and component crafting, Mekanism handles industrial processing. Neither obsoletes the other.

- AE2's Storage Bus on a Sophisticated Storage barrel makes early-mid game storage migration seamless. Place a Storage Bus on your barrel wall, configure it to show contents in the ME terminal, and transition gradually to full digital storage.
- **Sophisticated Storage Create Integration** and **Sophisticated Backpacks Create Integration** (both by P3pp3rF1y) enable full backpack and barrel functionality on Create contraptions — upgrades, automation features (pickup, magnet, feeding), and inventory settings all work while the contraption is in motion. Without these addons, Sophisticated inventories on contraptions are limited to basic item transport.
- **Applied Mekanistics** bridges AE2 autocrafting with Mekanism's chemical processing — ME Chemical Output Buses, ME Chemical Import Buses, and P2P tunnels for chemicals/gases enable autocrafting of Mekanism recipes (including fission/fusion products) directly through the AE2 crafting CPU. Without this mod, Mekanism chemical steps are manual bottlenecks.
- **Create: The Factory Must Grow** adds oil drilling, distillation towers, blast furnaces, and diesel engines — expanding Create's mechanical toolkit into heavy industry. Its steel replaces Mekanism's steel (TFMG's blast furnace is the intended steel source; Mekanism's steel recipe is disabled by the Mekanism TFMG Compat mod). TFMG aluminum and plastic feed into Mekanism's advanced processing chains.
- **Mekanism TFMG Compatibility** resolves 85+ recipe overlaps between TFMG and Mekanism — lead, sulfur, and steel are unified under TFMG's production lines, while Mekanism retains its chemical processing advantage. Both EMI item blacklist config and per-recipe toggles are available for pack tuning.
- **Create: Protection Pixel** adds steampunk power armor that bridges Create's brass-tier infrastructure with the power fantasy of Mekanism's MekaSuit. Brass-tier armor requires Create brass (early Phase 2); alloy-tier requires additional Create alloy processing (mid-Phase 2). The Combustion Reactor's blaze rod fuel can be farmed via Mekanism spawner systems or vanilla blaze grinders. Blaze automation is a natural Phase 2 milestone that unlocks full power armor capability. MekaSuit remains the Phase 3 ultimate upgrade — Protection Pixel fills the gap, it doesn't compete.

## Core Tech

### Create — Mechanical Automation

Create is about **rotational power (SU)**. Machines consume SU to run. If consumption exceeds generation, machines stop.

**Learning with Ponder:** Hold **W** while hovering over any Create block/item in EMI. An animated tutorial plays showing exactly how it works. **Use Ponder constantly** — it's better than any written guide.

### First Create setup:

1. Mine **andesite** — the most important early resource
2. Craft **andesite casing** (andesite + wood nugget in EMI)
3. Craft **cogwheel** and **large cogwheel** — transfer rotational power
4. Place a **hand crank** → attach cogwheel → attach **millstone** → right-click crank to grind wheat/ore manually
5. Build a **water wheel** in flowing water (2–3 block drop) → attach shaft → power 2–3 basic machines (256 SU)

**Brass progression** (mid-Create): Find zinc ore → mix zinc + copper in a **heated mixer** → brass ingots → brass casing → unlock:

- **Deployer** — places blocks/uses items from inventory. Automates crafting
- **Sequenced Crafter** — multi-step autocrafting sequence
- **Mechanical Arm** — moves items between inventories
- **Chute & Funnel** — vertical item transport with filtering
- **Train Station** — build tracks, assemble trains

**Blueprint system:** Craft **Blueprint and Quill**. Select an area, save as a blueprint. Place a **Deployer** with the blueprint to auto-build it. Or use the **Schematicannon** — load a blueprint, feed it materials from chests, and it builds the structure for you automatically. This is your “here’s a blueprint → build it” tool.

### Keybinds:

- Ponder: **W** (hold over block)
- Wrench: right-click
- Goggles: **G** (hold over machine) — shows SU, RPM, stress info
- Toolbox: **T** (hold)

**Goggles** are essential — craft early, check EMI.

## Mekanism — Industrial Processing

Mekanism v10 for 1.21.1 is split into two required modules:

- **Mekanism** (core) — machines, processing, cables, armor
- **Mekanism Generators** — all power generation (heat, wind, solar, gas-burning, bio, fission, fusion)

**Finding Osmium:** Generates at all underground levels, similar rarity to iron. Mine with stone pickaxe or better.

**Tier system:** Every machine has 4 tiers. Upgrade by shift-right-clicking with the tier upgrade item.

| Tier     | Color    | Speed | Upgrade                          |
|----------|----------|-------|----------------------------------|
| Basic    | Orange   | 1x    | Starting                         |
| Advanced | Red      | 2x    | Basic + osmium + redstone        |
| Elite    | Dark Red | 4x    | Advanced + diamond + obsidian    |
| Ultimate | Purple   | 8x    | Elite + refined obsidian + alloy |

### First Mekanism setup:

1. **Heat Generator** — place lava source below it for early power
2. **Metallurgic Infuser** — combine osmium + carbon (coal) → **steel ingots**
3. **Enrichment Chamber** — 1 ore → 2 dust → smelt → 2 ingots (2x processing)

### Processing multipliers:

- **2x**: Enrichment Chamber only
- **3x**: Enrichment Chamber → Crusher → Enrichment Chamber again
- **5x**: Full chemical line — Chemical Dissolution Chamber (sulfuric acid) → Washer → Crystallizer → 5 dust

### Gas infrastructure (for 5x processing):

- Hydrogen/Oxygen: Electrolytic Separator (split water)
- Sulfuric Acid: Chemical Infuser (sulfur + oxygen)
- Deuterium/Tritium: Fusion reactor fuel (late game)

Gases move through **pressurized tubes**. Use a **Rotary Condensentrator** to convert gas ↔ liquid.

**Digital Miner** (mid-game): Automated mining machine. Set radius (up to 64 blocks) and filters (whitelist ores). Needs significant power. Tip: set to mine only ores to avoid filling with stone.

**Fusion Reactor** (endgame): Millions of Joules per tick. Fuel chain: water → Electrolytic Separator → hydrogen → deuterium → tritium → feed both into reactor. Requires significant infrastructure — a Phase 3 project.

**Config**: `config/Mekanism/general.toml` → `allowChunkloading = false` (delegated to Chunk Loaders mod).

## AE2 — Applied Energistics 2

Digital item storage and autocrafting. Everything lives in an electronic network accessed from a single terminal.

### Finding Certus Quartz:

- **Meteorites** — circular surface depressions. Dig down to find certus quartz blocks + sky stone chest with free AE2 components
- **Ore veins** — certus quartz ore underground

**Processing:** certus quartz block → mine → item → craft into dust → smelt → crystal → charge in **AE2 Charger** → charged certus quartz

**Channel system:** AE2 limits devices per cable. This is the hardest concept for new players.

| Cable type           | Channels | Notes                    |
|----------------------|----------|--------------------------|
| ME Glass Cable       | 8        | No indicator             |
| ME Smart Cable       | 8        | Colored lines show usage |
| ME Dense Smart Cable | 32       | Thicker, for backbone    |

- Each ME Controller face provides 32 channels
- Each device (drive, terminal, bus) consumes 1 channel
- Exceed channels → devices show “no power”

**Use ME Smart Cable everywhere** — colored lines instantly show usage.

#### First network:

1. **Power:** Build a **vibration chamber** or **energy acceptor**, connect any power source (Mekanism generator works)
2. **Controller:** Craft ME Controller, place it, connect power. One controller = 32 channels
3. **Storage:** Craft **ME Drive + 1k storage cells** (charged certus + redstone + glass), place next to controller
4. **Terminal:** Craft **ME Crafting Terminal** — essential, lets you craft from stored items

#### Connecting to the world:

- **ME Import Bus** — pulls items from adjacent inventory into network
- **ME Export Bus** — pushes items from network into adjacent inventory
- **Storage Bus** — treats adjacent inventory as part of network (no import/export needed)
- **ME Interface** — exports AND accepts items for autocrafting

#### Autocrafting:

1. Craft **Patterns** (blank pattern + items in grid)
2. Put patterns in **ME Pattern Provider**
3. Place provider next to **ME Molecular Assembler**
4. Request item from terminal — AE2 pulls ingredients, crafts, returns result

For Mekanism/Create machines: use **ME Interface** instead of pattern provider. Put pattern in interface, it exports ingredients to adjacent machine and accepts the result.

**Subnetworks** (advanced): When you exceed 32 channels, add a second controller connected via **ME P2P Tunnel** or **ME Quantum Link** for wireless.

## Applied Mekanistics

Bridges AE2 autocrafting with Mekanism's chemical processing. Adds:

- ME Chemical Output Buses / Import Buses
- P2P tunnels for chemicals/gases
- Autocraft Mekanism recipes (including fission/fusion products) through AE2 crafting CPU

Without this mod, Mekanism chemical steps are manual bottlenecks.

---

## Create Addons

### Create Crafts & Additions

Electric motors and alternators create a direct **kinetic SU ↔ Forge Energy (FE)** bridge. Also adds:

- **Rolling mill** — processes metal into wires/rods
- **Silver and electrum** — new material tiers that integrate with Mekanism ore processing

**Use:** Connect an electric motor to your Create network to power it with FE from Mekanism. Or connect an alternator to generate FE from Create rotational power.

### Create: Power Grid

Physics-based electricity simulation — Ohm's law, AC/DC, cable loss, transformers. Bridges to FE so Power Grid can power Mekanism/AE2 and vice versa.

#### Key concepts:

- Generators produce voltage, cables have resistance, machines draw current
- AC travels farther but needs rectification for DC machines
- Use transformers to step voltage up for long runs, then step down locally
- Built-in FE conversion via FE Converter block

**When to build:** Mid-Phase 2, after you have basic Create power and have outgrown SU. Diesel engines from TFMG feed into Power Grid naturally.

### Steam 'n' Rails

Expanded train system:

- New track types: spruce, monorail, wide gauge
- Semaphores for signal control

- Conductor mob (rides trains, interacts with stations)
- Coupling/decoupling blocks

Essential for moving resources between MineColonies, Mekanism factories, and AE2 storage across a sprawling base.

## Create: The Factory Must Grow (TFMG)

Heavy engineering and oil processing. Drill for crude oil, refine it, burn fuels in diesel engines.

### Oil processing chain:

1. Find oil — surface seepage in desert/plains biomes, or underground pools
2. **Pumpjack** — extracts oil automatically (crafted from brass and steel)
3. **Distillation Tower** (multi-block, 4+ blocks tall) — refines crude oil into diesel, gasoline, LPG, lubricant, asphalt
4. **Cracker** — breaks heavy fractions into lighter fuels
5. **Desulfurizer** — cleans fuel for more efficiency

### Diesel engines:

| Engine       | Fuel            | SU Output  |
|--------------|-----------------|------------|
| Small Diesel | Diesel/gasoline | ~2,000 SU  |
| Large Diesel | Diesel/gasoline | ~8,000 SU  |
| Gas Turbine  | LPG             | ~16,000 SU |

Progression: water wheel → steam engine → diesel engine.

**Steel Blast Furnace:** Multi-block structure for industrial-scale steel. Faster than Mekanism's Metallurgic Infuser once you have oil power.

**Aluminum:** Bauxite ore (mid-to-high Y-levels) → multi-block smelter → aluminum ingots. Used in advanced TFMG machines and feeds into Mekanism alloy processing.

**TFMG Energy Converter:** Bidirectional bridge between TFMG electricity ↔ FE/AE2. Wire diesel generators into your ME system or Mekanism induction matrix. Requires Kotlin for Forge (auto-installed Wave 0 dep).

**Mekanism TFMG Compatibility:** Resolves 85+ recipe overlaps between TFMG and Mekanism. Steel, lead, sulfur unified under TFMG. Mekanism's steel recipe disabled — use TFMG Blast Furnace instead. Config in [config/mekanismtfgmcompat-common.toml](#).

## Create: Gunsmithing

Steampunk firearms crafted using Create machines. Animated guns with attachments and ammo types.

## Getting started:

1. Build a **Gunsmithing Table** (check EMI — uses Create brass + wood)
2. Craft **gun frames** — pistol, rifle, shotgun, each with different stats
3. Craft **barrels, grips, magazines** at the mechanical press
4. Assemble at the Gunsmithing Table
5. Craft **ammo** — each gun type uses different ammo (pistol rounds, rifle cartridges, shotgun shells)

**Attachments:** Scopes, extended magazines, suppressors, bayonets. Attach at the Gunsmithing Table.

**Compatibility:** Built-in Better Combat support — guns use appropriate animations.

**Expansion path — NTGL gun packs:** Create: Gunsmithing runs on NTGL, and the library supports *gun packs* (datapack-style content packs that add new guns, stats, and models via a [registry.json](#)). In practice, **no curated NTGL gun pack ecosystem exists** for 1.21.1 — the only published items are micro-fixes (renewable lead, breakable framed-glass trapdoors, an Alloyed compat pack, each a few hundred downloads at most). Authoring a custom gun pack is a DIY datapack task (NTGL wiki), not a drop-in download. Do not reach for TACZ/Scorched Guns to fill this gap — that violates the pack’s steampunk-only weapon rule.

**Dependencies:** NTGL (gun framework), GeckoLib (3D animation), Framework (utility library) — all auto-installed.

## Create Ornithopter Glider

Mechanical flapping glider powered by Create rotational power.

**How:** Craft the glider (check EMI — brass + leather + feathers). Equip in chest slot. Hold jump to flap (consumes SU from a connected power source or internal buffer). Boost-assisted flight for short bursts.

**Use:** Early flight before MekaSuit. Pairs with Create windmills or water wheels for power.

## Create Train Ecosystem

Five mods work together for a complete train network:

| Mod                       | Role                                                  |
|---------------------------|-------------------------------------------------------|
| <b>Steam ‘n’ Rails</b>    | Track types, semaphores, conductor, coupling          |
| <b>Train Utilities</b>    | Station architecture — sliding doors, platform blocks |
| <b>Railways Navigator</b> | Display boards, route management, schedule entries    |
| <b>Interiors</b>          | Train car furniture — chairs, tray tables, kelp seats |

| Mod         | Role                                                 |
|-------------|------------------------------------------------------|
| Train Parts | Crossing gates, sliding windows, train steps, slides |

Together: tracks → stations → navigation → interiors → detailing, with no overlap.

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## Mekanism Addons

### Advanced Finders

Ore detection devices with directional indicators for nearby ore veins.

**How:** Craft a finder (check EMI — tiers for different ores). Hold it — arrows point toward nearest vein, distance shown on HUD. Synchronized in multiplayer.

### Advanced Chimneys

Smoke ventilation systems for factories. Redirect emissions outdoors.

**How:** Craft chimney blocks, place above furnaces/generators. Smoke renders realistically and vents upward. Customizable appearance.

**Dependencies:** Both require ForgeEndertech library (auto-installed).

### Mekanism: Weapons

Phase 3 energy-weapon line sharing the “Meka” aesthetic: **Meka-Tana** (katana — high damage, teleport + sweeping modules), **Meka-Bow** (energy arrows, auto-fire, draw speed, gravity dampener), **Meka-Gun** (laser rifle with heat/cooldown system), plus exclusive modules (Weapon Attack Amplification — up to 4× damage, configurable; Looting V) and a Wireless Charger (block + optional Curios back-slot display). Requires only Mekanism (pinned 10.7.17.83), MIT license.

**Progression placement — naturally Phase 3, no extra gating needed:** every weapon recipe requires **Polonium Pellets** (fission-reactor line) + **Ultimate Control Circuits** + **Basic Induction Cells**; the Meka-Tana additionally requires **Atomic Alloy**, already gated behind MineColonies “Industrial Revolution” research in this pack. The weapons arrive exactly when the power fantasy intends them.

**Integration:** built-in Better Combat support (Meka-Tana uses the `bettercombat:katana` profile). Energy-based, so it does not trip the pack’s ballistic-firearms ban (TACZ/Scorched Guns doctrine protects Create: Gunsmithing’s role — a laser is a distinct category). Optional Curios integration (already in pack). Weapons are enchantable (config toggle, default on) — vanilla enchantments apply via the enchanting table.

**Config** — Mekanism-style SERVER config ([config/Mekanism/mekaweapons-server.toml](#), generated on first launch): per-weapon `base_damage`, `energy_usage`, capacity/charge rate; Meka-Tana teleport reach + energy; Meka-Gun beam length, heat, cooldown; enchantability toggles. Energy is in Joules (2.5 J = 1 FE). Leave at defaults first; tune damage down only if playtesting shows it outclassing endgame enchanted gear.

## How Wave 1 Mods Work Together

| Layer              | Mod                                           | What's Happening                                                     |
|--------------------|-----------------------------------------------|----------------------------------------------------------------------|
| Power generation   | Create + Mekanism Generators                  | Water wheels → windmills → steam → heat/gas/solar → fission → fusion |
| Ore processing     | Mekanism                                      | 2x→3x→5x ore multiplication                                          |
| Factory automation | Create                                        | Conveyors, deployers, sequenced crafters, mechanical arms            |
| Storage            | AE2                                           | ME drives, storage busses, crafting CPUs                             |
| Transport          | Create trains + AE2 P2P tunnels               | Bulk item movement across bases                                      |
| Endgame            | Mekanism fusion + MekaSuit + AE2 autocrafting | God-tier power and automation                                        |
| Oil/steel          | TFMG                                          | Distillation → diesel engines → blast furnace steel                  |
| Power bridge       | Crafts & Additions + TFMG Energy Converter    | Kinetic ↔ FE ↔ TFMG voltage                                          |
| Rail expansion     | Steam 'n' Rails + train ecosystem             | Full train network with stations and interiors                       |
| Colony defense     | Create Big Cannons (Wave 5)                   | Artillery for base perimeter                                         |

## Session Plan — Wave 1

| Session                            |  | Focus                                                                                                  |  | What to Build                         |  |
|------------------------------------|--|--------------------------------------------------------------------------------------------------------|--|---------------------------------------|--|
| <b>Session 1 — Create basics</b>   |  | Mine andesite. Hand crank + millstone. Water wheel. Grind wheat and ore                                |  | 256 SU power, basic ore doubling      |  |
| <b>Session 2 — Mekanism start</b>  |  | Find osmium. Heat Generator + Enrichment Chamber + Metallurgic Infuser. 2x ore processing              |  | Basic Mekanism line, first steel      |  |
| <b>Session 3 — AE2 discovery</b>   |  | Find certus quartz meteorite. Mine it. Craft dust → crystal → charge. Build AE2 Charger                |  | First certus, charge infrastructure   |  |
| <b>Session 4 — AE2 network</b>     |  | ME Controller + ME Drive + 1k cells + ME Crafting Terminal. Storage Bus on Sophisticated barrel        |  | First digital storage network         |  |
| <b>Session 5 — Mid Create</b>      |  | Find zinc. Make brass. Deployer + Sequenced Crafter + Mechanical Arm. Automate a simple recipe         |  | Component automation, brass-tier      |  |
| <b>Session 6 — Mekanism expand</b> |  | Add Crusher for 3x ore. Electrolytic Separator for gas. Upgrade to Advanced tier                       |  | 3x multiplication, gas infrastructure |  |
| <b>Session 7 — Integration</b>     |  | Import Bus on Enrichment Chamber → AE2. Export Bus to Create machines. Connect alternator to Mek power |  | Cross-mod automation bridge           |  |
| <b>Session 8 — TFMG start</b>      |  | Find oil (desert/plains pools). Pumpjack + Distillation Tower. Small Diesel Engine                     |  | Oil refining, massive SU upgrade      |  |

| Session                        | Focus                                                                                                  | What to Build                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Session 9 — TFMG expand</b> | Build Steel Blast Furnace. Mass-produce steel. Find bauxite for aluminum                               | Industrial steel, aluminum       |
| <b>Session 10+ — Scaling</b>   | Digital Miner. Expand AE2 drives. Add Create train between base and outpost. Work toward 5x processing | Automated mining, bulk transport |

**Pacing tip:** You don't need to finish Wave 1 before starting Wave 1.1 (megabuilding), Wave 1.5 (colony), or Wave 2 (exploration). Dip into exploration between factory sessions. The colony takes real-world days to grow — start it early. TFMG is mid-game — start after you have brass Create and basic Mekanism.

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# Wave 1.1 — Megabuilding

Schematics and blueprint tools for building at scale in survival. Import community-made structures, place them layer-by-layer with Forgematica's holographic overlay, and construct anything from starter homes to mega-factories without creative mode or guesswork. This wave unlocks the Phase 3 megabuild fantasy early — you can plan and start placing mega-structures from day one, even if the resources take weeks to gather.

## Megabuilding Mod

| Mod         | Role                                                                                                                                 |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Forgematica | Client-side schematic mod — import <code>.litematic</code> blueprints, display holographic overlay, place layer-by-layer in survival |

**Forgematica** is the unofficial NeoForge port of Litematica. It lets you load schematics (`.litematic` files) and shows a translucent hologram of the build in-world, complete with material lists per layer. You place blocks manually while the overlay guides every position — no auto-build, fully survival-legal. Essential for mega-scale projects that would be impossible to freehand.

- **Version:** 0.4.3+mc1.21.11 (NeoForge, 1.8 MB)
- **Environment:** Client-side only
- **Dependencies:** None (standalone — no MaLiLib required)
- **License:** LGPLv3
- **Downloads:** 1.6M+ total

**Shader compatibility:** Forgematica rendering may conflict with shader packs. If the schematic overlay looks broken, toggle shaders off temporarily (`K` by default in Iris). On 1.21.1 (this pack's version), Iris compatibility is better than newer MC versions. Switch back when done placing.

**Keybinds** (set in Options → Controls → Key Binds after installing):

| Action                      | Suggested Key             | Purpose                                              |
|-----------------------------|---------------------------|------------------------------------------------------|
| Open Schematic Manager      | <code>M</code>            | Load, manage, and switch between schematics          |
| Execute Schematic Placement | <code>Ctrl+Enter</code>   | Locks in the current layer after verifying alignment |
| Toggle Schematic Rendering  | <code>Ctrl+M</code>       | Show/hide hologram (useful for screenshots)          |
| Toggle Material List        | <code>Ctrl+Shift+M</code> | Show/hide required materials HUD                     |

**How it works:**

1. Download [.litematic](#) schematics and place them in your instance's [schematics/](#) folder.
2. Open Schematic Manager ([M](#)), select a schematic, choose placement position and rotation.
3. The holographic overlay appears at the target location. Use the material list to gather blocks per layer.
4. Place blocks manually following the overlay. Cycle through layers as you build up.
5. When a layer is complete, press [Ctrl+Enter](#) to lock it in and show the next layer.

No blocks are placed automatically — Forgematica is a guide, not a world editor. Every block must be placed in survival. This respects the pack's survival-only constraint while enabling builds at a scale that would normally require creative mode.

## Schematic Recommendations

These schematics complement the Driftwood progression. All are free downloads — 16 from Abfielder.com (one of the largest schematic libraries, 700+ schematics, all free) and 5 from PhantomMarket. Each schematic is picked to serve a specific purpose in your journey from Cozy Roots through Industrial Ascent to Perpetual Ascent; the block count (non-air blocks) and volume (bounding box X×Y×Z) show the material commitment before you start.

| # | Schematic                                 | Source · ID       | Phase | Purpose                                  | Style                   | Blocks (Volume) | Difficulty |
|---|-------------------------------------------|-------------------|-------|------------------------------------------|-------------------------|-----------------|------------|
| 1 | <a href="#">Stone Medieval Tiny House</a> | Abfielder · 13580 | 1     | Compact medieval starter with shop front | Stone & timber medieval | 968 (3,136)     | Starter    |
| 2 | <a href="#">Forest House</a>              | Abfielder · 2865  | 1–2   | Woodland home with room to grow          | Rustic forest lodge     | 4,568 (30,912)  | Moderate   |
| 3 | <a href="#">Ivy's Starter House</a>       | Abfielder · 2338  | 1     | Minimal first shelter — place in the     | Cozy cottage            | 651 (1,980)     | Starter    |

| # | Schematic                       | Source · ID       | Phase | Purpose                                                                   | Style                  | Blocks (Volume)  | Difficulty |
|---|---------------------------------|-------------------|-------|---------------------------------------------------------------------------|------------------------|------------------|------------|
|   |                                 |                   |       | first hour                                                                |                        |                  |            |
| 4 | Simple Medieval House           | Abfielder · 7986  | 1–2   | 3-floor home with animal pen and full interior                            | Timber-framed medieval | 3,654 (24,800)   | Moderate   |
| 5 | Ultimate Survival Base          | Abfielder · 2665  | 1     | Practical all-in-one survival base, easy-to-gather resources              | Rustic survival house  | 2,992 (10,200)   | Moderate   |
| 6 | Trading Hall with Basement      | Abfielder · 3396  | 2     | Villager trading hall — pairs with Guard Villagers and the colony economy | Grand stone hall       | 37,720 (176,900) | Large      |
| 7 | Large Farmhouse                 | Abfielder · 2539  | 1–2   | Farmstead base with room for crops and animals                            | Medieval farmhouse     | 2,871 (11,424)   | Moderate   |
| 8 | Survival Friendly Cherry House! | Abfielder · 13576 | 1–2   | Cherry-blossom shell — no interior,                                       | Cherry-blossom cottage | 2,614 (14,280)   | Moderate   |

| #  | Schematic                                                 | Source · ID       | Phase | Purpose                                                                  | Style                   | Blocks (Volume)  | Difficulty |
|----|-----------------------------------------------------------|-------------------|-------|--------------------------------------------------------------------------|-------------------------|------------------|------------|
|    |                                                           |                   |       | furnish it yourself                                                      |                         |                  |            |
| 9  | <a href="#">Medieval House – Fully Decorated Interior</a> | Abfielder · 13041 | 1–2   | Survival-friendly decorated home                                         | Stone & timber medieval | 3,544 (12,474)   | Moderate   |
| 10 | <a href="#">Oasis Sanctuary</a>                           | Abfielder · 13573 | 2–3   | Landmark desert settlement with towers and gardens — a long-term project | Sandstone desert oasis  | 16,634 (227,766) | Large      |
| 11 | <a href="#">Minecraft Wooden Mansion</a>                  | Abfielder · 12525 | 2     | Large mansion — Phase 2 upgrade from starter home                        | Grand timber estate     | 16,992 (146,080) | Large      |
| 12 | <a href="#">Medieval House – Fully Decorated Interior</a> | Abfielder · 12842 | 1–2   | Tall decorated fantasy-style home                                        | Stone & timber medieval | 4,238 (21,546)   | Moderate   |
| 13 | <a href="#">Cozy Mountain Chalet</a>                      | Abfielder · 13061 | 2     | Storage room, kitchen, brewing room and library in                       | Snowy mountain chalet   | 18,353 (66,048)  | Large      |

| #  | Schematic                                                 | Source · ID                              | Phase | Purpose                                       | Style                   | Blocks (Volume)   | Difficulty |
|----|-----------------------------------------------------------|------------------------------------------|-------|-----------------------------------------------|-------------------------|-------------------|------------|
|    |                                                           |                                          |       | an alpine shell                               |                         |                   |            |
| 14 | <a href="#">Medieval House – Fully Decorated Interior</a> | Abfielder · 12891                        | 1–2   | Compact decorated medieval home               | Stone & timber medieval | 4,855 (9,614)     | Moderate   |
| 15 | <a href="#">Modern House</a>                              | Abfielder · 4201                         | 1–2   | Easy-build modern home                        | Modern minimalist       | 5,956 (13,041)    | Moderate   |
| 16 | <a href="#">Modern Mansion</a>                            | Abfielder · 9485                         | 2     | Medium-sized modern mansion                   | Modern glass & concrete | 14,074 (67,375)   | Large      |
| 17 | <a href="#">Simple modern house!</a>                      | PhantomMarket · simple-modern-h-fe5bfqfp |       | Modern starter base                           | Modern minimalist       | 1,523 (6,188)     | Starter    |
| 18 | <a href="#">Solo Survival House!</a>                      | PhantomMarket · solo-survival-h-a7lmgelv |       | Compact solo survival starter                 | Modern compact          | 1,250 (10,472)    | Starter    |
| 19 | <a href="#">Luxury Waterfront Villa</a>                   | PhantomMarket · luxury-waterfro-0dd5wxfh | 2–3   | Waterfront villa with garage and private pool | Modern coastal villa    | 190,065 (379,600) | Mega       |
| 20 | <a href="#">Modern Stone and Wood House</a>               | PhantomMarket · casa-moderna-de-qdkqmx4d | 1–2   | Riverside home with terraces and balconies    | Modern stone & wood     | 5,126 (18,240)    | Moderate   |

| #  | Schematic        | Source · ID                            | Phase | Purpose           | Style               | Blocks (Volume)  | Difficulty |
|----|------------------|----------------------------------------|-------|-------------------|---------------------|------------------|------------|
| 21 | Modern Big House | PhantomMarket · casa-modernay-jjda2sk1 |       | Large modern home | Modern contemporary | 61,877 (412,528) | Mega       |

**Download:** For Abfielder schematics, visit <https://abfielder.com/Products/ProductDetails.php?id=<ID>> and click “Download” for the `.litematic` file. For PhantomMarket builds, open the post page and click Download — files ship as `.litematic` or `.schem`, both of which ForgeMatica loads. Place all files in your instance’s `schematics/` folder; ForgeMatica picks them up on next launch.

**Difficulty tiers** (by block count): **Starter** < 2,000 — a weekend project · **Moderate** < 10,000 — a solid base · **Large** < 50,000 — a multi-session build · **Mega** ≥ 50,000 — a long-term megabuild

**Finding more:** Browse <https://abfielder.com/Products/BrowseProducts.php?sort=trending&game=minecraft&type=Schematic> — filter by tag (House, Castle, Factory, Bridge, Mega Build, Farm, Medieval, Industrial) — or PhantomMarket’s explore page ([market.phantom-node.com/explore](http://market.phantom-node.com/explore)). Planet Minecraft ([planetminecraft.com/projects/tag/litematica/](http://planetminecraft.com/projects/tag/litematica/)) and MineMev ([minemev.com/explore/](http://minemev.com/explore/)) have thousands more schematics if you want alternatives. Search for tags: `medieval`, `survival`, `industrial`, `fantasy`, `steampunk`.

**Style guide:** When browsing additional schematics, stick to materials already present in the modpack — stone, deepslate, oak/spruce/dark oak, brick, terracotta, copper, iron, and Create decorative blocks. Avoid schematics that depend on mods not in this pack (looking for Quark, Chipped, or BYG blocks in the material list is a red flag). Medieval, steampunk, dark fantasy, and the modern builds recommended above all fit the Driftwood aesthetic; sci-fi and glass-heavy builds don’t.

## Interdependency Notes — Wave 1.1

- **ForgeMatica × Create:** Big interior shells — the Cozy Mountain Chalet ([13-Snowy Chalet-from-abfielder.litematic](#)) and Modern Mansion ([16-Modern Mansion-from-abfielder.litematic](#)) — are built to house Create machinery and Mekanism lines; they provide the structure, you wire the automation. The Large Farmhouse ([7-Large\\_Farmhouse-from-abfielder.litematic](#)) pairs well with Create water-wheel power for a farm-to-tech compound.
- **ForgeMatica × MineColonies:** The Oasis Sanctuary ([10-Oasis Sanctuary Schematic-from-abfielder.litematic](#)) makes a landmark-scale colony centerpiece; the Simple Medieval House ([4-use this-from-](#)

[abfielder.litematic](#)) and Large Farmhouse ([7-Large\\_Farmhouse-from-abfielder.litematic](#)) work as early colony housing, and the Trading Hall with Basement ([6-Trading-Hall-from-abfielder.litematic](#), ID:3396) anchors a villager economy beside the colony.

- **Forgemata × AE2:** The Cozy Mountain Chalet's ([13-Snowy Chalet-from-abfielder.litematic](#)) large storage room is built for pre-AE2 bulk storage — once you transition to ME drives, repurpose it as overflow or a decorative archive hall.
- **Forgemata × Enchanting:** The Cozy Mountain Chalet ([13-Snowy Chalet-from-abfielder.litematic](#)) has a built-in enchanting corner, and the Simple Medieval House ([4-use this-from-abfielder.litematic](#)), Ultimate Survival Base ([5-Ultimate Survival Base by Emik-from-abfielder.litematic](#)), Large Farmhouse ([7-Large\\_Farmhouse-from-abfielder.litematic](#)), and Modern Stone and Wood House ([20-casa-moderna-de-qdkqmx4d.litematic](#)) each include an enchanting table — base your enchanting setup on whichever room you like best.
- **Shader caveat:** Forgemata hologram rendering may flicker under shaders (Solas/Photon). Toggle shaders off ([K](#)) while placing schematics, then re-enable. The 1.21.1 compatibility is better than newer MC versions — distortion is usually minor.
- **Survival constraint:** Forgemata is purely a visual guide. No blocks are auto-placed. Every resource must be gathered and placed by the player in survival mode. This respects the pack's survival-only constraint.

**Install:** In XMCL, search [Forgemata](#) in the Mods tab, or download the [.jar](#) from CurseForge and drop into `mods/`. Create a `schematics/` folder in your instance directory, then drop downloaded `.litematic` (or `.schem`) files there. Forgemata will pick them up automatically on next launch.

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## Session Plan — Wave 1.1

| Session                           | Focus                                                                                                                                                          | What to Build                               |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Session 1 — Pick a starter</b> | Download 3 Phase 1 schematics. Place Stone Medieval Tiny House (ID:13580 — <a href="#">1-Stone Medieval Tiny House (RottaNog) -from-abfielder.litematic</a> ). | First schematic build, learn Forgemata flow |
| <b>Session 2 — Farm hub</b>       | Place Large Farmhouse (ID:2539 — <a href="#">7-Large_Farmhouse-from-</a>                                                                                       | Farm-to-table setup next to base            |

| Session                          | Focus                                                                                                                                                                                                                                                                    | What to Build                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                                  | <a href="#">abfielder.litematic</a> )<br>and Survival Friendly<br>Cherry House (ID:13576<br>— 8-Cherry House<br>without Interior-from-<br><a href="#">abfielder.litematic</a> ).<br>Plant Farmer's Delight<br>crops around them.                                         |                                            |
| <b>Session 3 — Trading hub</b>   | Place Trading Hall with Basement (ID:3396 — 6-<br><a href="#">Trading-Hall-from-abfielder.litematic</a> )<br>beside the colony and stock villagers.                                                                                                                      | Villager economy + Guard Villagers synergy |
| <b>Session 4+ — Mansion prep</b> | Download Phase 2 schematics (Minecraft Wooden Mansion — 11-<br><a href="#">house mouintain-from-abfielder.litematic</a> ,<br>Cozy Mountain Chalet — 13-<br><a href="#">Snowy Chalet-from-abfielder.litematic</a> ).<br>Scout a flat area for the big build.              | Site planning for industrial ascent        |
| <b>Ongoing</b>                   | As your tech tree advances, place larger schematics. Oasis Sanctuary (10-<br><a href="#">Oasis Sanctuary Schematic-from-abfielder.litematic</a> ) for a desert kingdom.<br>Luxury Waterfront Villa (19-<br><a href="#">luxury-waterfro-0dd5wxfh.schem</a> ) for a coast. | Megabuilding scales with progression       |

**Tip:** Schematics eat materials fast — the Minecraft Wooden Mansion ([11-house mouintain-from-abfielder.litematic](#)) alone runs ~17,000 blocks. Set up a Create tree farm and cobblestone generator early; you'll need them. Treat schematic building as a multi-session project, not a speedrun.

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# Wave 1.5 — Colony Management

MineColonies provides autonomous NPC workers that mine, farm, craft, and guard your base.

## Colony Automation

| Mod          | Role                                                                               |
|--------------|------------------------------------------------------------------------------------|
| MineColonies | Autonomous NPC colony — builders, miners, farmers, crafters, guards, research tree |

## Colony Addons

| Mod                      | Role                                                                                                  |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| Byzantine Styles Pack    | 3 build styles — Byzantine, Shogun, Nile. Visual variety for colony buildings                         |
| Stylecolonies            | Official MineColonies blueprint addon — 10+ styles (Steampunk, Frontier, High Magic, FairyTale, etc.) |
| Create: Colony Logistics | Bridges MineColonies warehouses to Create logistics — belts, chutes, trains for colony supply         |
| ColonyLink               | Connects MineColonies builders to AE2 — monitor resources, trigger autocrafting from ME terminal      |
| JadeColonies             | Jade tooltips for MineColonies — worker stats, hut info, Domum Ornamentum blocks                      |

**Style packs coexist:** Byzantine and Stylecolonies cover completely different themes (Byzantine/Shogun/Nile vs Steampunk/Frontier/FairyTale/etc.) — no overlap. Both can be installed simultaneously. FairyTale style requires Twilight Forest + Create (both in pack) and Quark (not in pack — that style simply won't appear).

**Create: Colony Logistics** and **ColonyLink** serve complementary roles — Colony Logistics handles Create-based colony restocking (belts, trains), while ColonyLink handles AE2-based resource monitoring and autocrafting. Both integrate MineColonies into the tech loop.

## Interdependency notes for Wave 1.5

- MineColonies guards patrol your base perimeter; coverage scales with Guard Tower level and skill research — defense grows with the colony.
- Colony Research soft-gates Phase 2: “Industrial Revolution” (Colony level 3, 5000 points) signals you have the infrastructure to support Mekanism and AE2. Hard recipe gating is not implemented in this pack — playtesting determines if soft gating is sufficient.

## Getting Started

1. Craft the **Town Hall** and place it on flat ground
2. Right-click → colony GUI opens
3. Hire a **Builder** from the Citizens tab
4. Place a **Builder's Hut** near the Town Hall
5. Put wood, cobblestone, and tools in the builder's chest
6. The builder starts constructing from blueprints

## Core Worker Huts (build in this order)

| Hut                | Resource                 | What it does                                    |
|--------------------|--------------------------|-------------------------------------------------|
| <b>Mine</b>        | Coal, iron, stone        | Digs shaft mine. Passive ore supply             |
| <b>Lumberjack</b>  | Wood                     | Chops trees, replants. Infinite wood            |
| <b>Farmer</b>      | Wheat, carrots, potatoes | Plants, grows, harvests. Feeds colony           |
| <b>Guard Tower</b> | Defense                  | Patrols base perimeter. Scales with colony size |
| <b>Crafter</b>     | Recipes                  | Crafts items from storage materials             |

## Research Tree

| Research              | Colony Level | Points | Unlocks                             |
|-----------------------|--------------|--------|-------------------------------------|
| Basic Industry        | 2            | 2000   | Better mines, furnace automation    |
| Advanced Smithing     | 3            | 4000   | Better worker tools                 |
| Industrial Revolution | 3            | 5000   | Atomic Alloy recipe (Mekanism gate) |
| Logistics             | 4            | 8000   | Warehouse, courier routes           |

**Tech gating:** Atomic Alloy (Mekanism ultimate-tier) requires “Industrial Revolution” research. This prevents skipping from Phase 1 straight to Phase 3.

## Guard Scaling

- 1 Guard Tower = 20-block radius coverage
- Each additional tower adds overlapping coverage
- Guard equipment upgrades with colony level (leather → iron → diamond)

## Integration With Your Tech Base

- **AE2 Storage Bus** on colony warehouse → all colony resources in your ME terminal
- **ColonyLink** on AE2 network → monitor builder resource needs, trigger autocrafting when supplies run low

- **Create belts/trains via Colony Logistics** → automated material delivery to builder huts and worker chests
- **Create: Colony Logistics** on colony warehouse → colony resources flow through Create's logistic network (belts, depots, trains, stockpile switches)
- **Farmer's Delight meals** supply colony kitchen for worker happiness bonuses
- **Chunk Loaders** keep colony running when you're in other dimensions

## Session Plan — Wave 1.5

| Session                          | Focus                                                                          | What to Do                              |
|----------------------------------|--------------------------------------------------------------------------------|-----------------------------------------|
| <b>Session 1 — Foundation</b>    | Town Hall, Builder's Hut, supply materials                                     | Colony established, first build started |
| <b>Session 2 — First workers</b> | Mine hut + Lumberjack hut                                                      | Passive coal/iron/wood income           |
| <b>Session 3 — Farm + Guard</b>  | Farmer hut + Guard Tower                                                       | Colony self-sufficient, base defended   |
| <b>Session 4 — Research</b>      | Basic Industry research, Crafter hut                                           | First tech unlocked                     |
| <b>Ongoing</b>                   | Check in every few sessions. Upgrade buildings, expand guards, unlock research | Atomic Alloy ready by Phase 2           |

**Tip:** The colony is a long-term background investment. Start it early, check periodically, let it grow.

# Wave 2 — Exploration (World & Dimensions)

## Overworld Worldgen

| Mod                                      | Role                                                                                                                                                                       |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Terralith</a> 2.4.6          | Overworld biome/terrain overhaul — 85+ biomes, new landforms, Canyons, Painted Mountains, Volcanic regions                                                                 |
| <a href="#">Tectonic</a> v9.0+           | Terrain overhaul — deeper oceans, abyssal trenches, taller mountains, cave/noise tuning. Requires TerraBlender                                                             |
| <a href="#">Streams Reflowing</a> 2.13.1 | Hydrology — terrain-following carved streams, lakes at varying elevations, directional river currents draining to oceans; ambience (spray, whitewater, debris, bank flora) |

**Streams Reflowing** carves terrain-following streams, lakes at varying elevations, and gives natural rivers a directional current that drains to the sea — boats, items, and Create water wheels ride the flow. Built for worldgen compatibility: Terralith and Tectonic terrain are first-class (Terralith’s Yellowstone is carve-blacklisted by default), carved banks use biome-native materials, and structures (YUNG’s, villages) are stepped around rather than cut through. Every feature is independently configurable. Zero dependencies.

 **New world recommended:** hydrology is worldgen — streams/lakes appear in newly generated chunks only, and network-shaping settings are best chosen *before* creating a world. Existing explored terrain keeps what it already has.

## Environment

| Mod                                        | Role                                                                              |
|--------------------------------------------|-----------------------------------------------------------------------------------|
| <a href="#">Serene Seasons</a>             | Seasonal foliage colors, temperature shifts, crop growth cycles                   |
| <a href="#">Serene Seasons Plus</a> v5.1.1 | Serene Seasons add-on — sub-season day/night speed + improved snow piling/melting |

## Underground & Ocean

| Mod                                                       | Role                                                                              |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------|
| <a href="#">TerraBlender</a> (NeoForge) — <b>REQUIRED</b> | Worldgen biome library — required by Darker Depths 2.1+ for biome placement       |
| <a href="#">Darker Depths</a>                             | Underground expansion — 3 new cave biomes, mobs, Forsaken Bronze, Living Crystals |
| <a href="#">Upgrade Aquatic</a>                           | Ocean expansion — Thrasher mob, Ocean Ravines, Prismarine Coral, renewable sand   |
| <a href="#">Hybrid Aquatic</a>                            | Marine creatures — sharks, crustaceans, otters, nautilus, diving armor, corals    |

## Structures

| Mod                                             | Role                                   |
|-------------------------------------------------|----------------------------------------|
| <a href="#">YUNG's Better Dungeons</a>          | Overhauled dungeons                    |
| <a href="#">YUNG's Better Mineshafts</a>        | Improved mineshafts                    |
| <a href="#">YUNG's Better Strongholds</a>       | Larger, more complex strongholds       |
| <a href="#">YUNG's Better Desert Temples</a>    | Multi-room desert temples              |
| <a href="#">YUNG's Better Jungle Temples</a>    | Jungle temples with traps              |
| <a href="#">YUNG's Better Ocean Monuments</a>   | Redesigned ocean monuments             |
| <a href="#">YUNG's Better Witch Huts</a>        | Expanded witch huts                    |
| <a href="#">YUNG's Better Nether Fortresses</a> | Improved nether fortresses             |
| <a href="#">YUNG's Better Caves</a>             | Larger, more varied caves              |
| <a href="#">YUNG's Bridges</a>                  | 15+ natural bridges across terrain     |
| <a href="#">YUNG's Extras</a>                   | Desert wells, obelisks, ruins, pillars |

## Nether Worldgen

| Mod                                                  | Role                                                                                                       |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <a href="#">Incendium</a> v5.4.4                     | Nether biome and structure overhaul — 8 biomes, 9+ structures, 25+ custom weapons, Hovering Inferno boss   |
| <a href="#">MNS - Moog's Nether Structures</a> 3.0.0 | Vanilla-style Nether structures — 40+ ruins, houses, bridges, towers, arenas, Mega Fortress, wrecked train |

**Incendium** is a server-side data pack (packaged as a NeoForge mod) by Starmute that completely revamps the Nether using only vanilla blocks. The Nether generation height is extended to 192. Eight new biomes (Quartz Flats, Withered Forest, Infernal Dunes, Ash Barrens, Toxic Heap, and more) replace vanilla Nether biomes with 3D terrain — jagged mountains, twisting caves, and themed landscapes. Nine new structures range from the massive Forbidden Castle (one of Minecraft’s largest random structures) to the Sanctum (illager fortress), Nether Reactor, and Quartz Kitchen. Over 25 custom weapons and items (Trailblazer flame bow, Greatsword of Sacrifice, Hefty Pickaxe, Hazmat Suit) are rewarded from mobs and structures. New mobs include Toxic Slimes, Restless Spirits, a tamable Ghastling, and the summonable Hovering Inferno boss. The optional [Sparkles](#) resource pack gives Incendium items custom textures — without it, they use vanilla sprites with custom behavior and remain fully functional.

 **New world or Nether reset required:** Incendium overhauls Nether terrain generation. Do not add to an existing world with an explored Nether — reset the Nether dimension or create a new world.

**Note:** Incendium is listed as an alpha/unsupported port for 1.21.x while the Stardust Labs team works on a full rewrite. The mod is mature (34M+ downloads across all versions) and the featured v5.4.4 build is well-tested. The upcoming rewrite may require a Nether reset to upgrade — plan accordingly.

**MNS - Moog’s Nether Structures** is a pure structure mod by finndog\_123 (FinnSetchell — same author as Moog’s End Structures) that fills the Nether with ~40 vanilla-style structures using only vanilla blocks and entities. Structures include blackstone/nether-brick ruins (circles, pillars, walls, rubble), abandoned houses (medium, large, warped variants), bridges, wells (crimson, warped, lava), towers (copper, nether), warped dome, giant/sandy skulls, sword landmarks, shrines, soul fire, grave yards, a **wrecked train wagon** (campfire + skulls + cracked bricks + hay — a ruin, not a vehicle), and three combat arenas (small, large, dragon) with custom mobs. v3.0.0 adds the **Mega Fortress** — a 207-piece sprawling fortress with loot, traps, dungeons under lava, and a chain-suspended second floor.

Requires **Moog’s Structure Lib** (already in the pack via Moog’s End Structures). Structures generate in Incendium biomes (both target `#minecraft:is_nether`). Official config pack available for density/loot tuning. Server-side OK. LGPL-3.0.

## Dimensions

| Mod                               | Role                                                                                                                       |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">The Aether</a>        | Sky dimension — safe islands, gear progression, dungeons with bosses                                                       |
| <a href="#">Deep Aether</a>       | Aether addon — 5 new biomes, Skyjade/Stratus gear, new mobs, poison swamps. Gives the Aether the “Nether Update” treatment |
| <a href="#">Umbral Skies</a>      | Aether × Twilight Forest fusion — Aether-themed structures in TF, Aether boss trophies, TF-material gloves                 |
| <a href="#">Deeper and Darker</a> | The “Otherside” dimension — deep dark themed, warden gear                                                                  |
| <a href="#">Twilight Forest</a>   | Classic dimension — 6+ bosses with progression gates, unique loot, enchanted forest biomes                                 |
| <a href="#">The Bumblezone</a>    | Bee dimension — honey/wax biomes, giant bees, cozy opt-in exploration; enter via bee nest/hive                             |

## End Access & Structures

| Mod                                          | Role                                                                                                                            |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">End Remastered</a> v6.3.0        | Portal progression overhaul — 16 custom Ender Eyes hidden across vanilla structures and bosses. Find 12+ to open the End portal |
| <a href="#">Moog's End Structures</a> v2.0.3 | New structures to fill the End dimension                                                                                        |
| <a href="#">YUNG's Better End Island</a>     | Overhauled dragon island — redesigned pillars, gateways, bell-tower summoning, new portal structure                             |

**End Remastered** replaces vanilla Ender Eyes with 16 custom eyes scattered across vanilla structures and bosses. The Cryptic Eye comes from enchanting, the Evil Eye from master clerics, the Wither Eye from the Wither — each eye has a unique source. Once you have 12+, follow them to a stronghold and insert all 12 to open the portal. This turns reaching the End from a grind (kill Endermen for pearls) into an exploration-driven scavenger hunt.

**Nullscape:** A data pack that transforms the End’s terrain into alien shapes (floating islands, spires, arches, voids) with unique sky colors per biome. Nullscape handles terrain shape and sky only — it does not add biomes. Install as a data pack in the world’s [datapacks/](#) folder.

**YUNG’s Better End Island:** Overhauls the main End island where the dragon fight takes place. Redesigned obsidian pillars, End gateways, spawn platform, and central portal structure. The dragon no longer spawns automatically — approach

the bell tower in the center to begin the fight. Resummoning uses crystals on four bedrock blocks in the tower wings (vanilla positions also work). Dragon AI and difficulty are unchanged — this is a visual and atmospheric upgrade only. Requires YUNG's API.

## Navigation

| Mod                                  | Role                                                                                                 |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <a href="#">Explorer's Compass</a>   | Craftable structure/biome finder                                                                     |
| <a href="#">Compass to Map 2.0.4</a> | Auto-creates color-coded JourneyMap waypoints from Explorer's Compass finds — no hand-copying coords |

## Mobility

| Mod                                       | Role                                                                                                                                                    |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Immersive Aircraft 1.4.6</a>  | Flyable aircraft — biplane, airship, cargo airship, warship, gyrocopter. Upgradable, colorable                                                          |
| <a href="#">Small Ships 2.0.0-b2.1</a>    | Sailable ships — brigantine, cog, drakkar, galley. Build, crew, and navigate; moveable bases on water. Beta (NeoForge)                                  |
| <a href="#">Hang Glider v21.1.0</a>       | Craftable hang glider — glide from any height. No fuel, no power. Phase 1 accessible exploration aid                                                    |
| <a href="#">Man of Many Planes 0.2.1</a>  | Immersive Aircraft addon — two Create-themed aircraft. Requires Immersive Aircraft                                                                      |
| <a href="#">Immersive Machinery 0.2.0</a> | Rustic utility machines — Copperfin submarine, Tunnel Digger, Redstone Sheep, Bamboo Bee item drone                                                     |
| <a href="#">Vehicle Upgrade v21.1.0</a>   | Vanilla mount/boat QoL — mounts swim, no wandering, full mining speed + sprint + reach while riding, translucent mount view, fixed passenger collisions |

## Field Guide

| Mod                                 | Role                                                                                              |
|-------------------------------------|---------------------------------------------------------------------------------------------------|
| <a href="#">Field Guide v1.14.0</a> | Spyglass exploration catalog — scan plants, animals, and monsters to fill an in-game encyclopedia |

Opens a new way to experience the world. Craft a spyglass, then look at any plant, animal, or monster to catalog it in your Field Guide. Each entry you discover adds to your collection with lore about the creature or plant. Works with modded mobs and biomes — automatically discovers content from every mod in the pack.

**Companion mods** — Immersive Overlays and Item Descriptions are **required dependencies** of the NeoForge 1.21.1 build: install them alongside Field Guide. **Mod Descriptions is an optional resource pack** — not a mod: it's language data only (no textures); it's listed in the [Resource Pack Load Order](#) (position irrelevant):

| Mod                                              | Feature Added                                       |
|--------------------------------------------------|-----------------------------------------------------|
| <a href="#">Immersive Overlays</a>               | Biome displays — shows spawn details for entities   |
| <a href="#">Item Descriptions</a>                | Entry descriptions — auto-fills entry lore          |
| <a href="#">Mod Descriptions</a> (resource pack) | Optional — lore descriptions for 200+ mods' entries |

## Configuration — Wave 2

### Aether ([aether-common.toml](#)):

- `"Use default Accessories' menu" = false` — Uses Aether's own Accessories Menu instead of Curios API menu.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Grants Book of Lore and Golden Parachutes on first entry.
- `"Show Patreon message" = false` — Removes Patreon nag screen.
- `"Add Temporary Freezing automatically" = false` — Disables Aether's temporary freezing datapack.
- `"Add Ruined Portals automatically" = false` — Disables ruined portal datapack.

### Tectonic ([config/tectonic.json](#)):

| Setting                              | Default | Recommended | Why                                                           |
|--------------------------------------|---------|-------------|---------------------------------------------------------------|
| <code>oceans.ocean_depth</code>      | -0.22   | -0.35       | Standard ocean floor ~45 blocks deep (vs vanilla ~28)         |
| <code>oceans.deep_ocean_depth</code> | -0.45   | -0.7        | Deep ocean trenches ~90 blocks deep — into deepslate at Y=-28 |
| <code>oceans.monument_offset</code>  | -30     | -55         | Ocean monuments sink to match deeper ocean floor              |

| Setting                           |  | Default | Recommended | Why                                                             |
|-----------------------------------|--|---------|-------------|-----------------------------------------------------------------|
| <code>caves.ore_fix</code>        |  | false   | true        | Redistributes ores when terrain extends below vanilla Y=-64     |
| <code>global_terrain.min_y</code> |  | -64     | -96         | Lower bedrock to give 68 blocks of crust below the trench floor |

**How depths work:** The noise value maps linearly —  $|value| / 0.5 \times 64 = \text{blocks}$  below sea level (Y=62). With `deep_ocean_depth: -0.7`:  $0.7 / 0.5 \times 64 = 90$  blocks deep, trench floor at Y=-28 (solid deepslate). With `ocean_depth: -0.35`:  $0.35 / 0.5 \times 64 = 45$  blocks deep, floor at Y=17. `monument_offset: -55` places ocean monument bases at Y=-16, hovering 12 blocks above the trench floor.

**Ore generation with `min_y: -96 + ore_fix: true`:** Lowering bedrock expands the vertical range ore must cover. `ore_fix` remaps ore distributions to the new depth envelope — diamond still clusters near the bottom, iron and coal fill the expanded crust. Without `ore_fix`, the extra stone between Y=-64 and Y=-96 would be barren.

 **New world required:** These are worldgen changes. Existing chunks keep their old terrain — explore fresh areas to see the new depths.

**Quick verification:** Stand at ocean surface (Y=62), look down with F3 open. Deep oceans should show the floor at Y=-28 (±5). Standard oceans at Y=17 (±5). Monuments should have their base at Y=-16 (±5) — floating ~12 blocks above the trench floor. If values are off, adjust in-game via the Mods screen → Tectonic → Configure.

**YUNG's Better Dungeons — Rarer Dungeons** ([datapacks/rarer-better-dungeons/](#)):

Rebalances YUNG's Better Dungeons density for a desolate, vanilla+ Overworld. The 1.21.1 NeoForge build no longer exposes separation in its config — dungeon spacing is set by the mod's structure-set data files, so the override ships as a small datapack (pack\_format 48).

| Dungeon        | Average separation (chunks)                              | Default | Recommended | Applied |
|----------------|----------------------------------------------------------|---------|-------------|---------|
| Small Dungeons | <code>spacing</code> in <code>small_dungeons.json</code> | 10      | 20-24       | 22      |

| Dungeon           | Average separation (chunks)                                 | Default | Recommended | Applied   |
|-------------------|-------------------------------------------------------------|---------|-------------|-----------|
| Spider Caves      | <code>spacing</code> in <code>spider_dungeons.json</code>   | 44      | 55–60       | <b>58</b> |
| Undead Fortresses | <code>spacing</code> in <code>skeleton_dungeons.json</code> | 46      | 55–65       | <b>60</b> |
| Catacombs         | <code>spacing</code> in <code>zombie_dungeons.json</code>   | 48      | 55–65       | <b>60</b> |

- **Spacing** = average distance between structures in chunks (the old `averageSeparation` value). **Separation** (minimum distance) is raised in step to ~50% of spacing, matching the mod's own default ratio.
- **Why**: at default spacing (10) the classic small dungeons feel common. Rarer finds make each dungeon feel earned and keep the Overworld desolate between landmarks.
- **Install**: copy `datapacks/rarer-better-dungeons/` (or its zip) into the world's `datapacks/` folder — or `%APPDATA%\minecraft\datapacks\` for all worlds. Applies to newly generated regions only.
- **Verify**: `/locate structure betterdungeons:small_dungeon` should show ~22-chunk average spacing with the datapack active.

## Overworld

### Terrain — Terralith + Tectonic

| Aspect    | What changed                                                                    | What to look for                                                |
|-----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Mountains | Taller, steeper. Painted mountains (orange/white layers), volcanic peaks        | Build in mountain valleys                                       |
| Oceans    | 90-block abyssal trenches, 45-block ocean floor, monuments floating in the deep | Use Small Ships (below) to cross, dive with Hybrid Aquatic gear |
| Caves     | YUNG's Better Caves — wider tunnels, deeper chasms                              | Easier to navigate, more impressive                             |
| Rivers    | Deeper valleys from Tectonic, shattered savannahs                               | Follow rivers for scenic travel                                 |

**Pro tip**: These worldgen mods work best in newly generated chunks. Every direction has something different.

### Environment — Serene Seasons

Four seasons with visual foliage changes and crop growth modifiers:

- **Spring/Summer/Autumn/Winter** — foliage colors shift realistically

- **Crop impact ON** — planting in wrong season reduces yield. Plan greenhouses for off-season crops
- **Temperature damage OFF** — no freezing or heat damage (chill rule preserved)

**Serene Seasons config** ([config/sereneseasons/seasons.toml](#)):

| Setting                        | Value              | Why                                                                  |
|--------------------------------|--------------------|----------------------------------------------------------------------|
| <code>generate_snow_ice</code> | <code>false</code> | Disables water freezing and snow accumulation in winter (chill rule) |

**Integration:** Farmer’s Delight crops are affected by seasons. Build glass greenhouses or use Season Sensor blocks to track optimal planting times.

**Serene Seasons Plus** (add-on): refines snow piling/melting and adds sub-season day/night speed — drives Better Days via its `[seasonalDaylightCycle]` config (below). Requires Serene Seasons + Better Days + Gabou’s Libs (all in pack).

**Serene Seasons Plus config** ([config/sereneseasonsplus-common.toml](#)):

Day/night length lives here, not in Better Days. For a fixed **40-minute cycle** (2× vanilla → ~20 min day + ~20 min night):

| Setting                                        | Value              | Why                                                                             |
|------------------------------------------------|--------------------|---------------------------------------------------------------------------------|
| <code>enableSeasonalDaylightCycle</code>       | <code>false</code> | Fixed length, not seasonal variation                                            |
| <code>enableBetterDaysDynamicTimeCompat</code> | <code>true</code>  | Let SSP write <code>daySpeed/nightSpeed</code> into Better Days (reflection)    |
| <code>customCycleLength</code>                 | <code>true</code>  | Use the custom day/night values below                                           |
| <code>customDayLength</code>                   | <code>0.5</code>   | Better Days day-speed ratio (1.0 = vanilla 10 min day) → 0.5 = 20 min day       |
| <code>customNightLength</code>                 | <code>0.5</code>   | Better Days night-speed ratio (1.0 = vanilla 10 min night) → 0.5 = 20 min night |

`customDayLength/customNightLength` are **speed ratios, not “ticks”** (the in-file comment is misleading): 1.0 = vanilla, lower = longer. Leave Better Days and Serene Seasons at their defaults — Serene Seasons Plus owns day/night length.

## YUNG’s Overhauled Structures

Every vanilla structure is expanded:

| Structure                | What's new                                  | Loot                                  |
|--------------------------|---------------------------------------------|---------------------------------------|
| Better Dungeons          | Multi-room with spawners, traps, treasure   | Early iron, gold, enchanted books     |
| Better Mineshafts        | Multi-level with support beams, loot caches | Rails, minecarts, ore samples         |
| Better Strongholds       | Deeper portal room, more libraries/prisons  | Enchanted books, mid-game exploration |
| Better Desert Temples    | Multi-room underground, traps, vault        | High-value vault loot                 |
| Better Jungle Temples    | Vertical temple, redstone puzzles           | Puzzle-solving loot                   |
| Better Ocean Monuments   | New rooms, more elder guardians             | Better sponge room loot               |
| Better Witch Huts        | Expanded hut with potion lab                | Brewing supplies, potions             |
| Better Nether Fortresses | Districts (blaze, wither skeleton, strider) | Nether wart, blaze rods, skulls       |
| Better End Island        | Redesigned pillars, bell tower, new portal  | Dragon egg, gateway access            |

## YUNG's Bridges & Extras

- **YUNG's Bridges** — 15+ types of naturally generated bridges with biome-specific variants, spanning rivers and ravines
- **YUNG's Extras** — small vanilla+ additions: improved Desert Wells (rare Wishing Wells with treasure), Desert Obelisks, Flame Outposts, scattered ruins and pillars

All require **YUNG's API** (already in the pack).

**What to do:** Explore! Mark interesting structures on your JourneyMap (fullscreen map → right-click to set waypoint). Return with an Explorer's Compass.

## Explorer's Compass — Navigation Tool

1. Craft: 1 compass + 4 iron ingots in a + pattern
2. Right-click → GUI opens
3. Search tab: type biome or structure name (partial matching — “cherry” finds all cherry biomes)
4. Select an entry → compass points to nearest instance, distance shown
5. Follow the arrow in real-time
6. Compass to Map auto-creates the JourneyMap waypoint (color-coded) — no hand-copying coords

**Configuration** (`config/explorerscompass-common.toml` → [General] section):

| Setting                         |  | Default | Recommended  | Why                                                                                                                                            |
|---------------------------------|--|---------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>maxRadius</code>          |  | 10000   | <b>5000</b>  | Halve search radius — you must explore ~5K blocks before the compass finds a structure. Prevents instant discovery from spawn                  |
| <code>maxSamples</code>         |  | 100000  | <b>50000</b> | Fewer search samples = lower search accuracy at range. Combined with reduced radius, structures won't be found until you're meaningfully close |
| <code>maxNextSearches</code>    |  | 100     | <b>10</b>    | Limit "find next" to 10 — once you've cleared the 10 nearest instances, explore further and search again                                       |
| <code>displayCoordinates</code> |  | true    | <b>true</b>  | Show X/Z coords and distance on HUD — essential QoL                                                                                            |
| <code>defaultXpLevels</code>    |  | 0       | <b>0</b>     | No XP cost for searching — chill pack, navigation shouldn't gate                                                                               |

**Effect:** With `maxRadius` = 5000, the compass scans a ~10K×10K area centered on you instead of 20K×20K. You'll need to explore new regions to find new structures — no more "find the nearest End City from spawn."

## Underground & Ocean

### Darker Depths — Expanded Underground

3 unique cave biomes deep underground:

| Biome                  | Depth        | What to expect                            | Key resources                                          |
|------------------------|--------------|-------------------------------------------|--------------------------------------------------------|
| <b>Molten Cavern</b>   | Deep (Y < 0) | Lava pools, geysers, amber clusters       | Living Crystals (feed diamond to grow), Crystal Melons |
| <b>Sandy Catacombs</b> | Mid (Y 0-20) | Petrified roots, aridrock, Void Soul mobs | Void Souls (bottle them), Forsaken Bronze              |
| <b>Crystal Cavern</b>  | Any depth    | Growing crystals, glowing ambience        | Crystal materials for decorative/utility               |

**Living Crystals:** Found dead in Molten Cavern. Feed a diamond → reawakens, grows, spreads to nearby diamond ore. Crystal Melons (off-hand) = unbreakable tools with +20% speed/damage for 5 minutes.

**Void Souls:** Kill a Void Soul mob → becomes invulnerable orb (60 seconds). Capture in glass bottle. Use amber to prevent expiration. Forsaken Bronze from void-touched materials has death-related abilities.

**Compatibility:** Darker Depths biomes generate inside YUNG's Better Caves systems — they layer without conflict.

**Required dependency:** TerraBlender NeoForge (auto-installed). Darker Depths 2.1+ crashes without it.

## Upgrade Aquatic — Richer Oceans

| Feature               | What it does                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------|
| <b>Thrasher</b>       | Fantasy shark-like mob. Uses sonar, thrashes prey. Drops thrasher teeth for trident crafting |
| <b>Ocean Ravines</b>  | Underwater ravines with Prismarine Coral — harvest for decorative blocks                     |
| <b>Nautilus</b>       | Passive mob. Drops shells for conduit crafting                                               |
| <b>Renewable sand</b> | Bubble columns below sandstone convert stone to sand                                         |

**Integration:** Thrasher teeth and prismarine coral integrate with Ocean's Delight (Wave 4) for seafood recipes. Compatible with YUNG's Better Ocean Monuments.

## Hybrid Aquatic — Marine Creatures

Expands underwater life with realistic marine creatures and diving equipment.

**Creatures added:**

| Category           | Examples                                         |
|--------------------|--------------------------------------------------|
| <b>Fish</b>        | Reef fish, clownfish, groupers, puffers          |
| <b>Sharks</b>      | Hammerhead, great white — neutral until provoked |
| <b>Crustaceans</b> | Crabs, lobsters, shrimp — drop food ingredients  |
| <b>Critters</b>    | Otters (tameable!), nautiluses, seahorses        |
| <b>Mini-boss</b>   | Deep sea guardian — found in ocean caves         |

**Blocks & items:**

| Item                 | Use                                                         |
|----------------------|-------------------------------------------------------------|
| <b>Diving Armor</b>  | Full set — extended underwater breathing, faster swim speed |
| <b>Sea Sponges</b>   | Absorb water, decorative blocks                             |
| <b>Fishing Hooks</b> | Specialty hooks for better catches                          |

| Item              | Use                                            |
|-------------------|------------------------------------------------|
| <b>Crates</b>     | Storage blocks with ocean aesthetic            |
| <b>Anemones</b>   | Decorative coral-like blocks                   |
| <b>New Corals</b> | Additional coral variants for ocean decoration |

### Getting started:

1. Explore oceans — creatures spawn naturally in rivers, swamps, and oceans
2. Kill crustaceans → food ingredients for Farmer's Delight recipes
3. Find deep sea guardian in ocean caves → challenging mini-boss fight
4. Collect coral and anemones → decorate your underwater base
5. Craft diving armor → extended ocean exploration becomes viable

**Integration with Starcatcher (Wave 4):** Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

**Integration with Ocean's Delight (Wave 4):** Hybrid Aquatic crustacean drops (crab meat, lobster tail) integrate as seafood ingredients. **Integration with**

**Hybrid Delights (Wave 4):** HybridLabs' official Farmer's Delight compat adds recipes from Hybrid Aquatic fish and crustacean drops — fish n chips and more in the FD kitchen.

## Mobility

### Immersive Aircraft

Vanilla-style flyable aircraft. Mid-game (requires iron + basic Create/Mek materials).

| Vehicle       | Fuel          | Seats | Cargo    | Best for                     |
|---------------|---------------|-------|----------|------------------------------|
| Biplane       | Coal/charcoal | 1     | None     | Fast personal flight, combat |
| Airship       | Coal/charcoal | 3     | 9 slots  | Exploration with friends     |
| Cargo Airship | Coal/charcoal | 1     | 27 slots | Material transport           |
| Warship       | Coal/charcoal | 3     | 9 slots  | Armed combat, defense        |
| Gyrocopter    | Coal/charcoal | 1     | None     | Agile short-range flight     |

**How:** Craft via EMI (iron + wool/leather), place on ground, right-click to enter. [W](#) to ascend, [S](#) to descend, [A/D](#) to rotate. Any furnace fuel works. With the Create Immersive Aircrafts data pack installed (below), recipes switch to Create materials and machines — check EMI for the changed recipes.

**Integration:** Aircraft work in the **Aether** — fly between sky islands without bridges. Park indoors or on high platforms (mobs damage parked aircraft).

**Create integration:** The [Create Immersive Aircrafts](#) resource pack (3.0) retextures every IA item and vehicle to match Create, and its companion [Recipes data pack](#) (3.0) rewrites IA crafting, upgrade, and aircraft recipes around Create materials. Install both together — textures and recipes are designed as a pair. The datapack goes in the world's [datapacks/](#) folder (see [datapacks/README.md](#)); recipes show in EMI.

## Small Ships

Buildable, sailable ships. **Phase 1 accessible** — requires wood, wool, iron.

| Ship       |  | Crew | Cargo    | Best for              |  |
|------------|--|------|----------|-----------------------|--|
| Brigantine |  | 1    | 18 slots | All-purpose sailing   |  |
| Cog        |  | 2    | 27 slots | Cargo transport       |  |
| Drakkar    |  | 2    | 9 slots  | Fast travel, combat   |  |
| Galley     |  | 1    | 9 slots  | Early personal vessel |  |

**How:** Craft hull (wood) + mast (wood + wool) + helm. Right-click water to place. Right-click helm to board. [W/S](#) to move, [A/D](#) to steer. Speed depends on wind direction. Dismount to park — ships float where left. Right-click with lead to tow.

**Note:** Beta for NeoForge (2.0.0-b2.1). Core features work; minor bugs expected.

## Hang Glider

Simple fabric-and-wood glider. **Phase 1 accessible** — requires only wood and leather, no fuel or power.

- Craft the glider (check EMI — wood + leather + string)
- Equip in your hand and jump off any height to glide
- Hold forward to maintain speed, steer with mouse
- Dismount by pressing sneak — you'll drop safely to the ground
- Requires **Puzzles Lib** (already in pack — Leave My Bars Alone)

**Progression:** The Hang Glider is the earliest flight option. It provides gentle gliding for early exploration before the player has Create power for the Ornithopter Glider (Wave 1) or fuel infrastructure for Immersive Aircraft. It cannot gain altitude — it's pure glide, not powered flight. Ideal for scouting Terralith biomes from mountain peaks.

**Config:** No changes needed — default behavior is ideal. Gliding is inherently chill (no fuel, no maintenance, no risk of crashing unless you fly into terrain).

## Man of Many Planes — Immersive Aircraft Addon

Adds two Create-themed aircraft to Immersive Aircraft. **Phase 2 accessible** — requires Immersive Aircraft (iron + wool/leather) plus Create materials (andesite alloy, brass).

- **Two new aircraft** — brass-and-mechanical themed designs
- Uses the same fuel, parking, and controls as base Immersive Aircraft
- Works in the Aether dimension like all Immersive Aircraft
- Craft recipes available in EMI
- **Create recipes companion (optional)** — the [Create Man of Many Planes datapack](#) (1.1) adds Mechanical Crafter recipes for both aircraft — assemble them on Create machines instead of a crafting table (see [datapacks/README.md](#))

**Integration:** These aircraft offer a visual and thematic bridge between Create (Wave 1) and Immersive Aircraft (Wave 2). A player deep in Create will appreciate aircraft that match their factory’s aesthetic.

## Vehicle Upgrade

QoL polish for **vanilla mounts and boats** (horses, donkeys, mules, camels, boats). Fully configurable.

- Saddled mounts stay put — no more wandering off when you dismount
- Mounts swim through water and ride through leaves without getting stuck
- Full mining speed, sprint, and an interaction-range bonus while mounted
- Translucent mount view when you look down; fixed passenger collisions (no more suffocating in ceilings)
- Open your inventory while riding; held items (maps) show while rowing

**Dependency:** Puzzles Lib (already in pack). Client + server required — a non-issue for single-player (the integrated server loads it automatically).

**Integration:** Pure vanilla-mount/boat QoL — no content, no worldgen. Pairs naturally with Small Ships for ocean travel. ⚠ **Small Ships watch:** Vehicle Upgrade’s boat features (step height, held items, collisions) target *vanilla* boats; Small Ships’ custom ships are a separate entity class and are unaffected — expect harmless layering, but verify once in-game if boat behavior ever looks doubled.

## Dimensions

### The Aether — Sky Dimension

Safe, beautiful sky islands. Gentle adventure curve.

**Portal:** 4×5 glowstone frame (like Nether portal) → fill with water bucket → step through.

**Progression:**

- **Phase 1:** Spawn on sky island. Collect **ambrosium** (Aether coal) and **zanite** (Aether iron — mining level increases as durability drops)

- **Phase 2:** Bronze dungeon → defeat **Slider** (hit when blue, avoid when red) → Golden Amber
- **Phase 3:** Silver dungeon → defeat **Valkyrie Queen** → Valkyrie tools (flight-enabled weapons), **Gravitite** (floats upward when right-clicked)
- **Phase 4:** Highest crystal islands → **Sun Spirit** — bring fire resistance

**Tip:** Not required for any tech progression. Visit when you want a change of pace. MekaSuit trivializes all Aether combat.

**Deep Aether expansion:** Installed alongside — it adds 5 new biomes (Aerglow Forest, Aervallender Fields, Yagroot Swamp, Golden Heights) that generate organically in the Aether. **Skyjade** tools/armor (stronger at full durability) appear in holystone. **Stratus** armor (slow-fall + dash — press R with full set) is crafted from Sterling Aerclouds found above Y=200. The Yagroot Swamp biome introduces poison liquid, Virulent Quicksand, and the Spooky Ring (night damage bonus). All content is purely additive — the base Aether progression is unchanged.

**Config** (`aether-common.toml`):

- `"Use default Accessories' menu" = false` — Uses Aether's own Accessories Menu instead of Curios API menu.
- `"Gives player Aether Portal Frame item" = false` — Player must build the portal manually.
- `"Gives starting loot on entry" = true` — Book of Lore + Golden Parachutes on first entry
- `"Show Patreon message" = false`
- `"Add Temporary Freezing automatically" = false`
- `"Add Ruined Portals automatically" = false`

## Twilight Forest — Progression Dimension

Dark, enchanted dimension with strict boss progression. Each boss kill unlocks the next area.

**Portal:** Find a 2×2 water pool → surround with 12 flowers (any type) → throw a **diamond** into the pool → lightning strikes → step into the water.

**Note:** One-way down. To return, craft a **Twilight Portal** block or use a Waystone (place one immediately on arrival).

### Boss progression:

| Order | Boss | Location       | Strategy                         | Unlock     |
|-------|------|----------------|----------------------------------|------------|
| 1     | Naga | Naga Courtyard | Strafe charges, hit tail segment | Lich Tower |

| Order | Boss                | Location      | Strategy                                                                | Unlock                                |
|-------|---------------------|---------------|-------------------------------------------------------------------------|---------------------------------------|
| 2     | <b>Lich</b>         | Lich Tower    | Dodge projectiles, hit during cooldown. Shield reflects projectiles     | Twilight Swamp                        |
| 3     | <b>Minoshroom</b>   | Labyrinth     | Find minotaur room, kill it, grab Meef Stroganoff                       | Firefly Forest                        |
| 4     | <b>Hydra</b>        | Hydra Lair    | Dodge fireballs, melee heads. Drops <b>Fiery Tears</b>                  | Ur-Ghast fight                        |
| 5     | <b>Ur-Ghast</b>     | Dark Tower    | Shoot tentacles, then main body. Dark Forest has blindness — bring milk | Snowy Forest + Highlands              |
| 6     | <b>Snow Queen</b>   | Aurora Palace | Push into pillars to break shield. Avoid ice arrows                     | Final Castle                          |
| 7     | <b>Final Castle</b> | Highlands     | Multi-boss fight. Needs weapons from all prior bosses                   | <b>Giant's Pick</b> (5×3 mining area) |

#### Key items:

- **Ironwood** gear — smelt ironwood ingots from iron + magic beans. Good starter armor
- **Fiery** gear — iron + blaze powder. Mid-tier weapons
- **Knightmetal** — from Lich Tower
- **Phantom armor** — creative flight for short bursts. From Ur-Ghast tentacles
- **Giant's Pick** — 5×3×5 mining area. From Final Castle

#### Tips:

- Always twilight — no day/night cycle. Mobs spawn any time
- Place **Waystones** frequently
- **Twilight Forest Maze Map** (paper + torch) reveals maze/castle layouts
- Bring **milk buckets** for Dark Forest blindness
- Your AE2 wireless terminal, Mekanism gear, and Create tools all work here

**Integration:** Twilight Forest bosses are buffed encounters intended to stay relevant even with late-game gear.

## Deeper and Darker — The Otherside

Deep-dark-themed dimension accessed through the Deep Dark.

### How to enter:

1. Find a **Deep Dark** biome (Y=-40 or below)
2. Locate an **Ancient City** (Explorer's Compass → search "ancient city")
3. Kill the **Warden** — drops a **Warden Heart**
4. Craft: Warden Heart + 8 sculk → **Otherside Portal Catalyst**
5. Build a 4×5 frame of **reinforced deepslate** → right-click catalyst on frame → step through

**Warden strategy:** Detects by sound. Use wool to silence footsteps. Sneak near sculk shriekers. Throw snowballs to distract. 500 HP, slow attacks — hit and run.

### What to expect:

- 4 new biomes — sculk forests, frozen deep, expanded ancient cities
- 8 new mobs — sculk variants, stalker (ranged), shriek worm (ambush), **Sculk Leech** (heals on damage)
- **Miniboss:** Shattered Warden — drops **Sculk Transmitter** components
- **Key loot: Sculk Transmitter** — Elytra upgrade for mid-flight teleportation. Combine at smithing table

**Gear:** Warden-themed armor/tools ≈ diamond tier with unique abilities (sound-based attacks, silence fields).

**Difficulty:** Harder than Aether, easier than Twilight Forest's Dark Forest. Come with iron-tier armor + decent weapon.

## The Bumblezone — Bee Dimension

A cozy bee-themed dimension entered directly from any **bee nest or beehive** — no portal structure to hunt for or build.

**How to enter:** Right-click a bee nest/hive (vanilla or modded) to teleport in. Leaving returns you to the Overworld.

### What to expect:

- Honeycomb caves, sugar-water seas, wax roads, and pollen fields — a warm, glowing, golden dimension
- Mobs: giant **Bumble Bees**, Honey Slime, Beehemoth, and bee-adjacent creatures
- Structures: Mite Fortress, Hanging Gardens, Candle Parkour, shrines and sanctums

- Key loot: **Royal Jelly**, bee bread, honey crystals, wax, the **Comb Cutter** enchantment (shear honeycombs from hives), and the **Crystalline Flower** (combine/transfer enchantments from books)
- A **Honey Compass** locates Bumblezone structures

**The bees' one rule:** they're peaceful until you take their honey — harvest and they turn hostile for a while, then calm down. Be a polite beekeeper.

### Integration:

- **Neo Bee Fix** (Wave 4) — its vanilla bee AI repairs apply inside the dimension too (Bumblezone's bees are vanilla bees at heart)
- **Farmer's Delight** — honey, royal jelly, and bee bread feed FD honey recipes (e.g., Honey Glazed Ham)
- **Field Guide** — auto-catalogs Bumblezone mobs and plants
- **Bundled resource packs** (from the mod, enable in Resource Packs):  
**Bumblezone - Shader Emissive** for glowing blocks under shaders (works with ETF/Continuity now in the pack); **Bumblezone - Anti Tripophobia** replaces the trippy honey patterns with plain textures — enable if they bother you

**Chill fit:** fully opt-in, no gates, no structure hunting — enter any hive whenever you like. Honey/wax builds a cozy supply chain; deeper structures (Mite Fortress) reward mid-game gear.

**Difficulty:** gentle by default — the only real threat is provoking the bees. Visit any time from Phase 1 onward.

## End Access — End Remastered

The End portal now requires a scavenger hunt instead of a grind.

**Old method:** Kill Endermen → craft Ender Eyes → find stronghold → insert 12 eyes.

**New method:** Find 16 unique custom Ender Eyes scattered across the world:

| Eye                | Source            | Eye                | Source                                  |
|--------------------|-------------------|--------------------|-----------------------------------------|
| <b>Old Eye</b>     | Desert Pyramids   | <b>Cursed Eye</b>  | Bastion treasure                        |
| <b>Nether Eye</b>  | Nether Fortresses | <b>Exotic Eye</b>  | Craft (aquatic items)                   |
| <b>Cold Eye</b>    | Igloos            | <b>Evil Eye</b>    | Master Cleric trade                     |
| <b>Rogue Eye</b>   | Jungle Pyramids   | <b>Undead Eye</b>  | Craft (Undead Soul from skeleton horse) |
| <b>Black Eye</b>   | Buried Chests     | <b>Cryptic Eye</b> | Random chance when enchanting           |
| <b>Magical Eye</b> | Evoker drop       | <b>Wither Eye</b>  | Wither drop                             |

| Eye                  | Source            | Eye                 | Source              |
|----------------------|-------------------|---------------------|---------------------|
| <b>Lost Eye</b>      | Mineshafts        | <b>Guardian Eye</b> | Elder Guardian drop |
| <b>Corrupted Eye</b> | Pillager Outposts | <b>Witch Eye</b>    | Craft (Witch Pupil) |

**How it works:** Find 12+ unique eyes. Hold one to locate the stronghold (like vanilla Ender Eyes). Insert all 12 into the portal frame to open it. If a vanilla eye is already in a frame, right-click with a custom eye to replace it.

**Tip:** This naturally paces your journey to the End — you'll need to explore deserts, swamps, nether fortresses, ocean monuments, woodland mansions, and bastions before you can fight the dragon. By the time you open the portal, you'll have gear, experience, and world knowledge that makes the End a natural next step rather than a rushed target.

## End Dimension

The End is now a layered destination — each mod handles a distinct piece:

| Mod                             | What it does                                                                        |
|---------------------------------|-------------------------------------------------------------------------------------|
| <b>End Remastered</b>           | How you get here — 16 custom Ender Eyes scattered across the world                  |
| <b>YUNG's Better End Island</b> | The main island — redesigned dragon arena with bell-tower summoning                 |
| <b>Nullscape</b> (data pack)    | The outer islands — surreal floating terrain, spires, arches, unique sky colors     |
| <b>Moog's End Structures</b>    | Structures on outer islands — towers, ruins, dungeons with better-than-vanilla loot |

**Ender's Delight** (Wave 4) adds food cooked from End ingredients — useful status effects for End exploration.

### Progression:

- Post-Dragon: Explore outer islands for new structures and loot
- Mid-End: Tackle Moog's structures for unique weapons
- Endgame: Build an End base — Nullscape terrain makes for dramatic architecture

**Tip:** Place a Waystone immediately. Build a small AE2 subnet with wireless terminal to access main base storage.

## Session Plan — Wave 2

| Session                              | Focus                                                                                                                                                                   | What to Do                        |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Session 1 — Scout</b>             | Craft Explorer's Compass. Search for cherry grove or redwood forest. Mark waypoints on JourneyMap. Loot a YUNG's dungeon. Craft a Hang Glider for early aerial scouting | First taste of transformed world  |
| <b>Session 2 — Nether</b>            | Build Nether portal. Scout Incendium biomes. Find a fortress. Rescue a Ghastling from an Abandoned Tower. Mark a Waystone at your portal                                | Nether foothold, ally, blaze rods |
| <b>Session 3 — Aether</b>            | Gather glowstone. Build Aether portal. Explore first sky island, collect ambrosium and zanite                                                                           | Safe new dimension                |
| <b>Session 4 — Twilight Forest</b>   | Find 2×2 pool, surround with flowers, throw diamond. Enter, place Waystone. Defeat the Naga                                                                             | First Twilight boss               |
| <b>Session 5+ — Deep exploration</b> | Work through Twilight Forest bosses. Visit Deeper and Darker (find Ancient City). Explore End post-Dragon                                                               | Boss loot, dimension resources    |
| <b>Session 6 — Aircraft</b>          | Craft a biplane or airship (iron + wool). Explore Terralith biomes from the air. Craft Man of Many Planes aircraft for Create-themed flight                             | Fast travel, scenic views         |

**Pacing tip:** Explore between factory builds. Grab your compass and walk in a random direction when waiting for Mekanism to process or Create to assemble.

## Wave 2.5 — Building Your Home

Building is a core pillar of this modpack — at any point you can stop optimizing and just build something beautiful. This guide works at any stage: no creative mode or artistic talent required, just a few principles that separate a “dirt hut” from a home you want to come back to.

### Finding Your Build Site

Before placing a single block, find a location that inspires you. You have tools for this:

| Tool                      | How                                                                                      | Why                                                |
|---------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Explorer’s Compass</b> | Craft: 1 compass + 4 iron (+pattern). Right-click → search “cherry”, “redwood”, “meadow” | Find specific biomes by name                       |
| <b>JourneyMap</b>         | Press <b>J</b> → explore the fullscreen map                                              | See terrain, biomes, and your marked waypoints     |
| <b>Small Ship</b>         | Sail along coastlines                                                                    | Discover waterfronts you’d miss on foot            |
| <b>Biplane</b>            | Fly low over terrain                                                                     | Spot valley builds, mountain perches, river mouths |

#### What to look for:

- **Waterfront** — river banks, ocean cliffs, lake shores. Water reflects builds beautifully with shaders
- **Mountain valleys** — enclosed, dramatic, defensible. Terralith makes these spectacular
- **Forest clearings** — surrounded by trees feels cozy and sheltered
- **Cliff faces** — cut into the hillside for a build that looks permanent, not placed
- **Cherry groves** — pink petals + mountain backdrop = instant atmosphere

**Pro tip:** Place a **Waystone** at your chosen site before you start building. You’ll want to teleport back with materials.

### The Block Palette Rule

The single biggest upgrade from “meh” to “wow” is using **3–5 complementary blocks** instead of one. Every great build you’ve ever seen uses a palette. Here’s the formula:

| Role             | Percentage | Purpose                                                    |
|------------------|------------|------------------------------------------------------------|
| <b>Primary</b>   | ~50%       | Dominant wall material — defines the build                 |
| <b>Secondary</b> | ~30%       | Complementary texture — similar tone, different surface    |
| <b>Accent</b>    | ~20%       | Detail material — corners, trim, highlights, pops of color |

## Finding Palettes: BlockPalettes.com

**BlockPalettes.com** — the single best resource for block combination inspiration. Browse hundreds of community-created palettes, each showing exactly which blocks work together. Filter by style, color, or mood.

### How to use it:

1. Open [blockpalettes.com/palettes](https://blockpalettes.com/palettes)
2. Browse featured palettes or filter by color/theme
3. Click a palette to see all blocks in the combination
4. Test it: place a 5×5 section in a flat area, step back, see if it works
5. If any block fights the others, swap it out

## Starter Palettes for This Modpack

These palettes use blocks you can gather in Waves 0–2. Each fits a different biome and vibe:

**Medieval Stone** — stone brick + cracked stone brick + mossy stone brick + cobblestone (corners/base)

**Cozy Cottage** — oak planks + spruce planks + cobblestone (chimney/base) + oak leaves (roofline trim)

**Desert Sandstone** — sandstone + smooth sandstone + cut sandstone + terracotta accents

**Nordic Cabin** — spruce planks + stripped spruce log + stone + dark oak stairs (roof)

**Cherry Zen** — cherry planks + cherry log + stone brick + pink petals (garden)

**Mountain Fortress** — stone brick + deepslate tiles + tuff bricks + copper (accents/patina)

**Tropical** — jungle planks + bamboo mosaic + mangrove propagules + sand

**Endstone Palace** — purpur block + purpur pillar + end stone bricks + chorus flower (End builds)

**Tip:** Open BlockPalettes.com in your browser while you build. When you're stuck on what goes together, browse palettes until something clicks.

## Depth and Layering

Flat walls are the second most common beginner mistake (after single-block builds). A flat wall looks like a placeholder. A layered wall looks **built**.

### Five Techniques That Transform Flat Walls

| Technique                   | How to do it                                                                               | Why it works                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Inset windows</b>        | Push glass back one block into the wall instead of flush with the surface                  | Creates shadow lines that frame windows                             |
| <b>Outcropped corners</b>   | Extend corner blocks one block forward from the main wall plane                            | Breaks up 90-degree edges, adds structural feel                     |
| <b>Alternating depth</b>    | Every 3–4 blocks along a wall, push one column back or pull one forward                    | Wall reads as textured rather than uniform                          |
| <b>Stair/slab detailing</b> | Upside-down stairs under windowsills, stairs as roofline trim, slabs as half-height ledges | Catches light and shadow in ways full blocks can't                  |
| <b>Overhanging roofline</b> | Always extend the roof 1–2 blocks past the wall on all sides                               | Creates shadow line that defines where building ends and sky begins |

**Fastest single improvement:** Add vertical **log pillars** every 3–5 blocks along exterior walls. Logs imply structural support and instantly break up flat surfaces. Works with every style.

## Roof Design

A bad roof ruins a good build. A good roof makes even a simple box look intentional.

### Three Roof Styles (Pick One)

| Style                    | Shape                  | Best for                   | How                                                                               |
|--------------------------|------------------------|----------------------------|-----------------------------------------------------------------------------------|
| <b>A-frame (pitched)</b> | Triangle cross-section | Cottages, cabins, medieval | Stairs ascending from each wall meeting at a ridge. Overhang 1 block on all sides |
| <b>Flat with parapet</b> | Flat top, raised edges | Modern, desert, fortress   | Slabs or full blocks, 1-block                                                     |

| Style                  | Shape                           | Best for                | How                                            |
|------------------------|---------------------------------|-------------------------|------------------------------------------------|
|                        |                                 |                         | wall around perimeter                          |
| <b>Stepped pyramid</b> | Layered squares getting smaller | Asian, temples, fantasy | Each layer 2 blocks smaller than the one below |

**Material rule:** Roof material should **contrast** with wall material. Dark roof on light walls, or vice versa. Spruce stairs on oak walls. Deepslate on stone brick. Blackstone on sandstone.

**Common mistake:** Roof too small. The roof should be wider and taller than feels “necessary.” An oversized roof looks dramatic; an undersized one looks like a hat that doesn’t fit.

## Interior Design

An empty interior is as bad as a bad exterior. Minecraft has no furniture mod yet (Wave 6 adds Handcrafted), but vanilla blocks create convincing rooms.

### Lighting (Critical)

**Never place torches on interior walls.** They look improvised. Better options:

| Light source           | Look                 | How                                                         |
|------------------------|----------------------|-------------------------------------------------------------|
| <b>Lanterns</b>        | Warm, architectural  | Hang from ceiling via fence post                            |
| <b>Glow berries</b>    | Organic, trailing    | Place on ceiling, let vines trail down                      |
| <b>Sea lanterns</b>    | Clean, neutral       | Modern or sci-fi builds                                     |
| <b>Jack o’lanterns</b> | Hidden under carpets | Floor lighting without visible sources                      |
| <b>Froglights</b>      | Subtle, colorful     | Pearlescent/verdant/ochre variants from Aether or Overworld |

### Functional Rooms

| Room           | Key blocks                              | Details                                                                           |
|----------------|-----------------------------------------|-----------------------------------------------------------------------------------|
| <b>Kitchen</b> | Smoker, campfire, barrels, chests       | Barrel = cabinets. Chest with food item frame = pantry. Cooking pot over campfire |
| <b>Bedroom</b> | Bed, crafting table, bookshelf, lantern | Banner on wall above bed as headboard. Carpet at foot of bed                      |
| <b>Library</b> | Bookshelves, lectern, quill             | Cover walls with bookshelves for instant warmth                                   |

| Room            | Key blocks                            | Details                                     |
|-----------------|---------------------------------------|---------------------------------------------|
| <b>Storage</b>  | Sophisticated Storage barrels, chests | Organized rows. Sign labels on each barrel  |
| <b>Workshop</b> | Crafting table, furnace, anvil        | Industrial feel — stone walls, iron accents |

**Pro tip:** Place **item frames** on walls with maps, banners, or shields inside. They function as paintings with more variety.

## Blending Into the Landscape

A build that looks dropped from the sky is unfinished, no matter how detailed the structure itself is.

**Key principles:**

- **Use biome-appropriate materials** — spruce in cold biomes, sandstone in desert, dark oak near jungle edges. The build should look like it was built from local materials, not imported
- **Never leave flat grass at the base** — add a slight slope toward the entrance, gravel/cobblestone paths leading away, small dirt mounds or rock outcroppings at corners
- **Plant trees near the build** — a cluster of 2–3 trees on one side breaks the silhouette and gives context. Use the biome’s native trees
- **Cut into hillsides** — a build emerging from a hillside looks permanent; one sitting on top looks temporary
- **Paths and surroundings** — gravel paths, stone brick walkways, and lantern posts leading away from the entrance sell the idea that people live here
- **Never run more than 7 blocks in a straight line** when shaping terrain — curves look natural, straight lines look artificial

## Modpack-Specific Building Tools

You have tools that vanilla players don’t. Use them:

| Tool                         | What it does for building                                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waystones</b>             | Place one at your build site. Teleport home for materials, teleport back. No more carrying 3 inventories of blocks across 500 blocks |
| <b>Sophisticated Storage</b> | Barrels near your build site hold materials organized by type. Label with signs                                                      |
| <b>Backpack</b>              | Carry your palette blocks in your backpack while building. Press <b>B</b> to swap materials on the go                                |

| Tool                           | What it does for building                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Create Blueprint</b>        | Craft Blueprint and Quill. Select your finished build → save as blueprint → use Schematicannon to auto-rebuild it elsewhere with materials from chests |
| <b>JourneyMap waypoints</b>    | Mark your build site, mark material locations, mark inspiration you find while exploring                                                               |
| <b>Dynamic Lights (Ctrl+L)</b> | Hold a lantern while building at night. No more stopping to place torches                                                                              |

## Quick-Start: Your First Real House

If you've never built anything beyond a dirt hut, follow these steps for your first proper home:

1. **Pick a palette** — Open [BlockPalettes.com](https://blockpalettes.com), find 4 blocks you like. Test in a 5×5 section
2. **Lay the foundation** — Cobblestone or stone brick ring, 7×9 interior minimum (small but not cramped)
3. **Raise the walls** — Use your primary block. Leave gaps for windows (2-block wide minimum)
4. **Add depth** — Push windows back 1 block. Add log pillars at corners and every 4 blocks
5. **Build the roof** — A-frame with stairs. Overhang 1 block. Use a contrasting material
6. **Add the door** — Center of one wall. Stairs as a small porch step in front
7. **Light it up** — Lanterns from ceiling, not torches on walls
8. **Furnish basics** — Bed, crafting table, furnace, 2–4 barrels for storage
9. **Blend the exterior** — Paths leading away, trees nearby, flowers at the entrance, leaves around the roofline
10. **Step back and admire** — You just built something worth coming home to

## Community Resources

| Resource                  | URL                                                                             | What it's for                                             |
|---------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>BlockPalettes</b>      | <a href="https://blockpalettes.com">blockpalettes.com</a>                       | Block color combinations and palette inspiration          |
| <b>Building Guide App</b> | <a href="https://buildingguide.app">buildingguide.app</a>                       | Step-by-step tutorials with materials lists (128+ builds) |
| <b>r/MinecraftBuilds</b>  | <a href="https://reddit.com/r/MinecraftBuilds">reddit.com/r/MinecraftBuilds</a> | Community showcase, inspiration, feedback                 |
| <b>r/MinecraftHouses</b>  | <a href="https://reddit.com/r/MinecraftHouses">reddit.com/r/MinecraftHouses</a> | House-specific builds and ideas                           |

| Resource                         | URL                                                                                         | What it's for                                      |
|----------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>r/MinecraftTutorials</b>      | <a href="https://reddit.com/r/MinecraftTutorials">reddit.com/r/MinecraftTutorials</a>       | How-to guides for specific techniques              |
| <b>Minecraft Wiki — Building</b> | <a href="https://minecraft.fandom.com/wiki/Building">minecraft.fandom.com/wiki/Building</a> | Official guides on terraforming, furniture, layout |

## Session Plan — Wave 2.5

| Session                                 | Focus                                                                                          | What to Build                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Session 1 — Site selection</b>       | Explore with Explorer's Compass. Find a biome you love. Place Waystone. Mark on JourneyMap     | Your future home location      |
| <b>Session 2 — Palette + foundation</b> | Browse BlockPalettes.com. Pick 4 blocks. Lay foundation ring. Raise walls to 4 blocks high     | Shell of your first real house |
| <b>Session 3 — Roof + depth</b>         | Build A-frame roof. Add log pillars. Inset windows. Add door                                   | Recognizable house shape       |
| <b>Session 4 — Interior</b>             | Lanterns from ceiling. Bed, crafting table, furnace. Barrels for storage. Item frames on walls | Livable interior               |
| <b>Session 5 — Landscaping</b>          | Paths, trees, flowers, leaf trim around roofline. Step back and see the difference             | Home in a world                |

**Tip:** Don't aim for perfection on your first build. Aim for "better than a dirt hut." You can always expand, renovate, or rebuild later. The skills transfer — your second build will be twice as good.

# Wave 3 — Equipment Magic & RPG

## Equipment

| Mod                                  | Role                                                                                                  |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| <a href="#">Simply Swords</a>        | 60+ unique melee weapons with passive/activated abilities                                             |
| <a href="#">Simply More</a>          | 10 new weapon types + 33 unique weapons — Simply Swords addon                                         |
| <a href="#">Too Many Bows</a>        | 30+ unique bows with special abilities and custom attributes                                          |
| <a href="#">Arsenal (RPG Series)</a> | 40+ loot-only legendary weapons with passive spells — same RPG Series family as Relics                |
| <a href="#">Relics (RPG Series)</a>  | Legendary accessories with unique abilities — levels up with use                                      |
| <a href="#">Immersive Armors</a>     | 40+ unique armor sets with custom effects — berserk, double jump, hostile radar, spikes, divine block |

## Enchanting

| Mod                                | Role                                                                                                                                                                                      |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Easy Magic</a> v21.1.4 | Enchanting-table QoL — items stay in the table after closing, easy re-rolls, enchantment hints in tooltips. Requires Puzzles Lib (in pack). Optional Enchantment Descriptions integration |

## RPG Progression

| Mod                                      | Role                                                                                           |
|------------------------------------------|------------------------------------------------------------------------------------------------|
| <a href="#">Improvable Skills</a> 21.0.2 | Buy permanent passive buffs + utility abilities with vanilla XP levels from the Book of Skills |

## Configuration — Wave 3

**Easy Magic** — stock defaults. No overrides. (Puzzles Lib already in pack; Enchantment Descriptions integration is automatic.)

**Improvable Skills** — stock defaults (plug-and-play by design). No overrides authored blind — skill costs are config-adjustable if tuning is ever wanted. [VERIFY] at first launch: XP levels spend on skills; Book of Skills obtainable (see EMI); no conflict with the vanilla XP bar / enchanting table.

## Interdependency notes for Wave 3

- **Spell Engine** is required by Relics (RPG Series) and Arsenal (RPG Series) — it was previously considered for removal but must be retained as a mandatory dependency of the RPG Series mods.
- **Ranged Weapon API** is shared between Relics (RPG Series) and Arsenal (RPG Series) — install once, used by all.
- **Shield API** is required by Arsenal (RPG Series) — small RPG-Series library; no standalone content.
- **Enchantability**: All Simply Swords, Arsenal, Too Many Bows, and Immersive Armors pieces are enchantable at the vanilla enchanting table (Easy Magic QoL). Relics accessories are not enchantable (accessory items, by design). Better Combat (Wave 5) is the shared animation layer for all four weapon mods.
- **Curios API** provides the accessory slot system used by Relics (RPG Series) and Immersive Armors. Adds an equipment GUI (keybind [Y](#)) with ring, amulet, belt, gloves, back, and charm slots. Auto-installed dependency — no separate config needed.
- **Improvable Skills** requires **HammerLib** (Zeith's library mod — same author, zero deps of its own; new dependency, auto-installs via XMCL from the Improvable Skills page, pinned 21.0.14 for 1.21.1). The optional JEI integration is skipped — the pack is EMI-only.

**Immersive Armors** adds 40+ unique armor sets with custom effects: Wooden (arrow/explosion protection), Skeleton (speed, Wither variant, skeleton neutrality), Divine (dyeable, blocks 1 attack/min), Heavy (high armor, anti-knockback, slow), Slime (bounce attackers), Warrior (berserk — more damage at low HP), Robe (dyeable wool, fire thorns, fireproof), Steampunk (hostile radar, enhanced UI, double jump, no fall damage), and Prismarine (spikes, depth strider). All sets are enchantable, repairable, and dyeable where applicable. Recipes in vanilla recipe book and EMI. All pieces enchant via the vanilla enchanting table (Easy Magic QoL). Mekanism Tools fills the material-tier progression (Osmium→Bronze→Steel→MekaSuit); Immersive Armors fills the unique-effect side grade slot — different armor sets for different situations rather than linear upgrades.

## Simply Swords — Unique Weapons

**Version pin (Aug 2026)**: Simply Swords **1.63.0** — pinned alongside Simply More 1.2.3 (both stable releases). The 1.70 line (Aug 24, 2026) is a save/config-breaking rewrite that forces Simply More onto its alpha 1.3.0 line; upgrade to 1.70.x + 1.3.0 once Simply More ships stable. Simply Tooltips (auto-installed dep) renders the dynamic tooltips with ability descriptions and stat previews.

60+ unique weapons with passive and activated abilities. Drop from mobs (~2% chance), chests, and bosses.

| Category        | Examples                                 | Playstyle                                              |
|-----------------|------------------------------------------|--------------------------------------------------------|
| <b>Heavy</b>    | Greathammer, Greatsword, Glaive          | Slow, high damage, knockback.<br>Two-handed            |
| <b>Medium</b>   | Longsword, Katana, Claymore              | Balanced speed/damage.<br>Versatile                    |
| <b>Light</b>    | Rapier, Dagger, Sai, Cutlass             | Fast, low damage per hit. Armor penetration on rapiers |
| <b>Polearms</b> | Spear, Halberd, Pike                     | Extended reach (2-3 blocks).<br>Throwable spears       |
| <b>Special</b>  | Scythe (AoE), Soul Harvester (lifesteal) | Unique mechanics                                       |

**Weapon abilities** (hold [Ctrl](#) while hovering to see):

| Ability   | Effect                               | Found on                |
|-----------|--------------------------------------|-------------------------|
| Whirlwind | Spin attack, hits all nearby         | Greathammer, Greatsword |
| Impale    | Charge + release for piercing thrust | Spear, Pike             |
| Backstab  | +300% damage from behind             | Dagger, Sai             |
| Lifesteal | Heal 10% of damage dealt             | Soul Harvester, Scythe  |
| Execute   | +50% damage to targets below 30% HP  | Axe variants            |

### Finding weapons:

- Mob drops: ~2% chance. Zombies = common, skeletons = ranged-adjacent
- Chests: Dungeon/mineshaft/YUNG's structure chests have higher rates
- Trading: Some from weaponsmith villagers
- Bosses: Higher guaranteed drops from Twilight Forest and Cataclysm bosses

## Too Many Bows — Unique Ranged Weapons

30+ unique bows with special abilities and custom attributes.

**How they work:** Each bow has a unique passive or activated ability. Check EMI for recipes and stats. Some shoot multiple arrows, some inflict status effects, some have special draw mechanics.

**Finding them:** Dungeon chests, mob drops, and boss loot. Enchant them at the enchanting table (Easy Magic QoL) for even more damage.

**Integration:** Simply Swords + Simply More covers melee (60+ unique weapons across 20+ types). Too Many Bows covers ranged. Better Combat (Wave 5) provides appropriate animations for all three.

## Simply More — Exotic Weapon Types

**Version pin (Aug 2026):** Simply More 1.2.3 (stable, Jun 10, 2026) — pairs with the pinned Simply Swords 1.63.0. The 1.3.0 line is alpha and tied to the Simply Swords 1.70 rewrite; hold until it goes stable.

A Simply Swords addon that adds 10 new weapon types — not just more swords, but entirely new combat styles. Weapons are balanced around Better Combat (Wave 5) for full animation support.

| Weapon Type           | Playstyle                                              |
|-----------------------|--------------------------------------------------------|
| <b>Great Katana</b>   | Two-handed, wide sweeping arcs                         |
| <b>Grandsword</b>     | Massive two-hander — disables enemy shields on block   |
| <b>Backhand Blade</b> | Fast, dual-wield friendly, close-range                 |
| <b>Lance</b>          | Mounted combat specialist — massive boost while riding |
| <b>Khopesh</b>        | Curved blade — bonus against unarmored targets         |
| <b>Dagger</b>         | Fastest attack speed, backstab bonus                   |
| <b>Pernach</b>        | Armor-piercing mace — shreds tough enemies             |
| <b>Quarterstaff</b>   | Extended reach blunt weapon, knockback-focused         |
| <b>Great Spear</b>    | Longest reach in the game — hit from 4+ blocks         |
| <b>Deer Horns</b>     | Dual-wield fist weapons — rapid flurry strikes         |

Plus 33 unique weapons with special abilities. All weapons use Simply Swords' unique drop system — loot from mobs, chests, and bosses.

**Dependencies:** Simply Swords (already in pack). Cloth Config API and Architecture API are already installed from Wave 0.5 — no new mods needed.

**Finding them:** Same as Simply Swords — mob drops (~2% chance), structure chests, boss loot. EMI shows all recipes and drop sources.

## Arsenal (RPG Series) — Loot-Only Legendary Weapons

40+ legendary weapons across 15 types, each with at least one built-in passive spell. **Not crafted — conquered:** drops from bosses and endgame dungeon chests only, matching the pack's loot-driven power fantasy.

**Integration:** Arsenal weapons are melee weapons, enchantable via the vanilla enchanting table (Easy Magic QoL). Passive spells run through Spell Engine — the same RPG Series infrastructure as Relics: no new magic systems, no crafting chain. Better Combat (Wave 5) animates them alongside Simply Swords.

**Dependencies:** Spell Engine, Ranged Weapon API, Shield API (auto-installed RPG-Series libraries) + GeckoLib (already in pack) — zero new frameworks.

**Finding them:** Boss drops + endgame dungeon chests (Twilight Forest, Cataclysm, YUNG's high-tier). EMI shows drop sources.

## Relics (RPG Series) — Legendary Accessories

Unique accessories for **Curios slots**. You have 6 slots: Ring (2), Amulet (1), Belt (1), Gloves (1), Back (1), Charm (1). Press Curios keybind (**Y** recommended) to open inventory.

| Relic                    | Slot   | Ability                               | Where to find               |
|--------------------------|--------|---------------------------------------|-----------------------------|
| <b>Monkey Talisman</b>   | Charm  | Double jump                           | Vanilla dungeon chests      |
| <b>Holy Water</b>        | Charm  | Right-click cleanses negative effects | Desert temples, churches    |
| <b>Lucky Coin</b>        | Charm  | +luck for better loot                 | Mineshafts, outposts        |
| <b>Captain's Hook</b>    | Gloves | Pull entities toward you              | Shipwrecks, ocean ruins     |
| <b>Sacred Wardstone</b>  | Amulet | Absorbs X damage then breaks          | Stronghold chests           |
| <b>Magic Mirror</b>      | Back   | Teleport to spawn/home                | End cities, libraries       |
| <b>Heart of the Void</b> | Ring   | +max HP, slow regen                   | Nether fortresses, bastions |
| <b>Blazing Feather</b>   | Back   | +15% movement speed                   | Jungle temples, pagodas     |
| <b>Drowned Belt</b>      | Belt   | Underwater speed boost                | Ocean monuments, shipwrecks |

**Stacking:** Two rings = two ring relics. Wear multiple different types.

**Finding strategy:** Use Explorer's Compass → find YUNG's structures → loot them. Relics have higher drop rates in YUNG's structures than vanilla ones.

## Easy Magic — Enchanting Table QoL

The enchanting table keeps vanilla behavior but loses the friction:

- **Items stay in the table** after closing the GUI — no re-inserting your sword every time.
- **Easy re-rolls** — cycle the offered enchantments without burning levels on bad rolls.
- **Enchantment hints** — see exactly what each option does in the tooltip; integrates with Enchantment Descriptions (in pack) for full descriptions.
- Requires **Puzzles Lib** (already in pack). Runs stock defaults — no config to manage.

Enchanting stays vanilla-capped (Protection IV, Sharpness V). The gear fantasy now comes from unique weapon/armor passives (Simply Swords, Arsenal, Relics, Immersive Armors) rather than max-level enchantment stacking.

## Improvable Skills — Vanilla-XP Skill Book

The pack's character layer — with no second XP system. Your vanilla levels ARE the skill points: every level earned by mining, fighting, breeding, fishing, crafting... is currency for permanent power stacked on top of the gear chase.

- **Vanilla XP = skill currency** — open the **Book of Skills** (right-click the item; see EMI for how to obtain it) and spend XP levels to permanently unlock skills. One XP pool serves enchanting, anvils, and character growth.
- **Skills** (permanent passive buffs): Extra Health (+20 HP), Strong Fist (melee damage), Sharp Arrow (ranged damage), Attack Speed, Mining / Digging / Cutting (tool speed), Dense Skin (armor), Leaper (jump height), Soft Landing (fall damage), Fast Swimmer, Soul Speed, Huckster (villager discounts), Enchanter (lower enchant level requirement), Experience Harvesting (+XP gain), Accelerated Furnace, Alchemist (brewing speed), Fast Growth, Luck of the Sea, Obsidian Skin, Ender Manipulator, Silent Foot, Treasure of the Sands.
- **Abilities** (utility, crafted as scrolls from Parchment Fragments): portable anvil / crafting table / enchanting table, Magnetism (item magnet), Auto XP Bank (stashes excess levels so you always hold exactly the amount you set).
- **Synergies** — GraveStone Mod keeps your XP on death, so skill-buying power survives too; Easy Magic re-rolls and skills both draw from the same levels, so you choose between enchanting and character upgrades.
- **Requires HammerLib** (Zeith's library mod, auto-installed, pinned 21.0.14). License ARR with explicit modpack permission; 5.3M+ CurseForge downloads.
- **No duplication** — Relics levels gear through use; Improvable Skills buys *character* power with vanilla XP. They layer; they don't overlap. (Magnetism vs Sophisticated Storage's Magnet Upgrade: opt-in ability — skip it if the backpack upgrade already covers you.)
- **Chill-safe** — skills are permanent passive buffs bought at your own pace; no item gating, no mob scaling, nothing hostile to idle play.

The skill GUI has **no keybind** — it lives on the Book of Skills item (right-click to open). There's no dark-theme resource pack for this mod; the GUI renders stock, consistent with other stock-default mods.

## Session Plan — Wave 3

| Session                              | Focus                                                                                      | What to Do            |
|--------------------------------------|--------------------------------------------------------------------------------------------|-----------------------|
| <b>Session 1 — First unique loot</b> | Check every gear drop and dungeon chest. Look for Simply Swords / Arsenal / Relics uniques | Learn the loot system |
| <b>Session 2 — Weapon hunt</b>       | Explorer's Compass → YUNG's dungeon → loot                                                 | First unique weapon   |

| Session                              |  | Focus |                                                                                                             | What to Do                                     |  |
|--------------------------------------|--|-------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------|--|
|                                      |  |       | for a Simply Swords or Arsenal weapon. Try Rapier (fast) or Greathammer (AoE)                               |                                                |  |
| <b>Session 3 — Enchanting setup</b>  |  |       | Build an enchanting table + bookshelves. Use Easy Magic re-rolls to get Sharpness IV / Protection IV        | First maxed gear                               |  |
| <b>Session 4 — Relic hunting</b>     |  |       | Monkey Talisman (dungeons) + Lucky Coin (mineshafts)                                                        | Accessory slots filled                         |  |
| <b>Session 5 — Endgame gear hunt</b> |  |       | Hunt Cataclysm / Twilight Forest / Aether bosses for top-tier unique drops                                  | Endgame gear progression                       |  |
| <b>Session 6 — Skill book</b>        |  |       | Obtain the Book of Skills (see EMI) and spend XP levels on mining/ combat/farming skills as they accumulate | Character progression alongside the gear chase |  |

**Tip:** Wave 3 gear changes how every other wave feels. A unique weapon with a strong passive makes Twilight Forest and Cataclysm bosses dramatically easier. Invest in gear before tackling tough content.

---

## Wave 4 — Food, Farming & Comfort

| Mod                                           | Role                                                                                                                                |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Farmer's Delight</b> 1.3.2                 | Cooking hub — cutting board, cooking pot, stove, new crops                                                                          |
| <b>Extra Mod Integrations</b>                 | Adds compat recipes for 50+ mods — Farmer's Delight, Create, Mekanism, AE2                                                          |
| <b>Brewin' And Chewin'</b> 4.5.0              | Drinks and fermentation — beer, wine, mead, whiskey                                                                                 |
| <b>Storage Delight</b> 1.3.0                  | Kitchen furniture — drawers, cabinets, shelves for ingredient storage                                                               |
| <b>Crate Delight</b> v26.07.01                | Bulk ingredient storage — crates (apples, berries, eggs, fish) and bags (cocoa, sugar, gunpowder)                                   |
| <b>More Delight</b> 26.04.18a+1.21-neoforge   | Extra meals, sandwiches, toasts, and knife variants                                                                                 |
| <b>My Nether's Delight</b> 1.10.2             | Nether-themed food ingredients and recipes                                                                                          |
| <b>Ocean's Delight</b> 1.0.4                  | Ocean-themed seafood dishes                                                                                                         |
| <b>Hybrid Delights</b> 1.2.1                  | Hybrid Aquatic × Farmer's Delight — HA fish, crustacean & more recipes                                                              |
| <b>Ender's Delight</b> v1.3.0                 | End-themed food for Farmer's Delight (moved from Wave 2)                                                                            |
| <b>Twilight's Flavors &amp; Delight</b> 3.2.2 | Twilight Forest × Farmer's Delight crossover — 35+ FD-style TF foods, 4 TF knives, maze stove, 5 potion effects, trophy dismantling |
| Neo Bee Fix                                   | Fixes bee AI bugs — bees leave hives, don't fall from sky, no NW wander bias                                                        |
| Comfortable Campfires                         | Campfires grant configurable status effects — sit by the fire, get a buff                                                           |
| Smarter Farmers                               | Farmer villagers replant correctly — incl. modded seeds (FD crops via tags)                                                         |
| Crops Love Rain 3.2.1                         | Crops grow faster while raining — crops, bamboo, saplings, sugar cane, cocoa, berries, melon/pumpkin stems (rain-only bonus)        |
| Create: Café                                  | Coffee, tea & boba drinks via Create machines — roasting, grinding, brewing                                                         |

| Mod                                                             | Role                                                                                                         |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Starcatcher</b>                                              | Fishing overhaul — 100+ fish, skill minigame, trophies, tournaments, catalogue                               |
| <b>Starcatcher's Delight</b> v3.0.1                             | Starcatcher × Farmer's Delight crossover — fish fillets, seafood dishes, sushi                               |
| <b>Better Fishtanks</b>                                         | Functional aquariums — fish bags, decorations, feeding system, Create/AE2 compat                             |
| <b>Spice of Life: Carrot Edition</b><br>solcarrot-1.21.1-1.16.6 | Food-variety reward — eating new foods permanently raises max health up to a cap. No punishment, only reward |

**Spice of Life: Carrot Edition** — the *gentle* variant of food variety: each distinct food eaten once permanently raises your max health (up to a configurable cap of hearts), and it never *reduces* nutrition for repeat meals (unlike the original Spice of Life). This pairs directly with the pack's Farmer's Delight ecosystem — 100+ distinct meals = steady, voluntary max-HP growth feeding the power-fantasy pillar, with zero chill penalty for eating the same thing. **Config lives in `saves/<world>/serverconfig/` (per-world), not the tracked global `config/` folder** — tune `solcarrot-server.toml` (milestone cap, heart counts) after first world launch; set the cap so the bonus stops short of feeling mandatory.

## Farmer's Delight — Cooking Hub

### New Crops

| Crop           | Grows on                                 | Use                       |
|----------------|------------------------------------------|---------------------------|
| <b>Tomato</b>  | Farmland. Bush grows, pick when red      | Soups, sandwiches, salads |
| <b>Cabbage</b> | Farmland. Full head when ready           | Coleslaw, cabbage wraps   |
| <b>Onion</b>   | Farmland. Pick when shoots show          | Savory dishes, soups      |
| <b>Rice</b>    | Shallow water (1 block deep on dirt/mud) | Hearty meals, sushi       |

**Wild crops:** Break wild variants in the world for seeds:

- Wild tomatoes — plains/forests
- Wild cabbage — beaches
- Wild onions — taiga
- Wild rice — swamps

## Cutting Board

Craft: 1 knife + 2 planks (horizontal row). Place on ground, right-click to open GUI.

Place ingredient in left slot → processed output in right slot.

| Input       | Output         | Use                |
|-------------|----------------|--------------------|
| Cooked meat | Meat strips    | Sandwiches, salads |
| Cabbage     | Cabbage leaves | Wraps, coleslaw    |
| Bread       | Bread slices   | Toast, sandwiches  |
| Carrot      | Carrot slices  | Salads, soups      |
| Apple       | Apple slices   | Desserts, snacks   |

**Tip:** Knife in off-hand while chopping lasts longer (uses knife durability). Iron knife = dozens of chops.

## Cooking Pot

Craft: 2 iron + 2 planks + 1 bowl. Place over a heat source:

- **Stove** (fastest) — 4 bricks + 4 iron nuggets
- **Campfire** (medium) — 3 logs + 3 sticks + 1 coal
- **Lava** (slowest) — pot must sit directly above lava source

### Cooking process:

1. Right-click pot → open GUI
2. Fill water slot (bucket or bottle)
3. Add up to 6 ingredients in any order
4. Wait for progress bar
5. Collect meal — most give **6 servings** with bowl stacking

### Meal categories:

- **Soups** — Tomato soup, beef stew. Good saturation, easy ingredients
- **Sandwiches** — Chicken sandwich, BLT. On-the-go eating
- **Salads** — Mixed salad, fruit salad. Good for hot biomes
- **Pasta** — Pasta with meat sauce. Filling, complex ingredients
- **Rice meals** — Fried rice. Requires rice farming

## Kitchen Storage

Craft **oak/pantry cabinets** (check EMI — varies by wood). Function as chests with kitchen aesthetics.

**Tip:** Set up kitchen near your AE2 network. Cook manually or stock ingredients in ME drives for autocrafting.

## Brewin' And Chewin' — Drinks & Fermentation

Alcohol brewing with aging mechanics. Grow hops, barley, grapes on farmland.

### Brewing Station

Craft (check EMI — barrel-like recipe). Right-click to open GUI.

1. Add **water** + main ingredient:
  - Barley → beer base (wort)
  - Grapes → wine base (must)
  - Honey → mead base
  - Potatoes → spirits base (mash)
2. Wait for fermentation
3. Collect **fermented mixture**

### Aging Barrel

Craft (check EMI — wooden barrel). Place fermented mixture inside, right-click to seal.

Ages one stage per in-game day: Young → Matured → Aged → Vintage (4 stages). Each stage increases status effect potency and duration.

**Optimal:** Let everything reach **Aged** (stage 3). Push to Vintage (stage 4) only for special occasions.

| Drink   | Matured effect       | Vintage effect                     |
|---------|----------------------|------------------------------------|
| Beer    | Resistance I (2 min) | Resistance II (5 min)              |
| Wine    | Haste I (2 min)      | Haste II (4 min)                   |
| Mead    | Speed I (2 min)      | Speed II (4 min)                   |
| Whiskey | Strength I (30 sec)  | Strength II (1 min)                |
| Vodka   | Night Vision (3 min) | Night Vision + Fire Resist (5 min) |

**Practical use:** Brew a batch of each, age to Matured/Aged, store in kitchen. Pop a drink before boss fights for free buffs. Mead stacks with Swiftiness potions.

**Bottle return:** Empty bottles are returned after drinking. Reuse at brewing station.

## Storage Delight — Kitchen Furniture

Decorative kitchen storage that works as functional inventory:

| Block   | Slots | Use                        |
|---------|-------|----------------------------|
| Cabinet | 27    | Counter-height storage     |
| Drawer  | 18    | Fits under counters        |
| Shelf   | 9     | Wall-mounted spice display |

**Tip:** Use for early-game ingredient storage before AE2. Later, place an ME Storage Bus on a cabinet to bridge into your digital network.

## Crate Delight — Ingredient Storage

Bulk storage for Farmer's Delight ingredients by axperty (same author as Storage Delight). Crates and bags keep your kitchen organized and look great.

| Type          | Examples                                                                      | Capacity                 |
|---------------|-------------------------------------------------------------------------------|--------------------------|
| <b>Crates</b> | Apples, Sweet Berries, Eggs, Raw Salmon, Raw Cod, Glow Berries, Golden Apples | Bulk single-item storage |
| <b>Bags</b>   | Cocoa Beans, Sugar, Gunpowder                                                 | Bulk single-item storage |

**Why it's useful:** Instead of a barrel full of mixed ingredients, you have a Crate of Apples next to your cooking station and a Bag of Sugar near your brewing setup. Thematic, organized, and satisfying to look at.

**Integration:** Same author as Storage Delight — identical art style, zero compatibility issues. All crates and bags are accessible via Create pipes and AE2 Storage Bus. Complements Storage Delight's furniture (cabinets/drawers/shelves) — Crate Delight handles bulk ingredient storage, Storage Delight handles kitchen furniture. Both are intentionally outclassed by AE2 in Phase 2 but remain useful for localized micro-storage near cooking and brewing stations.

## More Delight — Extra Meals & Knives

- **Extra meals:** Sandwiches, toasts, stuffed mushrooms
- **Knife variants:** Diamond and netherite knives (much longer lasting than iron)
- **New cooking pot recipes:** Stews, soups, pasta dishes

All recipes visible in EMI. Uses existing cutting board and cooking pot.

## My Nether's Delight — Nether Food

Nether-themed ingredients and recipes:

- **Nether crops:** Grow in soul sand or warped/crimson nylium
- **Ingredients:** Glowstone berries, soul berries, nether-root vegetables
- **Recipes:** Spicy stews (Fire Resistance), soul-force snacks (Night Vision)

**Tip:** Set up a Nether farm for combat-relevant status effects. Glow berry pie gives Haste II — useful before mining expeditions.

## Ocean's Delight — Seafood

Seafood dishes using ocean mob drops:

- **Ingredients:** Squid tentacles, fish fillets, seaweed, turtle eggs
- **Recipes:** Seafood stew, grilled fish, kelp salads, sushi rolls

- **Effects:** Water Breathing + Dolphin's Grace — perfect for ocean monument raids

**Tip:** Build a fish farm or farm kelp for steady supply. Combined with My Nether's Delight, prepare meals for any environment.

## Ender's Delight — End Food

End-themed food for post-Dragon exploration:

- **Ingredients:** Chorus fruit variants, end-themed crops
- **Recipes:** End stews and snacks
- **Effects:** Status boosts useful for End dimension exploration

**Integration:** End ingredients cook into Ender's Delight meals. Plan an End farming outpost with Farmer's Delight crops.

## Hybrid Delights — Hybrid Aquatic Cooking

The official HybridLabs bridge between Hybrid Aquatic (Wave 2) and Farmer's Delight (Wave 4) — FD-style recipes from Hybrid Aquatic's drops: fish and crustacean dishes, plus an FD-style knife (1.2.1 added fish n chips). **Zero new deps** — Farmer's Delight, Hybrid Aquatic, and Kotlin for Forge (Wave 0 dep) are all in the pack.

**Integration:** Routes Wave 2's Hybrid Aquatic drops into the Wave 4 kitchen — the crustacean path Ocean's Delight already uses, extended to the full roster. Complements (not duplicates) Ocean's Delight: OD covers generic ocean seafood, Hybrid Delights covers Hybrid Aquatic's specific creatures. Every distinct dish feeds Spice of Life: Carrot Edition's max-health rewards. Recipes show in EMI and store in AE2. Zero config needed — install and cook.

## Neo Bee Fix

Fixes bee AI — bees properly leave hives, don't fall from sky, no northwest wander bias. No config. Install and forget.

**Integration:** Bees produce honey for Brewin' And Chewin' mead. Farm bees near your kitchen for renewable brewing ingredients.

## Smarter Farmers — Villagers Replant

Fixes and upgrades villager farming: farmers replant the **correct** seed (no more wheat seeds on carrot rows) and can use **modded seeds** — Farmer's Delight crops included, via tag-based detection. A `harvest_blacklist` tag lets you exclude crops from auto-replanting.

**Integration:** Keeps village farms and FD crop plots self-sustaining — less manual replanting, more chill living. Works alongside MineColonies (separate villager system — no overlap). Requires Moonlight Lib (already in pack, Wave 6).

## Crops Love Rain — Rain-Powered Growth

Crops grow faster while it's raining — during rain, every random tick on any soil-based crop (vanilla + Farmer's Delight), bamboo, saplings, sugar cane, cocoa, sweet berry bushes, and melon/pumpkin stems gets a **1-in-10 chance of an extra growth stage** (`rain_growth_speed = 10`; lower = faster). Pure reward, zero maintenance — a chill-living garden bonus, nothing to set up or babysit.

**Config** (`config/cropsloverain.toml` — defaults are fine, no tracked override): `general.use_rain_growth_speed` (true), `general.rain_growth_speed` (10), plus per-type `individual.*` speeds (off by default). Native NeoForge file config — Mod Menu not required (the project page's "Requires Fabric API" banner applies to the Fabric build only; the NeoForge 1.21–1.21.5 build declares just `neoforge + minecraft`).

**Integration:** Rain harvests feed the kitchen — Farmer's Delight crops, Brewin' And Chewin' ingredients, and Spice of Life variety all benefit. Complements Serene Seasons (season-based growth modifiers — different axis, the bonus stacks). ⚠ Watch: Particle Rain replaces rain *rendering* only, not the weather state — the bonus still applies; verify once in-game.

## Create: Café — The Cozy Drinks Track

Coffee, tea, and boba milk tea made through **Create machines** — beans and leaves go through milling, crushing, and mixing processing (1.4.3, Jan 2026). Coffee is the one drink category Brewin' And Chewin' doesn't cover (B&C = fermentation/alcohol), so this is a second, non-overlapping drinks track that runs on Create power rather than the FD kitchen.

**Integration:** Feeds Create automation chains — roast/grind/brew as mechanical processing steps, pipeable and deployer-friendly. Phase 1 accessible (water wheel era). Drinks stack alongside FD meals and B&C drinks for buff-food variety. Only dependency is **Create** (already in pack) — zero new mods. [VERIFY] at test launch: drink effect strength (overpowered check), Create version compat, no runtime conflict with B&C.

## Starcatcher — Fishing Overhaul

A complete fishing overhaul with 100+ fish, skill-based minigame, progression system, and tournaments.

**Core mechanics:**

| Feature                 | What it does                                                     |
|-------------------------|------------------------------------------------------------------|
| <b>Fishing Minigame</b> | Skill-based reel-in — timing matters, not just patience          |
| <b>Catalogue</b>        | Records every catch — biome, weather, time conditions documented |

| Feature            | What it does                                                             |
|--------------------|--------------------------------------------------------------------------|
| <b>Trophies</b>    | Mount your best catches as wall decorations                              |
| <b>Tackle Box</b>  | Store hooks, bobbers, baits, and fish — portable fishing station         |
| <b>Aquariums</b>   | Multi-block fish tanks — place caught fish inside, scaled by size/weight |
| <b>Tournaments</b> | Compete against friends for biggest catch                                |
| <b>Selling Bin</b> | Sell fish for emeralds — price based on size, weight, rarity             |
| <b>Cosmetics</b>   | Fisherman hats, rod skins — found in shipwrecks and treasure             |

### Getting started:

1. Craft a **fishing rod** (3 sticks + 2 string) — Starcatcher replaces vanilla fishing
2. Cast near water → minigame triggers on bite
3. Complete the minigame → fish caught, added to catalogue
4. Craft a **Tackle Box** (check EMI) → store fish and accessories
5. Find **baits** in loot chests or craft them → attract specific fish types

**Fish conditions:** Each fish has specific requirements:

| Condition          | Examples                           |
|--------------------|------------------------------------|
| <b>Biome</b>       | Ocean fish, river fish, swamp fish |
| <b>Weather</b>     | Rain-only, clear-only, storm-only  |
| <b>Time of day</b> | Day fish, night fish, dawn/dusk    |
| <b>Elevation</b>   | Surface vs. deep water             |

**Golden Fish:** Rare variants of every fish — higher value, trophy-worthy. Hunt these for the ultimate collection.

**Configuration:** In `config/starcatcher-server.toml`, set `giveGuideOnFirstJoin = false` to prevent the fishing guide book from being handed to the player on first login. The guide is still craftable (check EMI for recipe) — just not auto-given.

**Integration with Hybrid Aquatic (Wave 2):** Starcatcher automatically adds Hybrid Aquatic fish as catchable entries. No config needed — more fish to catch, more variety in your catalogue.

## Better Fishtanks — Aquariums

Functional aquariums for displaying your catches. Build the tank of your dreams.

**Core blocks:**

| Block                 | Function                                                                        |
|-----------------------|---------------------------------------------------------------------------------|
| <b>Fishtank</b>       | Glass tank — house fish, configurable water levels                              |
| <b>Aquarist Table</b> | Crafting station for tanks, decorations, and supplies                           |
| <b>Fish Bag</b>       | Catch and transport fish — right-click fish in world, right-click tank to place |
| <b>Food Container</b> | Store fish food in bulk — flakes, kelp, shrimp, pellets                         |
| <b>Decorations</b>    | Coral, shipwrecks, stones, branches, towers, kelp — 30+ variants                |

### Getting started:

1. Craft an **Aquarist Table** (check EMI — iron + glass + planks)
2. Craft a **Fishtank** (glass + iron ingots) — place it down
3. Craft a **Fish Bag** (leather + string) → right-click any fish in the world to catch it
4. Right-click the Fishtank with a filled Fish Bag → fish placed inside
5. Add **decorations** from the Aquarist Table → customize the tank

**Fish Eggs:** Tanks passively produce fish eggs in 6 rarity tiers (Common → Legendary). Crack them in the Aquarist Table for loot.

**Aquarist Villager:** A new villager profession. Trades fish supplies and rare eggs. Assign by placing an Aquarist Table near an unemployed villager.

### Tech integration (Create, AE2):

- Fishtank and Aquarist Table inventories are **fully accessible** via Create pipes
- Use fluid pipes to fill tanks with water automatically
- Automate fish feeding with Create mechanical systems
- Connect to AE2 network for remote inventory access

### Config options:

| Setting                  | Default              | What it does                                               |
|--------------------------|----------------------|------------------------------------------------------------|
| <b>Require Food</b>      | ON                   | Fish need feeding to survive — disable for pure aesthetics |
| <b>Fish Bag Captures</b> | Normal fish only     | Enable to capture Guardians, Drowned, Whales               |
| <b>Rendering Mode</b>    | Vanilla Waterlogging | Toggle experimental entity rendering for water             |

**Integration with Starcatcher:** Better Fishtanks displays what Starcatcher catches. Starcatcher provides the fishing gameplay; Better Fishtanks provides the aquarium showcase. Different systems, complementary purposes.

## Starcatcher's Delight — Seafood Cooking

Starcatcher × Farmer's Delight crossover by wdiscute (same author as Starcatcher). Every fish you catch becomes a cooking ingredient.

| Food Type               | Ingredients                               | Notes                                                        |
|-------------------------|-------------------------------------------|--------------------------------------------------------------|
| <b>Fish Fillets</b>     | Any Starcatcher fish → Cutting Board      | Fish quality (Common → Legendary) determines fillet quality  |
| <b>Sushi</b>            | Fish fillet + rice + seaweed              | Rice from Farmer's Delight, seaweed from oceans              |
| <b>Fish Stew</b>        | Fish fillet + FD vegetables + Cooking Pot | Hearty meal with saturation buff                             |
| <b>Fried Fish</b>       | Fish fillet + FD oil + stove              | Quick crispy meal                                            |
| <b>Legendary Dishes</b> | Legendary fillet only                     | Always grants absorption hearts — save these for boss fights |

**Quality system:** The rarity of the fish you use determines the quality of the food. A legendary fish is cut into legendary fillets, which cook into legendary dishes. Each food has unique hand-written descriptions that change based on quality tier.

### Getting started:

1. Catch any fish via Starcatcher's minigame
2. Use a Farmer's Delight Cutting Board with a knife to produce fillets
3. Combine fillets with FD ingredients for meals
4. Hunt for legendary fish → legendary dishes with absorption hearts → boss fight food

**Integration:** All Starcatcher's Delight recipes appear in EMI. Fillets and dishes can be stored in AE2. Quality-based dishes stack separately (Common Sushi ≠ Legendary Sushi). Zero config needed — install and fish.

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## Wave 5 — Combat & Mobs

| Mod                                       | Role                                                                                                                                     |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Better Combat                             | Melee combat animations — weapon reach, sweeping, attack speed                                                                           |
| L_Ender's Cataclysm                       | Epic boss fights — Leviathan, Netherite Monstrosity, Ender Guardian (CurseForge, NeoForge 1.21.1)                                        |
| Cataclysmic Combat                        | Better Combat movesets for L_Ender's Cataclysm weapons — gauntlets, Incinerator, Void/Infernal forges, Annihilator, Soul Render          |
| Enchantment Descriptions                  | Shows enchantment effects in tooltips                                                                                                    |
| Create Big Cannons                        | Artillery engineering — siege cannons for colony defense and spectacle                                                                   |
| Create Big Cannons: Advanced Technologies | CBC addon — twin/heavy autocannons, rocket pods & rails, muzzle brakes, silencers, rifled barrels                                        |
| CBC Enchanced Shells [Create Big Cannons] | CBC addon — 86 tiered shells (Mk.I–V), powder charges, doomsday shells, rocket system                                                    |
| Torchmaster                               | Territory pacification — Mega Torch suppresses hostile spawns in a radius; each torch gated behind a Cataclysm boss conquest (see below) |

### Better Combat — Weapon Feel

Replaces Minecraft's single swing with weapon-specific movesets:

| Weapon       | Speed     | Reach      | Animation               | Best for                   |
|--------------|-----------|------------|-------------------------|----------------------------|
| Daggers      | Very fast | 2 blocks   | Quick stab              | Debuff stacking, fast crit |
| Swords       | Fast      | 3 blocks   | Horizontal arc          | General combat             |
| Rapiers      | Fast      | 3.5 blocks | Thrust (ignore shields) | Shield-breaking            |
| Greatswords  | Slow      | 4 blocks   | Wide sweep              | Group clearing             |
| Axes         | Medium    | 3 blocks   | Overhead chop           | Armor penetration          |
| Spears       | Medium    | 4.5 blocks | Forward thrust          | Kiting                     |
| Hammers      | Slow      | 3.5 blocks | Overhead slam           | Knockback, stun            |
| Greathammers | Very slow | 4 blocks   | Ground pound            | AoE knockback              |

**Config:** Mods screen → Better Combat → Config. Tweak reach, sweep radius, attack speed, visual effects per weapon type.

**Integration:** Simply Swords weapons have built-in Better Combat data files — each category uses correct animation, reach, and speed automatically. Too Many Bows and Simply More also integrate.

## Enchantment Descriptions

Hover over any enchantment in tooltip → detailed description of what it does. Works in enchanting table GUI, on items, and in books. No config.

## L\_Ender's Cataclysm — Epic Boss Fights

Hand-animated bosses with unique mechanics. **Opt-in** — you must build a summoning structure or find a specific location.

| Boss                  | Location               | How to Summon                                              | HP   | Recommended Gear                        |
|-----------------------|------------------------|------------------------------------------------------------|------|-----------------------------------------|
| Leviathan             | Deep ocean (Y<40)      | Prismarine + sea lantern altar + Prismarine Shard          | ~200 | Diamond, Depth Strider, door trick      |
| Netherite Monstrosity | Nether wastes          | Enter central arena in nether brick lair                   | ~300 | Full diamond/netherite, fire resistance |
| Ender Guardian        | End (outer islands)    | Eye of Ender on pedestal after Dragon                      | ~400 | Endgame gear, Elytra + Slow Falling     |
| Ancient Remnant       | Desert                 | Enter summoning room in large pyramid, insert Nether Star  | ~350 | Blast protection, blocks to pillar      |
| Ignite                | Nether (basalt deltas) | 5×5 hollow obsidian sphere + Blaze Rod center, wait 10 sec | ~250 | Fire resistance, water buckets          |

### General strategy:

1. Scout the arena first
2. Bring food buffs (Farmer's Delight + Brewin' And Chewin' stack)
3. Set a Waystone nearby

#### 4. AE2 wireless terminal for mid-fight resupply

**Leviathan:** Phase 1 — dodge charges, attack body, jump shockwaves. Phase 2 (50%) — avoid whirlpools, attack head between spawns. Drops Leviathan Blade (trident-like water weapon).

**Netherite Monstrosity:** Phase 1 — attack legs to slow. Phase 2 (60%) — enrages, faster attacks + fire breath. Sprint behind pillars. Drops Monstrosity tools (efficient netherite-level mining).

**Ender Guardian:** Phase 1 — dodge tracking energy bolts, hit wall crystals to stun. Phase 2 (50%) — teleports, summons endermite, break all crystals to lower shield. Phase 3 (20%) — energy nova, pillar up to avoid ground AOE. Drops high-damage weapons + teleportation gear.

**Ancient Remnant:** Phase 1 — sandstorm blinds, attack during cooldown. Phase 2 (40%) — sinks into sand, pops up elsewhere, shoot exposed weak point. Drops tools with unique enchantability.

**Ignite:** Phase 1 — fireball barrage, block with shield or dodge behind pillars. Phase 2 (40%) — absorbs lava from arena, stay on high ground, attack when emerging. Drops fire weapons (permanent fire aspect).

## Cataclysmic Combat — Better Combat for Boss Weapons

Bridges L\_Ender's Cataclysm weapons into Better Combat with proper movesets, so boss drops animate correctly instead of falling back to the vanilla swing:

| Cataclysm weapon    | Better Combat profile |
|---------------------|-----------------------|
| Gauntlet of Guard   | Fist — dual-wieldable |
| Gauntlet of Bulwark | Fist — dual-wieldable |
| The Incinerator     | Claymore — two-handed |
| Void Forge          | Hammer — two-handed   |
| Infernal Forge      | Hammer — two-handed   |
| The Annihilator     | Mace                  |
| Soul Render         | Glaive                |

**Dependency:** Better Combat + L\_Ender's Cataclysm (both in pack) — zero new mods. Install the NeoForge 1.21.1 build from CurseForge (the Modrinth listing is the old Forge/1.20.x line).

## Create Big Cannons — Artillery Engineering

Functional cannons built from Create materials, using rotational power for assembly and gunpowder for ammo.

## Getting started:

1. Craft cannon components: barrel (bronze/steel), mount (Create materials), charger
2. Build: place mount, attach barrel, connect charger
3. Arm: insert powder charges + cannonballs into GUI
4. Fire: right-click or redstone signal

| Ammo             | Effect                                | Use            |
|------------------|---------------------------------------|----------------|
| Solid cannonball | High damage, block-breaking explosive | Siege, defense |
| Canister shot    | Spread of small projectiles           | Anti-personnel |
| Grape shot       | Medium spread                         | Crowd control  |
| Incendiary       | Fire on impact + lingering fire       | Area denial    |
| Smoke shell      | Smoke screen                          | Concealment    |

**Integration:** Gated behind bronze/steel (Create progression) + gunpowder. Mount on colony walls for defense. Redstone-controlled pulse extender for automatic salvo fire.

**Dependency:** Ritchie's Projectile Library (auto-installed) — provides custom trajectories and hit detection. No separate config.

## Create Big Cannons: Advanced Technologies — Bigger Guns

An addon for Create Big Cannons adding 110 new cannon blocks and new ammunition, using the same casting/assembly workflow as base CBC.

| Addition                 | What it does                                                                                                                                             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muzzle Brake             | Reduces cannon recoil                                                                                                                                    |
| Silencer / Suppressor    | Reduces noise and recoil                                                                                                                                 |
| Fume Extractor           | Hides the muzzle smoke cloud — good for concealed emplacements                                                                                           |
| Rifled Barrel            | Tighter spread than a normal barrel                                                                                                                      |
| Twin Autocannon          | Double-barreled rapid fire (vertical variant can be hidden in leaves)                                                                                    |
| Heavy Autocannon         | Bigger-caliber autocannon for heavier rounds                                                                                                             |
| Rocket Pod / Rocket Rail | Launches rockets; the medium rocket rail handles four medium rockets                                                                                     |
| New ammunition           | Autocannon & heavy autocannon cartridges/caseless rounds, rockets (AP / HE / HEI / Flak), medium rockets (AP / HE / HEAT / HEF), fluid shells, grapeshot |

All tiers (bronze → cast iron → steel → nethersteel) follow the standard CBC casting workflow with cast moulds. No config required.

**Dependency:** Create Big Cannons (above) + Create. Install the 1.21.1 NeoForge build (0.1.4c, CurseForge/Modrinth).

## CBC Enhanced Shells — Tiered Munitions

Massive arsenal expansion for Create Big Cannons — 86 items across 5 progressive tiers (Mk.I → Mk.V) with distinct color codes and a linear upgrade chain:

| Category        | What's added                                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| Standard shells | 10 types × 5 tiers (HE, AP, Solid Shot, AP Shot, Shrapnel, Smoke, Drop Mortar, Grapeshot, Mortar Stone, Shell Holder) |
| Powder charges  | 5 upgraded propellants (Mk.I–V) with rising muzzle velocity                                                           |
| Doomsday shells | Nuclear Mk.I/II, H-Bomb, Mini H-Bomb (240+ block radius — late-game)                                                  |
| Shader shells   | 5 exotic projectiles with screen-warping post-process effects                                                         |
| Rocket system   | Redstone-fired launchpad + cascade-loading breech                                                                     |

Tiering maps to the progression curve: Mk.I = iron → Mk.V = netherite, spanning Create Phase 2 into Phase 3. Standard-shell explosive power is capped at ×3 for balance.

**Dependency:** Create Big Cannons (above) + Create 6.0+ — both in pack, zero new mods. Install the 1.21.1 NeoForge build from CurseForge.

**Watch — doomsday shells:** nukes are player-fired and opt-in (chill-clean), but a 240+ block radius will flatten your own base. Check the config on first launch and tame/disable the doomsday shells if they feel too destructive for colony defense.

## Torchmaster — Conquering the Land

The **Mega Torch** suppresses hostile mob spawning in a 64-block radius — but each torch is **earned**: crafting requires an Ignitium Ingot from a Cataclysm boss (Kube)S gate, [kubejs/server\\_scripts/wave5\\_torchmaster.js](#)):

| Item       | Gated Recipe                                                              | Gate                                       |
|------------|---------------------------------------------------------------------------|--------------------------------------------|
| Mega Torch | Ignitium Ingot (Cataclysm Ignite drop) + Create brass ingots + blaze rods | Phase 2 — first Cataclysm boss + brass age |

**The conquest loop:** scout a hostile region by air/ship/train → clear its structures → defeat the Cataclysm boss for the Ignitium Ingot → craft the Mega Torch → plant

it. The area is permanently peaceful. Conquest is renewable — Cataclysm arenas re-summon bosses (Eye item), so there's no world exhaustion.

**Config** (mod defaults — no override needed; `config/torchmaster-common.toml` is generated on first launch):

| Setting                             |  | Default           | Notes                                                                                 |
|-------------------------------------|--|-------------------|---------------------------------------------------------------------------------------|
| <code>megaTorchRadius</code>        |  | <b>64</b>         | Cube $\pm 64 \rightarrow 129 \times 129 \times 129$ zone: village + farms + perimeter |
| <code>feralFlareRadius</code>       |  | <b>16</b>         | Cube $\pm 16 \rightarrow 32 \times 32 \times 32$ zone, capped at 255 invisible lights |
| <code>blockOnlyNaturalSpawns</code> |  | <code>true</code> | Spawner + structure spawns exempt — YUNG's dungeons and Cataclysm content untouched   |

### Why these values, and why not bigger:

- **Mega Torch 64** — the radius is a *cube* (each direction), so 64 covers a village plus its hinterland. Each torch costs one Cataclysm boss kill (KubeJS gate), and the conquest loop is designed to repeat (arenas re-summon bosses). A 96+ cube would let a single kill pacify an entire region and collapse the loop; the original 48 draft underpays the conquest.
- **Feral Flare 16** — it's a *clearing aid*, not a territory tool: it sprinkles invisible light (level 15) in its cube until it hits the 255-light hardcap. At 16 the lights sit  $\sim 5$  blocks apart, keeping block light  $\geq 10$ ; at 24 they thin to  $\sim 7.5$ -block spacing and dark gaps reappear — worse than 16. Raising the hardcap to compensate trips the mod's own world-corruption warning (badly compressed packet errors).
- **Spawner exemption** — the Mega Torch never blocks `SPAWNER/STRUCTURE` spawns, so YUNG's dungeons and Cataclysm arenas keep their mobs at *any* radius. Torches pacify the surface, never the content.

## Session Plan — Wave 5

| Session                          | Focus                                                                                                                      | What to Do        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Session 1 — Combat feel</b>   | Better Combat works automatically. Craft/find a Simply Swords weapon. Test daggers (fast stab) vs greatswords (wide sweep) | Learn your weapon |
| <b>Session 2 — First dungeon</b> | Explorer's Compass $\rightarrow$ YUNG's Better Dungeons:                                                                   | Dungeon complete  |

| Session                                 |  | Focus                                                                                                                                                                     |  | What to Do               |  |
|-----------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|--|
|                                         |  | clear a dungeon or spider cave                                                                                                                                            |  |                          |  |
| <b>Session 3 — First Cataclysm boss</b> |  | Easiest = Leviathan (deep ocean) or Netherite Monstrosity (Nether). Prepare: good weapons, fire resistance, healing food                                                  |  | Epic boss beaten         |  |
| <b>Session 4 — Cannons</b>              |  | Craft bronze (zinc + copper in heated mixer). Build Small Cannon. Test fire. Mount near base entrance                                                                     |  | Artillery online         |  |
| <b>Session 5+ — Boss gauntlet</b>       |  | Remaining Cataclysm bosses. Each drops unique endgame materials. Revisit Twilight Forest bosses with Wave 5 gear                                                          |  | Full endgame gear        |  |
| <b>Session 6 — First conquest</b>       |  | Scout a region by plane, clear its structures with guns/artillery, defeat the Cataclysm boss, craft the Mega Torch from its Ignitium Ingot, plant it. The region is yours |  | First pacified territory |  |

**Tip:** Better Combat changes melee fundamentally. Don't skip Session 1 even as a ranged player.

---

# Wave 6 — Building & Decoration

## Core Decoration

| Mod                                       | Role                                                                                                                          |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Rechiseled</a>                | 600+ decorative block variants with connected textures                                                                        |
| <a href="#">Supplementaries</a>           | Vanilla+ decoration — jars, signposts, faucets, weather vanes                                                                 |
| <a href="#">Macaw's Bridges</a>           | Bridges in all wood types                                                                                                     |
| <a href="#">Macaw's Doors</a>             | Vanilla-style doors for every wood                                                                                            |
| <a href="#">Macaw's Roofs</a>             | Roofing blocks                                                                                                                |
| <a href="#">Macaw's Windows</a>           | Window variants                                                                                                               |
| <a href="#">Macaw's Fences and Walls</a>  | Fences, gates, and walls in every wood & stone material                                                                       |
| <a href="#">Macaw's Lights and Lamps</a>  | Lamps, torches, street lamps, paper lamps in every wood                                                                       |
| <a href="#">Macaw's Stairs</a>            | New stair styles, handrails, balconies                                                                                        |
| <a href="#">Macaw's Paintings</a>         | Custom hand-drawn paintings for your walls                                                                                    |
| <a href="#">Handcrafted</a>               | Furniture — chairs, tables, shelves, curtains                                                                                 |
| <a href="#">Fetzi's Displays</a>          | Item displays — glass cloches, katana stands, wall holders, racks                                                             |
| <a href="#">Laser Bridges &amp; Doors</a> | Laser platforms, bridges, doors, and fences powered by redstone — length scales with signal strength, dye-customizable colors |
| <a href="#">Diagonal Fences</a>           | Connect vanilla fences, fence gates, and walls diagonally — seamless corners                                                  |
| <a href="#">Create Deco 2.1.3</a>         | Industrial decoration — catwalks, train hulls, sheet metal, shipping containers, cage lamps, brick sets, decals               |

## Rechiseled Addons

| Mod                                               | Role                                                                 |
|---------------------------------------------------|----------------------------------------------------------------------|
| <a href="#">Rechiseled: Create</a>                | 100+ decorative variants of Create blocks + mechanical chisel        |
| <a href="#">Rechiseled: Applied Energistics 2</a> | AE2 autocrafting for Rechiseled blocks via Chiseling Pattern Encoder |

## Rechiseled — Decorative Block Variants

600+ decorative block variants with connected textures.

### How:

1. Craft **chisel** (iron + 2 sticks, check EMI)
2. Right-click with chisel → chiseling screen
3. Place base block in input slot → browse all variants
4. Click variant to preview (cycles through single block, row, 3×3 grid)
5. Take result from output slot
6. **Left-click** placed blocks with chisel to cycle through random variants in-world

### Key features:

- **Connected textures** — toggle with bottom-right button
- **Stairs & slabs** — every variant has stair/slab forms
- **No durability** — chisel never breaks

### Block types (30+ variants each):

- Stone (bricks, tiles, carved, pillars, panels, runed, mosaic)
- Glass (all colors + connected clear, borders, panels)
- Wood (grain variants, carved panels, ornate borders)
- Wool & Carpet (woven, knitted, cross-stitched)
- Concrete (smooth, paneled, tiles, bordered)
- Nether materials (glowstone, nether bricks, blackstone)

### Addons:

- **Rechiseled: Create** — 100+ decorative Create block variants + mechanical chisel
- **Rechiseled: AE2** — autocraft Rechiseled blocks via AE2's Chiseling Pattern Encoder

## Supplementaries — Vanilla+ Functional Decoration

Decoration blocks that serve a purpose:

| Block           | Function                                    | How                                                                           |
|-----------------|---------------------------------------------|-------------------------------------------------------------------------------|
| <b>Jar</b>      | Stores 1 item visually                      | Right-click with item to put in, empty hand to take out                       |
| <b>Signpost</b> | Directional sign pointing to named location | Right-click with tag item to set destination name. Arm points toward Waystone |
| <b>Faucet</b>   | Fluid flow decoration                       | Place above fluid tank. Right-click to toggle flow                            |

| Block               | Function                       | How                                                                         |
|---------------------|--------------------------------|-----------------------------------------------------------------------------|
| <b>Weather Vane</b> | Shows wind direction/speed     | Place outside. Rotates in real-time                                         |
| <b>Item Shelf</b>   | Wall-mounted display (3 items) | Right-click with item to place, sneak-right-click to remove                 |
| <b>Rope</b>         | Climbable, place in mid-air    | Right-click to place knot, add more rope below. Breaks when top knot broken |
| <b>Flute</b>        | Craftable instrument           | Right-click to play note. Villagers react                                   |

### Practical builds:

- Kitchen: Item shelves for ingredients, jars for flour/sugar, faucet over sink
- Entrance: Signposts to “Mine”, “Nether Portal”, “Village”
- Garden: Weather vane, rope swing from tree
- Study: Jars with glowing items as lamps

## Macaw’s — Complete Building Set

Eight modules covering structural elements in every wood type.

**Bridges:** 6 slabs + 1 stick in bridge pattern. Right-click to extend, shift-right-click to shorten. Straight or rope variants. All 10+ wood types.

**Doors:** Special doors per wood — glass, slatted, stable (top half opens independently), cottage, shop, paper. Craft with matching wood + relevant material.

**Roofs:** Roofing blocks — slopes, ridges, gutters. 3 slabs in triangle pattern for slope piece. Auto-connects corners.

**Windows:** Diamond pane, framed, barred, tropical, stained glass. Craft with glass + wood.

**Fences & Walls:** Picket, stockade, horse, wired, and highley fences + matching gates, plus stone/brick walls. Each a distinct look.

**Lights and Lamps:** Vanilla-styled lamps, torches, street lamps, paper lamps, and wall lamps — one for every wood type. Craft with matching wood + glowstone/lantern material.

**Stairs:** New stair styles (Terrace, Skyline, Compact, Bulk, Loft), handrails for stairs, and balconies/platforms.

**Paintings:** Custom hand-drawn paintings to fit your world — 35 new in 1.1.0 (farm scenes, the Creaking, pets, coral reefs) + 22 updated.

**Building tip:** Macaw's roofs + Rechiseled stone + Macaw's windows = cohesive aesthetic.

## Diagonal Fences — Seamless Corners

Connects vanilla fences, fence gates, and walls **diagonally** — the classic corner-post look without gaps. Client-side rendering, no config.

**Integration:** Complements Macaw's building set (Macaw's fences are separate blocks, unaffected) and Forgematica schematics — diagonal corners render correctly in-place. Requires Puzzles Lib (already in pack).

## Create Deco

Industrial decoration for Create-native builds — from catwalks to shipping containers.

- **Block sets** in 6 metals (andesite, zinc, iron, industrial iron, copper, brass): support blocks/wedges, train hulls, catwalks + railings + stairs, cage lamps (4 light colors), sheet metal, windows/panes, bars, chain-link fences, coin stacks, doors, ladders
- **Shipping containers** — all 16 dye colors, function like Create Item Vaults with a fresh look
- **Bricks** — 8 colors (red, scarlet, blue, umber, dean, dusk, verdant, pearl), each with stairs, slabs, and cut variants
- **Decals** — hazard / restricted-area signs

**Placement:** Decoration available from Phase 1 materials (iron/zinc/copper/brass). Pairs with Create (Wave 1) and Rechiseled: Create for industrial detailing. All recipes in EMI; no config needed.

## Handcrafted — Furniture

| Furniture        | Function                          | Craft                    |
|------------------|-----------------------------------|--------------------------|
| <b>Chairs</b>    | Right-click to sit                | Slabs + planks           |
| <b>Tables</b>    | Right-click to place items on top | Varies by type           |
| <b>Shelves</b>   | Wall-mounted item display         | Wood + relevant material |
| <b>Curtains</b>  | Right-click to open/close         | Wool + stick             |
| <b>Desks</b>     | Table + drawer storage            | Slabs + planks           |
| <b>Cupboards</b> | Wall-mounted aesthetic storage    | Wood + glass             |
| <b>Sofa</b>      | Multi-block seating               | Wool + wood              |
| <b>Dresser</b>   | Tall drawer storage               | Wood + relevant material |

**Tip:** Place in your most-used rooms — kitchen, dining, bedroom. Use shelves to display favorite loot (Twilight Forest trophies, unique drops, rare Simply Swords weapons).

## Fetzi's Displays — Item Showcases

Display your hard-earned items in style. Craft a **Workbench** (check EMI) to make all display types.

| Display Type                  | What it does                       | Best for                              |
|-------------------------------|------------------------------------|---------------------------------------|
| <b>Glass Cloches</b>          | Glass dome on pedestal — 16 colors | Trophy items, rare drops, gems        |
| <b>Katana Stands</b>          | Vertical weapon rack               | Simply Swords weapons, boss drops     |
| <b>Wall Holders + Plaques</b> | Wall-mounted display with name tag | Named weapons, “GOD SWORD” vibes      |
| <b>Racks</b> (4 types)        | Horizontal/vertical item holders   | Tool organization, collection display |

**Scale, rotation, position:** Items in displays can be scaled, rotated, and repositioned. Check the [Wiki](#) for exact controls.

**Integration with Wave 3 gear:** Glass cloches and katana stands are ideal for showcasing unique drops. Mount your best Simply Swords / Arsenal / Relics finds as trophies. Wall holders with plaques let you name your legendary weapons — “Flamebane, Bane of the Ender Dragon.”

## Lili's Pottery — Decorative Earthenware

Part of the [Let's Do](#) cozy mod series. Adds pottery blocks for interior decoration.

**What it adds:**

| Category                 | Variants                                   |
|--------------------------|--------------------------------------------|
| <b>Clay</b>              | 6 new clay types found in various biomes   |
| <b>Pots &amp; Vases</b>  | 7 variants in old and new clay styles      |
| <b>Urn &amp; Amphore</b> | Functional storage — save items from death |

**How:**

1. Find **new clay types** in world (beaches, rivers, various biomes)
2. **Knead** clay at crafting table → raw pot forms
3. **Bake** in furnace → hardened pottery
4. **Paint** patterns → finished decorated pottery

**Integration with Handcrafted:** Both add decorative blocks but cover different aesthetics. Handcrafted is rustic furniture (chairs, tables, shelves). Lili's Pottery is earthenware (pots, vases, urns). Together they provide a full interior decoration toolkit.

## Laser Bridges & Doors — Energy Barriers

Futuristic laser barriers created by powering Laser Source blocks with redstone.

**What it adds:**

| Block                           | Function                                                    |
|---------------------------------|-------------------------------------------------------------|
| <b>Laser Source Block</b>       | Projects a horizontal laser bridge when powered by redstone |
| <b>Laser Fence Source Block</b> | Projects a vertical laser barrier                           |

**How it works:**

1. Craft a **Laser Source Block** — End Crystal + 7 iron ingots + glass (check EMI)
2. Power with **redstone** — laser length scales with signal strength (max 15 blocks by default, configurable)
3. **Dye** with right-click to customize color — any dye works, dispenser-compatible too
4. Convert between laser bridge ↔ laser fence in a crafting grid

**Key features:**

- **Dynamic length** — changes in real-time as redstone signal strength varies (use comparators, levers at different levels)
- **Blocks cut through** — configurable list of blocks lasers can replace. Default: air only. Tune in [config/laserbridges.json](#)
- **Dispenser dyeing** — automate color changes with redstone and dispensers

**Recipe gate:** Requires an End Crystal — this is post-End-dragon content. Appropriate for Phase 2+ base building.

**Integration with Redstone & Tech:**

- **AE2 level emitters** — emit redstone when item quantities hit thresholds → auto-close laser gates when storage is low or auto-open when conditions are met
- **Create redstone links** — wireless redstone across your base to control laser arrays from a central panel
- **Comparators** — use container fullness to control bridge length dynamically

**Chill living fit:** Purely decorative/functional — no combat interaction, no progression pressure. Dye-customizable to match base aesthetics (warm amber lasers for cozy builds, cool blue for tech areas).

## Session Plan — Wave 6

| Session                                |  | Focus                                                                                                                                                        |  | What to Build            |  |
|----------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|--|
| <b>Session 1 — Rechiseled basics</b>   |  | Craft chisel. Convert stone, wood, glass into decorative variants. Try connected textures                                                                    |  | Decorative block library |  |
| <b>Session 2 — Structural elements</b> |  | Craft Macaw's roofs, windows, doors in your base wood type. Build a cohesive room                                                                            |  | Cohesive room aesthetic  |  |
| <b>Session 3 — Furniture</b>           |  | Handcrafted chairs, tables, shelves in kitchen and bedroom. Item shelves for loot display                                                                    |  | Functional interior      |  |
| <b>Session 4 — Details</b>             |  | Supplementaries jars, signposts, weather vanes. Laser Bridges & Doors for energy barriers and decorative lighting. Rechiseled: Create for factory aesthetics |  | Decorative details       |  |

# Wave 7 — Space Exploration

## Core Space Mod

| Mod                          | Role                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Northstar Redux</b> 0.6.4 | Create-based space exploration — player-built rockets, science unlocks, oxygen + temperature systems, Moon/Mars/Mercury/Venus |

Northstar Redux (by Astronauts-of-Create) is the maintained continuation of Create: Northstar — space exploration built on the Create mod. Rockets are player-built, superglued Create contraptions; progression runs through a science system (Telescope → Astronomical Readings → Astronomy Table → Space Atlas) that unlocks planets. Oxygen and temperature management, Titanium/Martian Steel/Tungsten materials, and four reachable planets: Moon, Mars, Mercury, Venus.

**No worldgen:** Northstar generates no Overworld structures — the launch infrastructure is entirely player-built, and planets are separate dimensions. The “Create: Northstar - Structures” addon (planet-surface craters/dungeons) exists but is **not installed** in this pack.

**Dependencies:** none new — Create 6.0+ and GeckoLib are already in the pack (Wave 1). No Potentials API or Sky Aesthetics needed (those shipped with Stellaris only).

**Update policy:** Northstar is in active early development (0.6.x). Updates may change item IDs and invalidate Return Tickets. Pin 0.6.4; back up worlds before updating; after an update, re-run the quest validation pipeline ([tools/ftbquests/generate.py](#) + [validate.py](#)) so stale item IDs surface immediately.

## Integration with Tech Mods

| Layer   | Northstar Redux ↔ Tech Mod Interaction                                                 |
|---------|----------------------------------------------------------------------------------------|
| Power   | Mekanism cables can power Northstar machines. Create rotational → Mekanism → Northstar |
| Energy  | Mekanism universal cable connects to Northstar machines (energy API compat)            |
| Storage | AE2 ME drives store Northstar materials; autocrafting builds rocket components         |

| Layer          | Northstar Redux ↔ Tech Mod Interaction                                                |
|----------------|---------------------------------------------------------------------------------------|
| Ore Processing | Mekanism 5x ore multiplication works on Northstar ores (titanium, martian iron, etc.) |
| Fuel           | TFMG diesel/gasoline compatibility for rocket fuel (native compat)                    |
| Food           | Farmer's Delight meals eatable in space when oxygen is present                        |

## Northstar — Interplanetary Travel

Create-native space exploration. Build a rocket from Create blocks and Northstar parts, glue it together, fuel it, and fly.

### Core concepts:

| Concept                | What it means                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Rocket Station</b>  | Assembles the rocket and sets the destination (unlocked via science)                                        |
| <b>Rocket Controls</b> | Interface with the rocket's systems — also accepts redstone for launch automation                           |
| <b>Science</b>         | Telescope → Astronomical Readings → Astronomy Table (spend as science points) → Space Atlas unlocks planets |
| <b>Oxygen</b>          | No atmosphere off-world. Separator + Filler + Sealer keep you and your base alive                           |
| <b>Heat</b>            | Heat shielding scales with rocket size; Temperature Regulator + Martian Steel suit for hot planets          |
| <b>Planets</b>         | Moon → Mars → Mercury → Venus. Each requires more science and better shielding                              |

### Phase 1: Getting to Space

#### Step 1 — Gather resources:

- Collect iron, copper, redstone, glass, and Create materials (andesite alloy, brass)
- Process **Titanium** (raw titanium ore → ingot) for advanced parts — check EMI for the chain

#### Step 2 — Build the launch site:

| Machine                 | Function                                        | Priority |
|-------------------------|-------------------------------------------------|----------|
| <b>Rocket Station</b>   | Assembles the rocket and picks the destination  | First    |
| <b>Rocket Controls</b>  | Rocket system interface (redstone-controllable) | Second   |
| <b>Oxygen Separator</b> | Extracts breathable oxygen                      | Third    |
| <b>Oxygen Filler</b>    | Refills suits and contraptions with oxygen      | Fourth   |

| Machine       | Function                 | Priority |
|---------------|--------------------------|----------|
| Oxygen Sealer | Pressurizes sealed rooms | Fifth    |

### Step 3 — Build the rocket:

1. Build the hull from Create blocks + **Rocket Thruster** + **Rocket Combustion Chamber**
2. Superglue the whole assembly into one contraption
3. Place **Rocket Controls** on board
4. Use the Rocket Station to assemble and set the destination

### Step 4 — Science unlocks:

1. Craft a **Telescope** and take **Astronomical Readings**
2. Spend them at the **Astronomy Table** as science points
3. The **Space Atlas** shows unlocked planets — Moon is first

### Step 5 — Launch:

1. Equip the **Iron Space Suit** (full set; fill oxygen at a Filler)
2. Enter the rocket, launch via the Rocket Controls
3. Land on the Moon — watch your oxygen

## The Moon — First Destination

Safest first destination. No atmosphere, low gravity, no mobs.

### What to do:

- Mine **moon stone** and **lunar sapphire** (shards → polished)
- Find **glowstone ore** and rich veins of standard ores
- Place a **Waystone** immediately; set up a sealed base (Oxygen Sealer + Filler + Temperature Regulator if needed)
- Return with a **Return Ticket** or by launching from the Moon

## Other Planets

| Planet         | Hazards                        | Key Resource        | Notes                            |
|----------------|--------------------------------|---------------------|----------------------------------|
| <b>Moon</b>    | None (oxygen required)         | Lunar Sapphire      | First destination. Safe practice |
| <b>Mars</b>    | Worms, toads, cobras, moths    | Martian Steel, Iron | Best permanent off-world base    |
| <b>Mercury</b> | Extreme heat, raptors, roaches | Tungsten, Titanium  | Heat shielding required          |
| <b>Venus</b>   | Acid plumes, mimics, scorpions | Volcanic Ash, ores  | Hottest planet — endgame target  |

## Off-World Machines

| Machine                         | Function                       | Why you need it                     |
|---------------------------------|--------------------------------|-------------------------------------|
| <b>Oxygen Filler</b>            | Refills suits/contraptions     | Required on every planet            |
| <b>Oxygen Sealer</b>            | Pressurizes sealed rooms       | Keeps bases breathable              |
| <b>Oxygen Detector</b>          | Reads oxygen levels (redstone) | Automation for life support         |
| <b>Temperature Regulator</b>    | Regulates heat/cold            | Mercury, Venus, and any cold base   |
| <b>Atmospheric Concentrator</b> | Concentrates breathable air    | Denser life support for large bases |
| <b>Rocket Waypoint</b>          | Precise landing coordinates    | Land at your base, not wherever     |

## Instance swap checklist (Stellaris → Northstar Redux)

In the game instance:

1. Remove the `stellaris` jar, `potentials` jar, and `sky-aesthetics` jar (Stellaris-only dependencies)
2. Add the `northstar-redux` 0.6.4 jar (Create 6.0+ and GeckoLib are already installed)
3. Do **not** install “Create: Northstar - Structures”
4. Existing worlds: Stellaris structures remain as inert blocks and its items vanish from inventories — a fresh world is recommended for the swap

## Integration With Tech Base

| System                | How it connects                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Power</b>          | Mekanism generators at Moon base. Universal cables power Northstar machines                                        |
| <b>Ore processing</b> | Bring planet ores (titanium, martian iron, tungsten) back to Mekanism line. 5x works on Northstar ores             |
| <b>Storage</b>        | AE2 ME drives store Northstar materials. Autocraft rocket components. Quantum link or P2P tunnel to off-world base |
| <b>Transport</b>      | Create trains move rocket materials to the launch site. Conveyors feed the Rocket Station                          |
| <b>Fuel</b>           | TFMG diesel/gasoline compatibility for Northstar rocket fuel (native compat)                                       |
| <b>Backpacks</b>      | Netherite-upgraded backpacks carry oxygen tanks and supplies                                                       |
| <b>Food</b>           | Farmer’s Delight meals eatable in space with oxygen present                                                        |

| System          | How it connects                                                                            |
|-----------------|--------------------------------------------------------------------------------------------|
| <b>Gear</b>     | Space suit pieces are enchantable via the vanilla enchanting table                         |
| <b>MekaSuit</b> | Pending compat bridge — modules make the MekaSuit a space suit once it tracks Redux 0.6.4+ |

## Progression Placement

Space belongs in **Phase 2** (hours 40–200). Prerequisites:

- Create rotational power (brass age — rockets and machines need it)
- Mechanism basic processing (Enrichment Chamber)
- Small AE2 network
- Good weapons/armor (Wave 3/5 gear helps with planet mobs)

Start with Moon missions and a sealed Moon base, then Mars. Mercury and Venus are late Phase 2.

## Session Plan — Wave 7

| Session                         | Focus                                                                                                   | What to Build              |
|---------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Session 1 — Ground prep</b>  | Build Rocket Station + Rocket Controls. Process Titanium. Assemble and superglue the rocket             | Launch infrastructure      |
| <b>Session 2 — First launch</b> | Craft Telescope, take Readings. Craft Iron Space Suit, fill oxygen. Launch to the Moon                  | First off-world base       |
| <b>Session 3 — Moon base</b>    | Sealed base: Oxygen Sealer + Filler. Waystone. Mine lunar sapphire. Chunk Loader                        | Sustainable Moon operation |
| <b>Session 4 — Mars</b>         | Smelt Martian Steel. Land on Mars, mine martian iron and titanium. Temp base + weapons for hostile mobs | New resources              |
| <b>Session 5+ — Deep space</b>  | Mercury (tungsten, heat). Venus (endgame).                                                              | Interplanetary logistics   |

| Session |  | Focus                                                        |  | What to Build |  |
|---------|--|--------------------------------------------------------------|--|---------------|--|
|         |  | Martian Steel Space Suit.<br>Space Atlas + Return<br>Tickets |  |               |  |

**Tip:** Space is the most resource-intensive activity. Don't rush it. Each planet should feel like a meaningful expansion. Note: the MekaSuit is **not** a Northstar space suit by default — you still need the Iron/Martian Steel suits for oxygen. The *Mekanism × Create: Northstar* compat mod adds MekaSuit space-suit modules; it's queued for the pack once it tracks Redux 0.6.4+.

---

# Create World

All mods installed and every wave configured — here's how to start a proper world. These settings keep the pack survival-pure, the chat clean, and the worldgen fully modded.

## Before Creating

2. **Optional data packs** — for the structure-spacing tweak, place [rarer-better-dungeons/](#) in `saves/<world>/datapacks/` after creating the world (see [datapacks/README.md](#)). The Create aircraft datapacks ([Create Immersive Aircrafts Datapack 1.21.1 - 3.0.zip](#) + [Create-Man-of-Many-Planes-1.1.zip](#)) go in the same folder to enable Create-machine aircraft crafting.

## World Creation Settings (Singleplayer → Create New World)

| Setting                        | Value           | Why                                                                                                                                                                                                   |
|--------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Game Mode                      | <b>Survival</b> | The pack is survival-only — every block placed and item crafted must be earned                                                                                                                        |
| Difficulty                     | <b>Normal</b>   | Easy-to-standard curve. The real difficulty ramp is Wave 5 combat mods (Cataclysm) and dimension boss fights, not vanilla mobs                                                                        |
| <b>Allow Commands (Cheats)</b> | <b>OFF</b>      | Survival purity — no <code>/gamemode</code> escapes, no creative shortcuts; quest rewards stay earned. Cheats can't be enabled later without opening the world to LAN, so set everything you need now |
| World Type                     | <b>Default</b>  | Terralith + Tectonic overhaul the overworld automatically. Don't pick Superflat, Single Biome, or Amplified — they break modded worldgen                                                              |
| Seed                           | Leave empty     | Random is fine — worldgen mods guarantee interesting terrain either way                                                                                                                               |
| Structure Generation           | <b>ON</b>       | YUNG's collection, Waystones, and every dimension need it                                                                                                                                             |

**Game Rules → Chat → disable all options** (Command Feedback, Command Block Output, Reduced Debug Info, Show Death Messages, Show Recipe Messages, Announce Advancements). Keeps the chat clean — progression popups still appear via Fancy Toasts.

## After First Launch

1. First world load takes a few extra minutes while Terralith, Tectonic, YUNG's, and Streams Reflowing generate the initial chunks — that's normal. On world creation, Streams Reflowing queries terrain height across the spawn area first; the loading bar may sit at 0% for up to ~1 minute — normal, not a hang. Chunk loading may also be slower for the first couple minutes of play while height data settles.

## Mod Count Summary

| Wave                | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wave 0 — Foundation | 11   | 10   | 22    | Dependencies (12), Performance & Rendering (12) + FastWorkbench<br>–Sodium Extras –Sodium Options API (Aug 2026 — broken dep chain: sodiumoptionsapi→RSO→Sodium 0.8.12+→incompatible with sodiumoptionsapi), Infrastructure (3) + Simple Recall Potion (travel) — all the nuts & bolts that make the game run (+1 shaderpack, +26 resource packs). Wave 0.5 UI swap: –Immersive UI –Smooth GUI –Tooltip Overhaul –Controlling –First Person Model +Modern UI; Aug 2026 Wave 0.5 review: –Easy Anvils (hard conflict with anvil mixins) –Searchables (orphan |

| Wave |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|------|--|------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      |  |      |      |       | <p>library — does nothing standalone) +Prism (Item Borders required dep)</p> <p>+Puzzles Lib (dep documented — 5 consumers) +Dark Mode Everywhere→Mindful Darkness swap +Stylish Effects –Status Effect Bars Reforged (Aug 2026 swap — SEBR’s duration bars natively covered)</p> <p>–Dynamic Crosshair (vanilla-coupled — context features unreliable on modded worlds), Aug 2026</p> <p>–JadeColonies (moved to Wave 1.5), Aug 2026;</p> <p>–Traveler Tool Belt (removed per user request, Aug 2026);</p> <p>–EMIffect +JEED +MEED (effect-wiki swap + curated content, Aug 2026); –Mindful Darkness → Dark Coffee GUI + Modded Coffee GUI resource packs (full swap, Aug 2026 — GUI dark mode is now a resource pack, not a mod); +Fast Noise (replaces NoisiumForked — archived predecessor, Aug 27 2026); +C2ME (NeoForge) — parallel chunk gen, pairs with Fast Noise + Lithium, alpha backport flagged (Aug 27 2026); –Immersive</p> |  |

| Wave                         |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|--|------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |  |      |      |       | <p>Armor HUD +Detail<br/>Armor Bar<br/>Reconstructed, Aug 2026;<br/>–Advancement Plaques<br/>–Toast Control +Fancy<br/>Toasts +Fancy Notify, Aug<br/>29 2026 (2:2 swap,<br/>counts unchanged);<br/>+Grassier Grass 1.4.5 (Sep<br/>2026 — Wave 0 Visual<br/>QoL, animated grass<br/>blades, client-only, zero<br/>deps); –Countered’s<br/>Smooth F5 (removed per<br/>user request, Sep 2026 —<br/>BTP keeps third-person<br/>rotation, zero deps<br/>removed);<br/>+JustEnoughBreeding<br/>3.2.1 (Sep 2026 — EMI<br/>breeding lookup, client-<br/>only, zero deps)</p> |
| Wave 0.5 — Resource<br>Packs |  | 5    | 0    | 5     | <p>EMF + ETF + Continuity +<br/>Sinytra Connector + Cull<br/>Leaves — client-side<br/>support mods for Fresh<br/>Animations (EMF/ETF),<br/>Overlay’s (Continuity/<br/>Connector), Better Leaves<br/>perf (Cull Leaves); +11<br/>resource packs –Patricx<br/>+Faithful –FA Extensions<br/>(Aug 2026 resource pack<br/>review; removed —<br/>Details addon corrupted<br/>animal models, user<br/>report) –Armored Legacy<br/>(removed per user<br/>request, Sep 2026)</p>                                                                                               |

| Wave                 |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|--|------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wave 1 — Tech        |  | 21   | 5    | 26    | Create + 13 addons (incl. Power Grid, Gunsmithing, Protection Pixel, Ornithopter Glider, Train Utilities, Railways Navigator, Interiors, Train Parts, Central Kitchen), Mekanism + Generators + MekaJadeUpgrades, AE2, Advanced Finders, Advanced Chimneys + NTGL, GeckoLib, Framework, ForgeEndertech, Kleiders deps, Mekanism: Weapons (Phase 3 energy weapons — polonium/ atomic-alloy gated)    |
| Wave 1.5 — Colony    |  | 6    | —    | 6     | MineColonies + 5 addons: Byzantine Styles, Stylecolonies, Create: Colony Logistics, ColonyLink, JadeColonies                                                                                                                                                                                                                                                                                        |
| Wave 2 — Exploration |  | 38   | 7    | 46    | YUNG's (12), Terralith/ Tectonic, Serene Seasons + Serene Seasons Plus + Gabou's Libs, Darker Depths, Upgrade Aquatic, dimensions (3), End overhaul (3 + Nullscape dp + 5 deps), navigation, aircraft, ships, hang glider, MoMP addon, Vehicle Upgrade (mount/ boat QoL), Incendium (Nether biome overhaul), <b>MNS (Nether structures)</b> , Immersive Machinery (utility machines), Field Guide + |

| Wave                     |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|--|------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |  |      |      |       | 2 required companions (Immersive Overlays, Item Descriptions), Bumblezone (bee dimension) + Resourceful Lib, Streams Reflowing (hydrology — carved streams/lakes + river currents, Terralith/ Tectonic first-class; restores the Dynamic Waters role removed Aug 27) +Compass to Map 2.0.4 (Sep 2026 — auto JourneyMap waypoints from Explorer's Compass finds, zero required deps)                                                                                                                                                     |
| Wave 3 — Equipment Magic |  | 8    | 9    | 18    | Simply Swords, Simply More, Too Many Bows, Arsenal (RPG Series), Relics, Immersive Armors, Easy Magic + 9 deps (Placebo, Patchouli, Simply Tooltips, Fzzy Config, Ranged Weapon API, Spell Engine, Shield API, Curios API, HammerLib) — -Project MMO -PMMO × Farmer's Delight Compat (Aug 27 2026 — reverted; not what the user wanted) -Skills Mastery Reimagined -Pufferfish's Skills -Pufferfish's Attributes (Aug 27 2026 — skills dropped; no tech-vibe tree, user decision) -Apotheosis -Apothic Compats -3 Apothic modules +Easy |

| Wave                    |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|--|------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |  |      |      |       | Magic (Aug 26 2026 removal), pins SS 1.63.0 + SM 1.2.3 +Tyz's Skills→Improvable Skills (Aug 31 2026 — skills restored then swapped same-day: Improvable Skills 21.0.2 + HammerLib 21.0.14 — vanilla-XP skill book replaces Tyz's separate XP system per user request)                                                                      |
| Wave 4 — Food & Farming |  | 20   | —    | 20    | Farmer's Delight + 9 addon mods + Neo Bee Fix + Comfortable Campfires + Smarter Farmers + Create: Café + Starcatcher + Starcatcher's Delight + Better Fishtanks + Spice of Life: Carrot Edition + Crops Love Rain + Hybrid Delights (Aug 2026 review; count corrected 14→18 — fishing trio + one Delight addon never reflected in summary) |
| Wave 5 — Combat & Mobs  |  | 8    | 5    | 13    | Better Combat, L_Ender's Cataclysm, Enchantment Descriptions, Create Big Cannons + Advanced Technologies addon, Torchmaster, CBC Enchanced Shells, Cataclysmic Combat                                                                                                                                                                      |
| Wave 6 — Building       |  | 15   | 3    | 18    | Rechiseled, Supplementaries, Macaw's (8), Handcrafted, Fetzi's Displays, Lili's                                                                                                                                                                                                                                                            |

| Wave                       |  | Mods       | Deps      | Total      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|--|------------|-----------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |  |            |           |            | Pottery, Laser Bridges & Doors, Diagonal Fences + Create Deco + Rechiseled: Create, Rechiseled: AE2 + Moonlight, Resourceful, Fusion –Building Wands (removed per user request, Aug 28 2026) ...                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Wave 7 — Space Exploration |  | 1          | 0         | 1          | Northstar Redux (Create 6.0+ and GeckoLib already in pack — no new dependencies; no worldgen structures) ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Total</b>               |  | <b>183</b> | <b>45</b> | <b>228</b> | All confirmed NeoForge 1.21.1 (Aug 2026 Wave 0.5 review: –Easy Anvils –Searchables +Prism +Rhino; Dark Mode Everywhere→Mindful Darkness swap + Puzzles Lib dep documented, Aug 2026; –Stellaris –Potentials API –Sky Aesthetics –CC:Tweaked +Northstar Redux, Aug 2026; +Real Camera –Real Camera (removed per user request, Sep 2026 — mods –1, deps unchanged); Day Counter Plus reverted — Day Counter + Hud Texts restored; +BetterGrassify + Forgified Fabric API + Sky Aesthetics; +Serene Seasons Plus + GlitchCore + Gabou’s Libs - Chunky - When Dungeons Arise - Structory: Towers, Aug 2026; +Smarter Farmers |

| Wave |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|------|--|------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      |  |      |      |       | <p>(Sodium Extras + Sodium Options API removed Aug 2026 — broken dep chain on NeoForge 1.21.1) + Diagonal Fences + Create: Café, Aug 2026 mod review; +Arsenal (RPG Series) + Shield API, Enchanting Runes dep corrected Bundle API→FLIB, stable pins SS 1.63.0 + SM 1.2.3, classes TODO closed (Aug 25 2026 Wave 3 review);</p> <p>–Enchanting Runes –FLIB (duplicates the pack’s socketing role; unproven 1.21.1), Aug 25 2026;</p> <p>–Mindful Darkness → Dark Coffee GUI + Modded Coffee GUI resource packs (full swap, Aug 2026 — mods –1, deps unchanged, +2 resource packs → 24);</p> <p>+Fa...; –Apotheosis</p> <p>–Apothic Compats –3 Apothic modules –Loot Beams –Nirvana Library</p> <p>–Common Network +Easy Magic (Aug 26 2026);</p> <p>–Dynamic Waters (worldgen — falling off the world, user report, Aug 27 2026); +Fast Noise (replaces NoisiumForked — archived predecessor, Aug 27 2026); +C2ME (NeoForge, parallel chunk gen — alpha backport,</p> |  |

| Wave |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|------|--|------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      |  |      |      |       | <p>Aug 27 2026); –Skills Mastery Reimagined</p> <p>–Pufferfish’s Skills</p> <p>–Pufferfish’s Attributes (Aug 27 2026 — skills dropped; no tech-vibe tree, user decision);</p> <p>–Project MMO –PMMO × Farmer’s Delight Compat (Aug 27 2026 — reverted; not what the user wanted) +CBC</p> <p>Enchanced Shells</p> <p>+Cataclysmic Combat (Aug 28 2026 — Wave 5 review) +Macaw’s Fences and Walls (Aug 28 2026 — Wave 6 5th Macaw’s module) –Building Wands (removed per user request, Aug 28 2026)</p> <p>+Macaw’s Lights and Lamps +Macaw’s Stairs</p> <p>+Macaw’s Paintings (Aug 28 2026 — Wave 6 6th–8th Macaw’s modules, all zero-dep); –Immersive Armor HUD +Detail</p> <p>Armor Bar</p> <p>Reconstructed, Aug 2026 (1:1 swap, zero deps, counts unchanged);</p> <p>–Advancement Plaques</p> <p>–Toast Control +Fancy Toasts +Fancy Notify, Aug 29 2026 (2:2 swap, counts unchanged);</p> <p>+Tyz’s Skills + Tyz’s Skills</p> <p>Dark Theme, Aug 31 2026 (skills restored — zero-dep; +1 mod, +1 resource</p> |  |

| Wave |  | Mods | Deps | Total | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
|------|--|------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      |  |      |      |       | pack → 25) → swapped Aug 31 to Improvable Skills + HammerLib<br>–Tyz’s Skills Dark Theme (vanilla-XP skill book per user request; deps +1 → 43, resource packs 25 → 24); +Streams Reflowing (Sep 2026 — Wave 2 hydrology, zero deps); +Upscaled Mekanism 32x v1.5 (Sep 2026 — Mekanism 32x layer, zero deps, resource packs 24 → 25); +Grassier Grass 1.4.5 (Sep 2026 — Visual QoL animated grass blades, client-only, zero deps); –Countered’s Smooth F5 (removed per user request, Sep 2026 — mods –1, deps unchanged); +AttributeFix (Sep 2026 — Wave 0.5 bar engine, lifts vanilla attribute caps for Overflowing Bars; +Prickle +Bookshelf deps); –Real Camera (removed per user request, Sep 2026); Dark Loading Screen Neoforge 1.0.1 trialed and reverted Sep 2026 (user report: doesn’t theme the initial/early loading screen — DLS restored, counts unchanged); +JustEnoughBreeding 3.2.1 (Sep 2026 — EMI breeding lookup, client- |  |

| Wave |  | Mods | Deps | Total | Notes                                                                                                                                                 |  |
|------|--|------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|      |  |      |      |       | only, zero deps; mods +1, deps unchanged);<br>+Compass to Map 2.0.4 (Sep 2026 — compass→JourneyMap glue, zero required deps; mods +1, deps unchanged) |  |

## Keybindings Reference

| Keybind               |  | Suggested                           |  | Mod                     |  | Wave |
|-----------------------|--|-------------------------------------|--|-------------------------|--|------|
| View Recipes          |  | <a href="#">R</a>                   |  | EMI                     |  | 0    |
| View Uses             |  | <a href="#">U</a>                   |  | EMI                     |  | 0    |
| Open Fullscreen Map   |  | <a href="#">J</a>                   |  | JourneyMap              |  | 0    |
| Toggle Minimap        |  | <b>Check default</b>                |  | JourneyMap              |  | 0    |
| Open Waystone Menu    |  | <a href="#">Ctrl+Shift+W</a>        |  | Waystones               |  | 0    |
| Open Backpack         |  | <a href="#">B</a>                   |  | Sophisticated Backpacks |  | 0    |
| Zoom                  |  | <a href="#">C</a>                   |  | Sodium                  |  | 0    |
| Open Action Center    |  | <a href="#">Ctrl+K</a>              |  | Modern UI               |  | 0.5  |
| Toggle Dynamic Lights |  | <a href="#">Ctrl+L</a>              |  | LambDynamicLights       |  | 0    |
| Open Curios Inventory |  | <a href="#">Y</a>                   |  | Curios API              |  | 0    |
| View Weapon Ability   |  | <a href="#">Ctrl</a> (hold)         |  | Simply Swords           |  | 3    |
| Open Create Ponder    |  | <a href="#">W</a> (hold over block) |  | Create                  |  | 1    |
| Open Town Hall GUI    |  | Right-click                         |  | MineColonies            |  | 1.5  |

**Pro tip:** Open **Options** → **Controls** → **Key Binds** and browse categories at the bottom. Every mod's binds are grouped under its name. Set the ones above first, then explore others as you install new waves.

**Modern UI note:** it also binds zoom to [C](#) (OptiFine-style). Disable it in the Action Center ([Ctrl+K](#)) so Sodium's zoom keeps the key.

# Progression Through All Waves

| Phase                         | What you'll be doing                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 2</b> (hours 40–200) | Full Create factory (conveyors, deployers, sequenced crafters, trains). Mekanism 3x–5x ore processing. AE2 ME network with autocrafting. Explore Twilight Forest bosses. Unique weapon/armor drops from dungeons and bosses. First Cataclysm boss. TFMG oil refining and diesel engines. Build with Rechiseled, Macaw's, Handcrafted. Small Ships for ocean travel. Immersive Aircraft for aerial exploration. Complete quest chapters. Northstar Redux Moon and early planets |
| <b>Phase 3</b> (hours 200+)   | Mekanism fusion reactor + MekaSuit. AE2 full autocrafting CPU chains. Endgame unique drops + max-enchanted gear. Twilight Forest final bosses. Cataclysm endgame bosses. End dimension exploration. Mars and Venus space bases. Monumental building projects. Create mega-factories. Perpetual optimization                                                                                                                                                                    |

## General Tips

### Finding Help

- **EMI (R/U)** — always your first stop. Hover any item, press R for recipe, U for uses
- **Ponder (W over Create blocks)** — interactive animated tutorials. Better than any wiki
- **Mods screen** — every installed mod's config button (Better Modlist adds badges/search)
- Most mods link to their wiki or Discord from their Mods screen entry

### Losing Progress

You won't. GraveStone Mod preserves your items on death. No creeper grief data pack prevents base damage. The pack is designed so you can walk away for a month and come back to exactly the world you left.

## Java & Performance

Minecraft 1.21.1 requires **Java 21**. See [Wave -1](#) for JDK installation and JVM arguments.

### Performance tips:

- ZGC garbage collector with generational mode (`-XX:+UseZGC -XX:+ZGenerational, -Xms = -Xmx`) — full argument set and RAM table in [Wave -1](#)
- Distant Horizons handles far rendering — keep vanilla render distance at 10
- ModernFix `dynamic_resources = true` for fastest startup
- If you see missing textures, disable `dynamic_resources` temporarily

## Cross-Wave Integration Cheat Sheet

| If you want to...                 | Combine these mods                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------|
| Auto-build a blueprint            | Create Schematicannon + Blueprint and Quill                                         |
| Power Create from Mekanism        | Create Crafts & Additions electric motor/ alternator                                |
| Bridge TFMG oil to AE2            | TFMG Energy Converter + Mekanism universal cable                                    |
| Store colony resources digitally  | AE2 Storage Bus on MineColonies warehouse                                           |
| Feed colony kitchen automatically | Create conveyor → builder's chest + Farmer's Delight meals                          |
| Autocraft Mekanism chemicals      | Applied Mekanistics + AE2 Pattern Provider                                          |
| Cook from your ME network         | AE2 Export Bus on Farmer's Delight cooking pot                                      |
| Get combat buffs before bosses    | Farmer's Delight meal + Brewin' And Chewin' drink (effects stack)                   |
| Find the best loot                | Explorer's Compass → YUNG's structure → loot → unique weapons and enchantment books |
| Build a train network             | Steam 'n' Rails tracks → Train Utilities stations → Railways Navigator boards       |
| Fly between Aether islands        | Immersive Aircraft airship + Aether portal                                          |
| Defend colony perimeter           | Create Big Cannons on walls + Guard Towers                                          |
| Explore oceans efficiently        | Small Ships cog → Ocean's Delight ingredients → YUNG's Ocean Monuments              |
| Process space ores                | Northstar titanium/martian steel → Mekanism 5x line → AE2 storage                   |

| If you want to... |  | Combine these mods                                                                                                     |  |
|-------------------|--|------------------------------------------------------------------------------------------------------------------------|--|
| Build a megabase  |  | Create Blueprint + Schematicannon<br>(structure) + Rechiseled blocks (aesthetic) +<br>Handcrafted furniture (interior) |  |